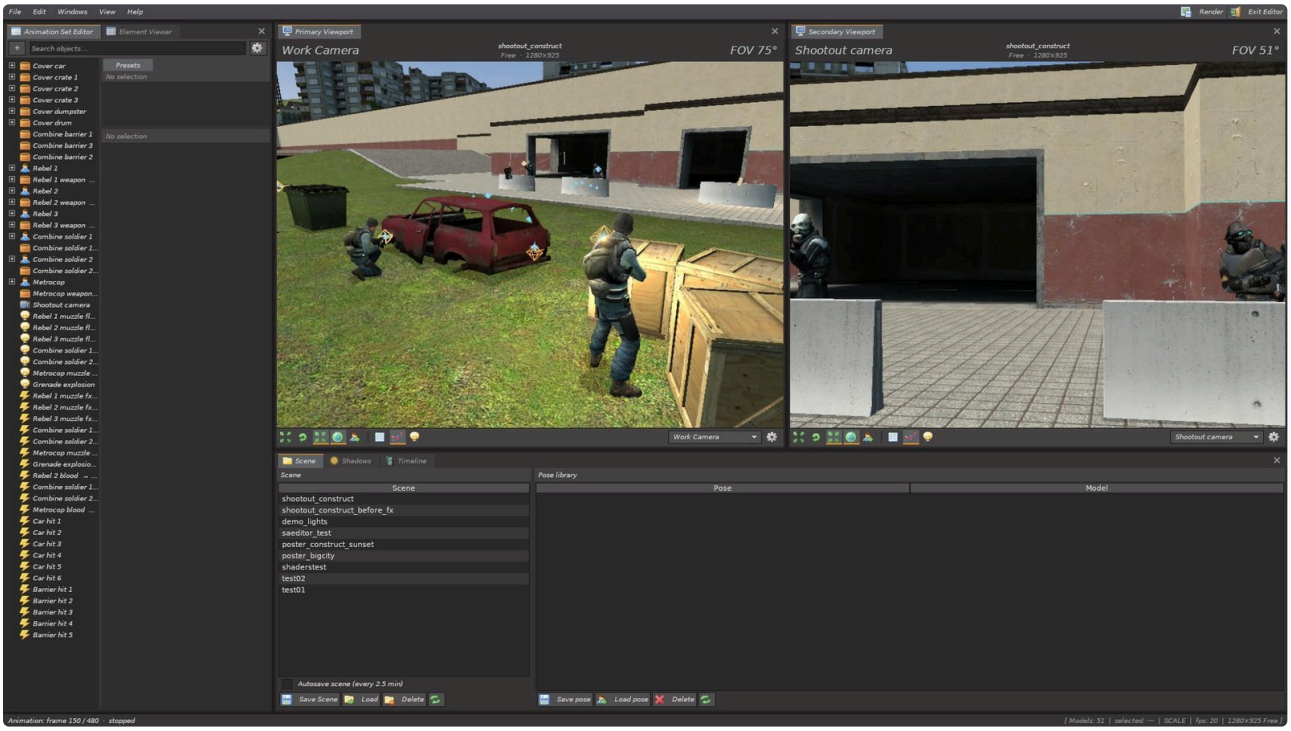
# SPeditor

## The complete guide



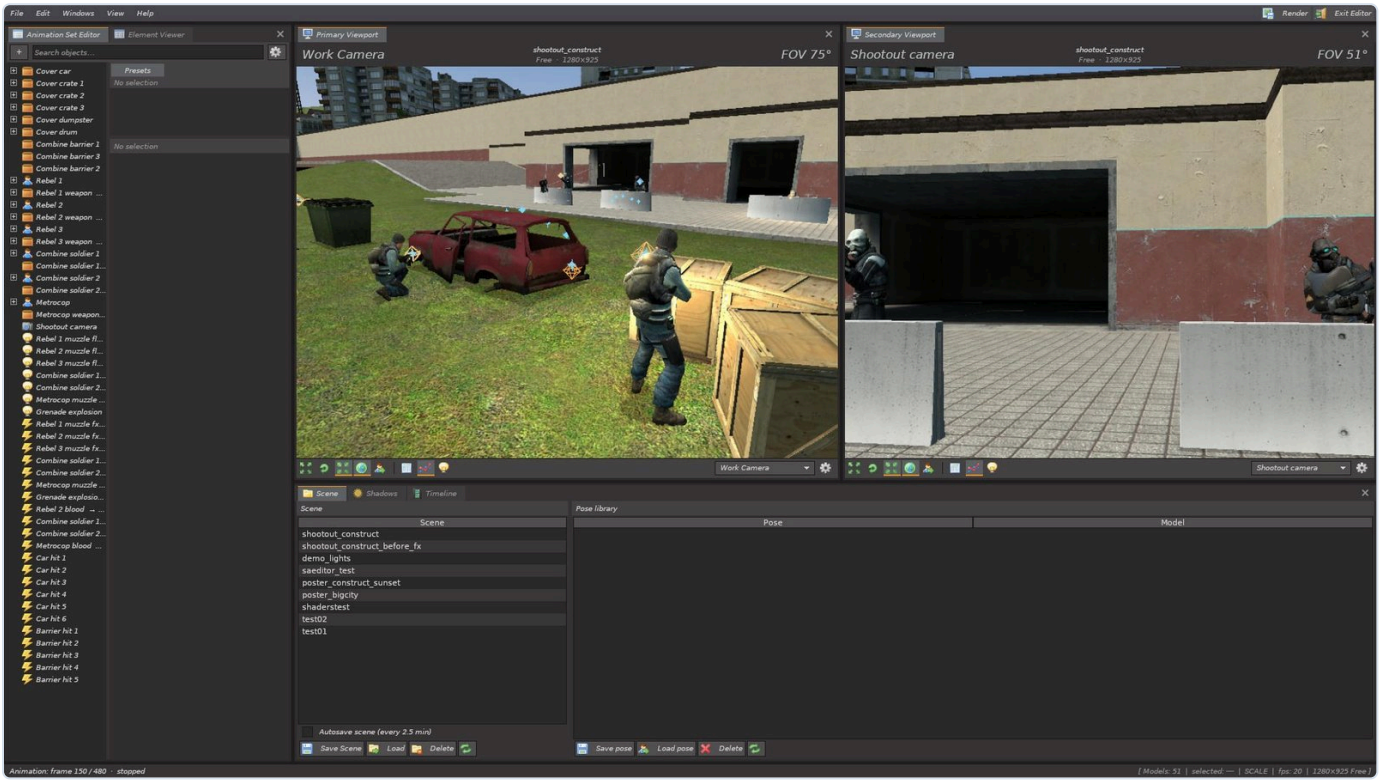

Poster, scene and animation studio for Garry's Mod · version a0.7.1 · October 2026
steamcommunity.com/sharedfiles/filedetails/?id=3817852435

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# SPeditor Wiki

SPeditor turns Garry's Mod into a small film studio: posters, still scenes and short animations, built on any map with the working habits of Source Filmmaker. This wiki explains every window, button and hotkey.




**The editor.** Animation Set Editor on the left, the Primary and Secondary viewports in the middle and on the right, the Scene / Shadows / Timeline panel at the bottom. The scene is the bundled example `shootout_construct`.

## Where to start

### Installing and opening

Subscribe, open the editor, server access.

### Your first poster

From an empty map to a PNG in five steps.

### The workspace

Windows, tabs, docking and layouts.

### Posing characters

Gizmo, bones, IK, faces, pose library.

### Lights

Spot and point lights, 19 light kinds, soft shadows.

### Rendering

Up to 3840 px, depth of field, camera effects.

### Animation (SAeditor)

Keys, timeline, camera cuts, recording gameplay, video.

### Problems and answers

The questions players ask most.

## What SPeditor can do

- **An SFM-like workspace:** dockable windows, tabs you can drag out, saved layouts, ten interface themes, your own accent colour and font.
- **Two viewports** with a frame gate, composition guides and fixed exposure; a free work camera and scene cameras you can look through.
- **Posing:** a gizmo for objects and for every single bone (fingers too), inverse kinematics that keeps the feet planted, bones locked to other bones, face flexes, eye target, skins and bodygroups, a pose library.
- **Objects:** colour and opacity, texture replacement, stretching, attaching a weapon to a hand, wearing clothing that bends with the body.
- **Lights:** point and spot lights, soft edges and penumbra, glow and a visible beam, light masks, 19 ready-made kinds, a three-point rig.
- **Effects:** explosions, fire, smoke, blood and sparks from the game and from Workshop addons, frozen at any moment.
- **Render** to PNG or JPEG up to 3840 px with smoothing, depth of field, soft shadows, camera effects and Workshop shaders.
- **SAeditor:** keyframe animation of everything in the scene (up to an hour long), a curve editor, the Motion Editor (an edit over a time selection with falloff), procedural breathing and blinking, bone keys with ghosts, camera cuts and an in-game player, recorded gameplay (machinima), lip sync, imported model animations and Stop Motion Helper saves, sounds and voices, video export at any frame rate up to 120 fps with motion blur.
- **Objects made with other tools** — Ragdoll Mover poses, bonemerged models, lamps of other addons — can be taken into the scene.
- Scenes, poses and layouts are saved; undo and redo for every edit; English and Russian interface, other languages as community language packs.

# Made with SPeditor






This wiki describes SPeditor **a0.7.1** (alpha); the changes of this version are marked. The screenshots show the English interface; switch the language in the top right corner of this page.

# Installing and opening

SPeditor is a normal Workshop addon. It works in singleplayer and on servers, on any map.

## Installing

1. Open the SPeditor page in the Steam Workshop and press **Subscribe**.
2. Start Garry's Mod (or restart it if it was running) and load any map. `gm_construct` and `gm_flatgrass` are good for a start.
3. Optional: subscribe to GShader Library if you want Workshop shaders in your renders (see Shaders).

## Opening the editor

There are three ways. All of them toggle: the same action closes the editor again.

### C menu

Hold `C` and click the **SPeditor** icon on the desktop of the context menu.

### Q menu

Hold `Q`, open the **SPeditor** tab on the right, the **SPeditor** item, then **Toggle editor**.

### Console

Type `speditor_toggle`. Bind it to a key to open the editor in one press: `bind f6 speditor_toggle`.




**Q menu → SPeditor.** The **Language** and **Theme** items next to it choose the interface language and look.

To leave the editor press ▸ **Exit Editor** in the top right corner, use ▸ **File → Exit Editor**, or run `speditor_toggle` again. Your scene stays on the map.

## Interface language

Q menu → SPeditor → **Language**, or the console: `speditor_language en` / `speditor_language ru`. The language is applied the next time the editor is opened.

## On a server

By default only admins may use the editor on a server. The server owner changes this with `speditor_access`:

| VALUE | WHO MAY OPEN THE EDITOR |
|---|---|
| `admin` | Admins and superadmins (default) |
| `superadmin` | Superadmins only |
| `all` | Every player |

Each player has a limit of models, lights, cameras and effects. A player without access sees the message “You do not have access to SPeditor on this server”.

## Where your files are

Everything SPeditor saves is in `garrysmod/data/speditor/`:

| FOLDER | CONTENTS |
|---|---|
| `renders/` | Pictures, frame sequences and videos you rendered |
| `scenes/<SteamID>/` | Saved scenes and autosaves (on a server they are kept on the server) |
| `poses/` | The pose library |
| `layouts/` | Window layouts; `_last.json` is the layout at the moment you closed the editor |
| `sounds/` | Your own WAV, MP3 and OGG files for SAeditor |
| `windows.json` | Window properties such as “Always on top” |

> **!** Garry's Mod lets addons write only a few file types. Videos are therefore saved as `name.avi.dat`: remove `.dat` from the name and it is a regular AVI file.

# Your first poster in five steps

A short walk through the whole workflow. Every step links to the page that explains it in detail.

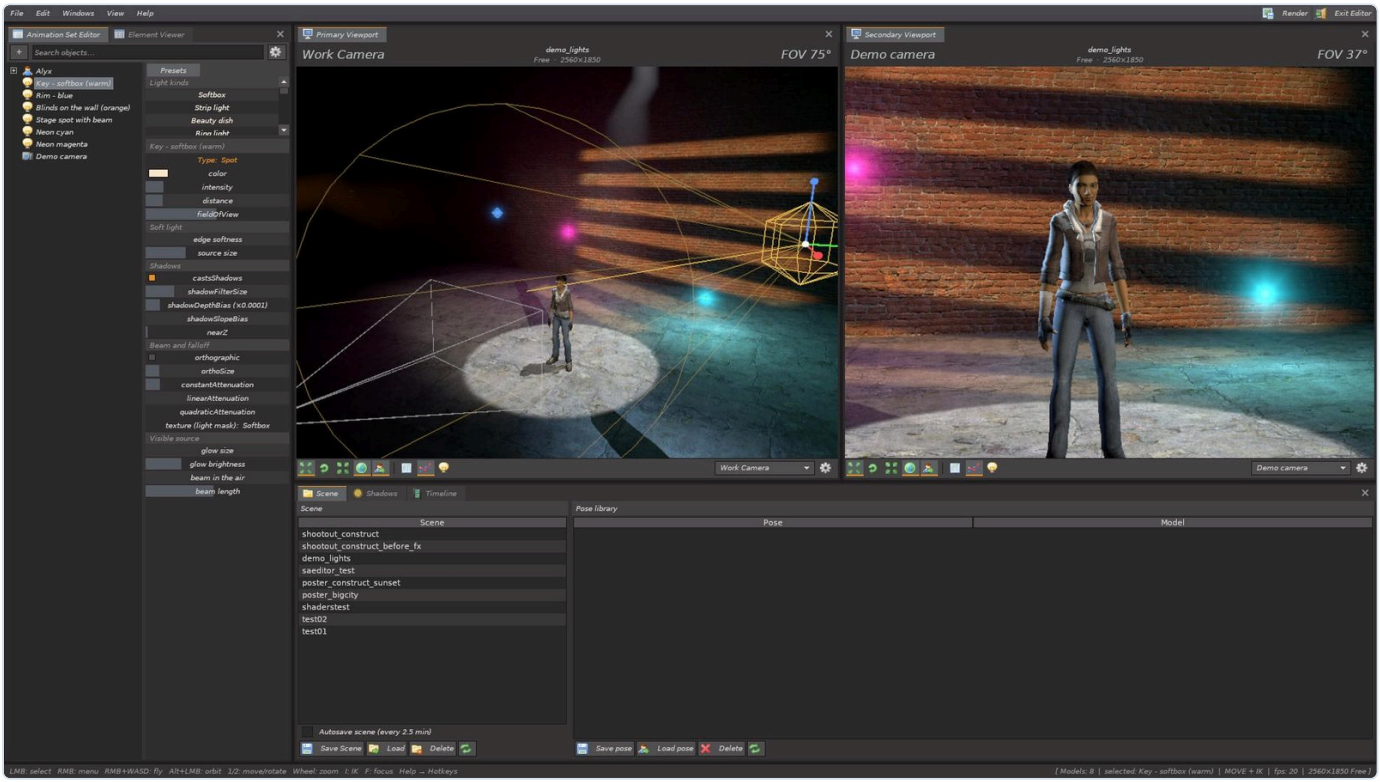
1. **Add a character.** Press **+** at the top of the Animation Set Editor and choose ▸ **Model...**. Find a model in the Model Browser (for example `alyx`) and double-click it. It appears in front of the work camera.

2. **Pose it.** Click the model in the viewport. Press `1` to move, `2` to rotate. Click a bone marker (a small cube on a joint) to pose a single bone; press `I` for inverse kinematics and drag a hand. See Posing.

3. **Light it.** **+** → ▸ **Soft light** → ▸ **Softbox** for the key light, then a ▸ **Rim light** behind the character. Or select the model and use ▸ **Create 3-point light rig**. See Lights.

4. **Frame it.** Fly the work camera to a good view (hold `RMB` and use `W` `A` `S` `D`), then **+** → ▸ **Spawn Camera**. The camera is created exactly where you look. Choose a frame aspect such as ▸ **2:3 (poster)** in the ▸ **View** → **Frame aspect** menu. See Cameras.

5. **Render.** Press ▸ **Render** in the top right corner, pick your camera, the size (up to 3840 px) and the format, and press ▸ **Render**. The file appears in `garrysmod/data/speditor/renders/`. See Rendering.



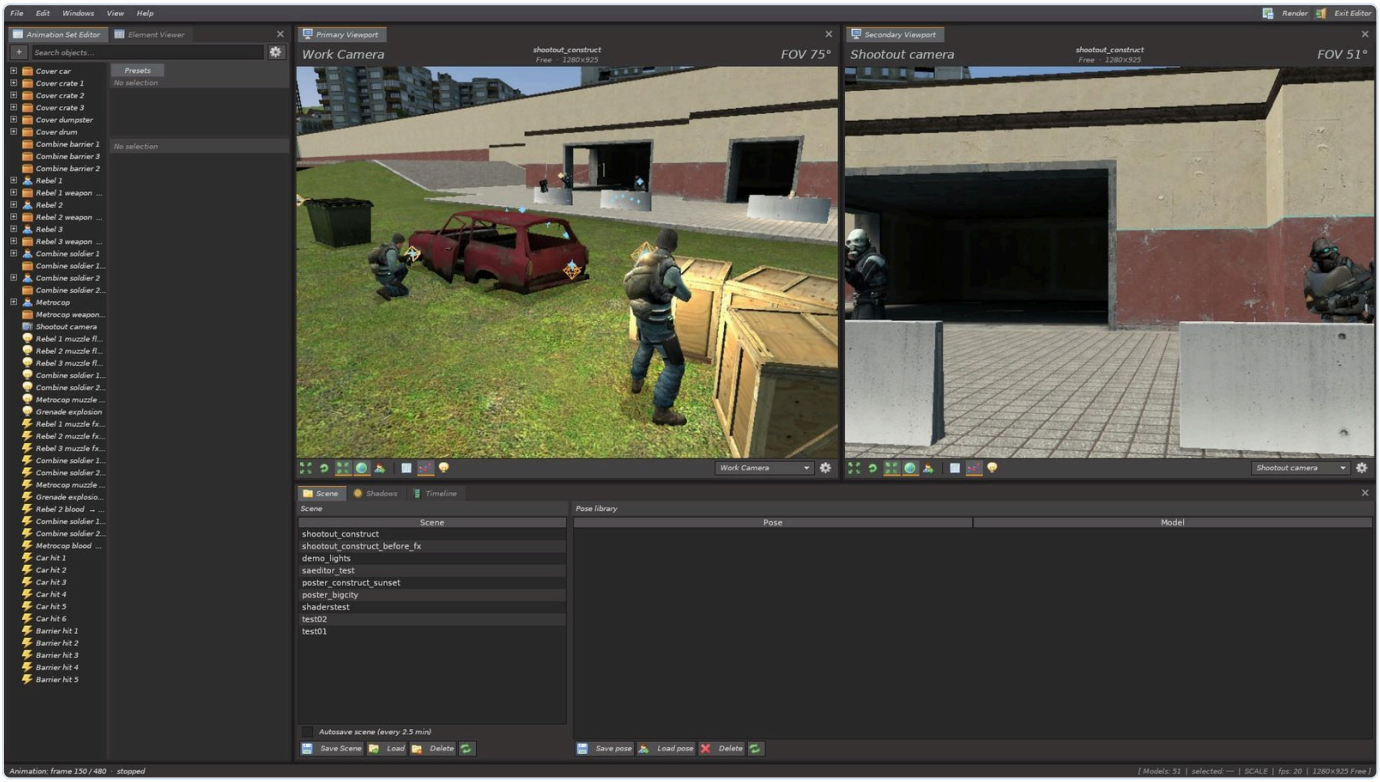

**The result of steps 1–4.** Left: the work camera shows the light cones and the selected softbox with its gizmo. Right: the Secondary Viewport looks through the scene camera — this is what the render will show.

Save often: ▸ **File → Save Scene...**. A scene keeps the objects, the poses, the lights, the camera effects, the frame settings and the animation. Switch on ▸ **Autosave scene** in the same menu for a backup every 2.5 minutes.

Want to see a finished scene first? Open the **Scene** tab at the bottom and load `shootout_construct` (on the map gm_construct). It is an animated shootout with effects and sounds that comes with the addon.

# The workspace

The editor covers the screen with dockable windows, like Source Filmmaker. Each window holds one or more tabs; tabs can be moved between windows or torn off into a window of their own.




**The default layout (“SFM default”).**

## Parts of the screen

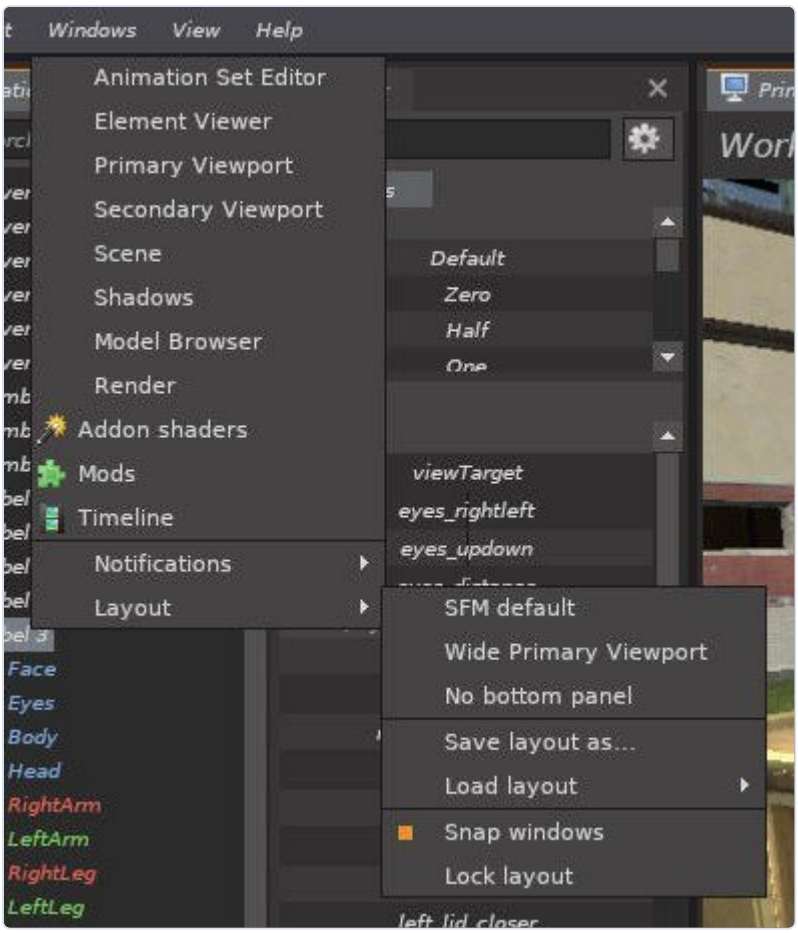
| PART | WHAT IT IS FOR |
| --- | --- |
| Menu bar | **File, Edit, Windows, View, Help** on the left; **Render** and **Exit Editor** on the right. |
| Animation Set Editor | The object tree and the controls of the selected object: bones, face, eyes, light and camera settings. See Objects. |
| Element Viewer | Every property of the selected object as a list of numbers you can type in. |
| Primary Viewport | The main 3D view where you select and edit. Shows the work camera or a scene camera. |

| | |
|---|---|
| Secondary Viewport | A second view, usually through the scene camera, so you see the final frame while you work. |
| Scene · Shadows · Timeline | The bottom window: scene and pose library, shadow settings, and the SAeditor timeline. |
| Status bar | Hints on the left; on the right the number of models, the selection, the tool, FPS and the render size. |

## Windows and tabs

- **Move a tab:** drag it by its tab bar and drop it on the edge of another window (it docks beside it) or on another window's tab bar (it becomes a tab there). Hold `Ctrl` while dropping to keep it floating.
- **Maximize:** double-click the tab bar. Double-click again to restore.
- **Resize:** drag the gap between two docked windows; neighbours follow.
- **Window properties:** right-click the tab bar → ▸ **Window properties** → ▸ **Always on top** keeps the window above the others (the Model Browser is on top by default).
- **Reopen a closed window:** the ▸ **Windows** menu lists all of them.

# Layouts



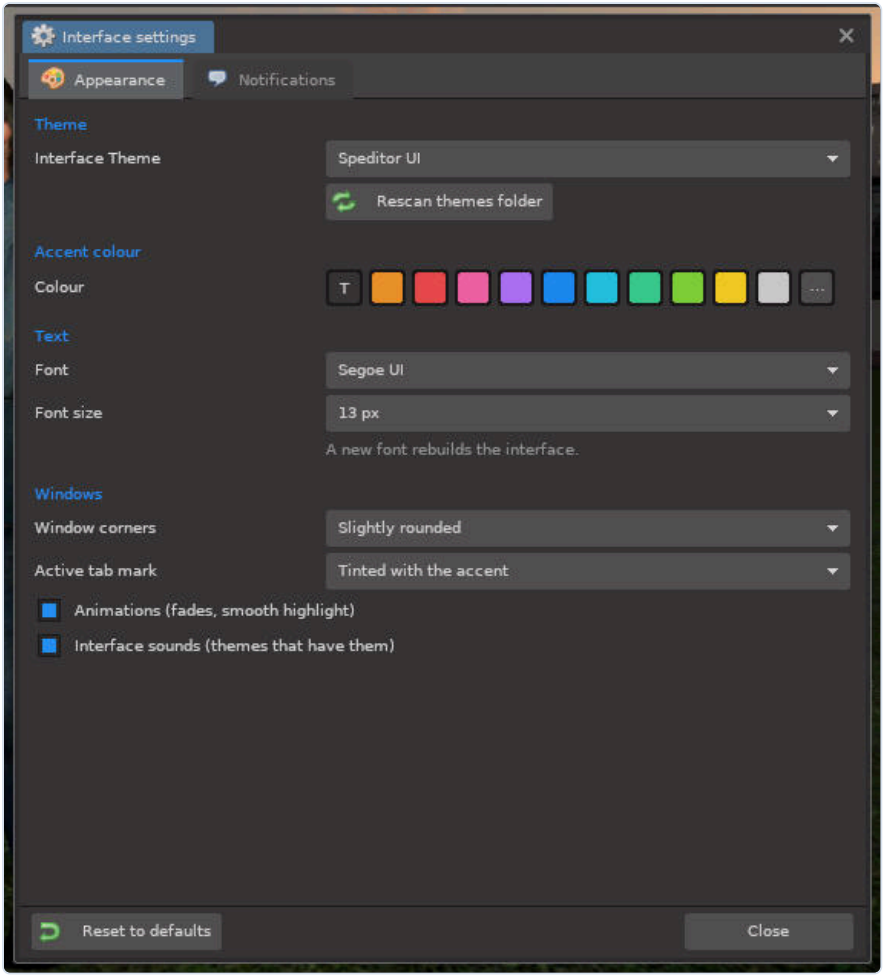

▸ **Windows → Layout** has three presets: **SFM default**, **Wide Primary Viewport** (one large viewport) and **No bottom panel**. ▸ **Save layout as...** keeps your own arrangement under a name; ▸ **Load layout** brings it back.

**Snap windows** makes floating windows stick to edges. **Lock layout** stops windows from being moved by accident.

The editor remembers the layout you had when you closed it and restores it next time.

> On screens narrower than about 1430 px (1366×768, 1280×720, 1024×768) the default layout used to break: the left panel covered the viewport and windows became floating. Since a0.4 the layout fits to any screen, and a saved layout that does not fit the current screen is replaced by the default one.

# Interface settings




▸ **View → Interface settings...** gathers everything about the look of the editor. The **Appearance** tab:

### Theme

The same list as ▸ **View → Interface Theme** (Themes).

### Interface scale

From 50 % (more room for the viewports) to 200 % (2K / 4K screens); by default it follows the screen (1440 lines — 125 %, 2160 — 200 %). The windows keep their places when it changes.

### Accent colour

The colour of selections, active tabs and highlights. *T* takes the theme's own colour, … picks any colour.

### Font · Font size

Any font installed on your computer; a new font rebuilds the interface at once.

### Window corners · Active tab mark

Square or rounded windows; how the tab you are on is marked.

### Animations · Interface sounds

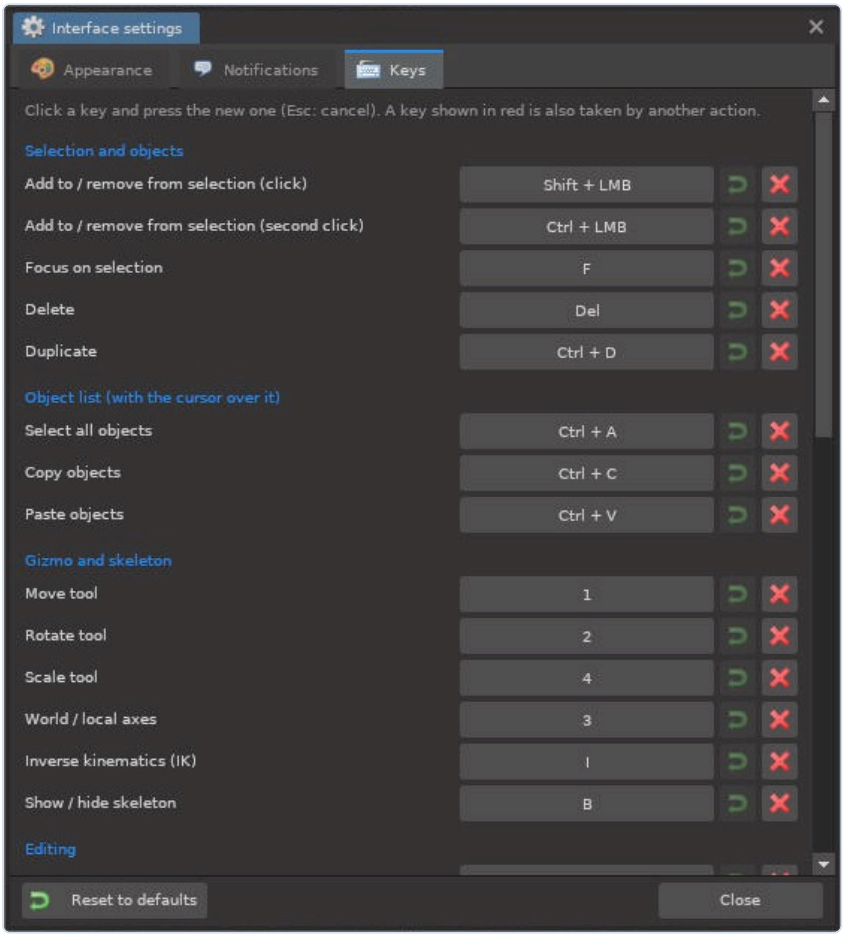
Windows and menus fade in and highlights glide; sounds play in themes that have them.

### Click sound · Window close sound NEW

Your own interface sounds for any theme: listen, or the folder button opens the sound browser — game and addon sounds, or your files from `garrysmod/data/speditor/sounds/`. Empty: the theme's own.

The **Notifications** tab is described on the Themes and notifications page. *Reset to defaults* brings back the stock look.

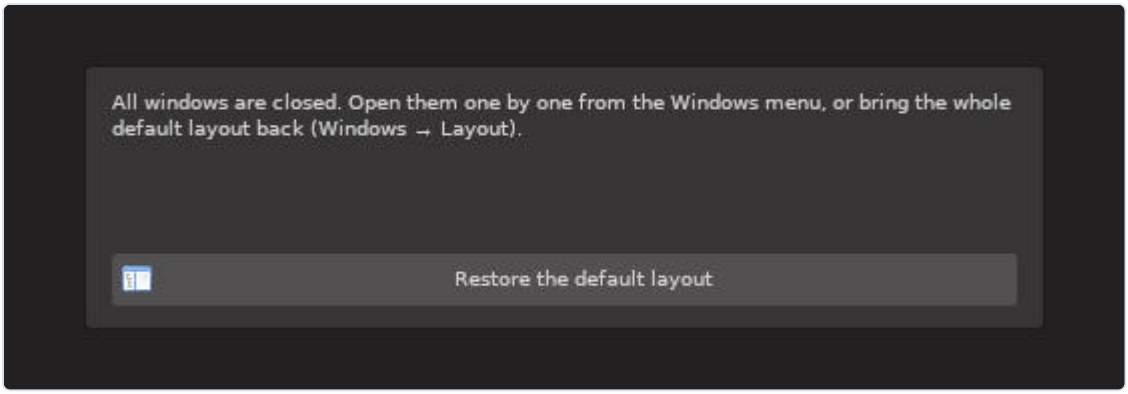
## Your own keys




The **Keys** tab lists every hotkey of the editor and of the mods (SAeditor too). Click the key of an action and press the new one, with `Ctrl` / `Shift` / `Alt` if you like; `Esc` cancels. For a click (adding to the selection) hold `Shift` or `Ctrl` and click.

- A key shown in **red** is also taken by another action — change one of them.
- The arrow button returns the default key, the cross leaves the action without a key; *Reset all keys* is at the bottom.
- Keys of the *Object list* group work only with the cursor over the object list, so `Ctrl` + `C` copies objects there and a pose in the viewport.
- The keys are kept in `garrysmod/data/speditor/keys.json`; ▸ **Help → Hotkeys** always shows the current ones.

## When every window is closed

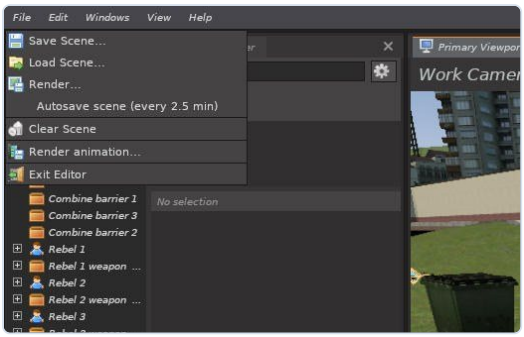



An editor with every window closed no longer looks broken: it shows a card with *Restore the default layout*.

# Menus

A reference of every item in the menu bar.

## File




### Save Scene...

Save the scene under a name.

### Load Scene...

Opens the Scene tab with the list of scenes.

### Render...

Opens the Render window.

### Autosave scene

A backup every 2.5 minutes; several older copies are kept.

### Clear Scene

Deletes every object (asks first).

### Take objects from the map

Ragdolls, props and lamps made with other tools come into the scene (more).

### Stop Motion Helper

Added by SAeditor: import and export of SMH saves (more).

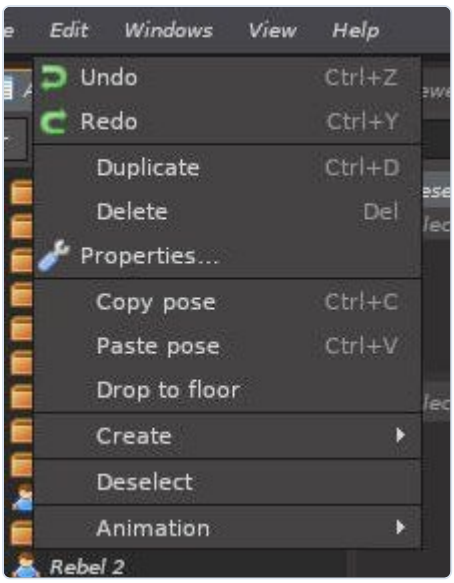
### Render animation...

Added by SAeditor: the animation render.

### Exit Editor

Closes the editor.

# Edit




### Undo / Redo

`Ctrl` + `Z` / `Ctrl` + `Y`. Every edit can be undone, including deleting objects.

### Duplicate / Delete

`Ctrl` + `D` / `Del` for the selection.

### Properties...

The Object Properties window.

### Copy / Paste pose

`Ctrl` + `C` / `Ctrl` + `V`, also between different models with the same skeleton.

### Drop to floor

Puts the object on the surface below it.

### Create

Model, camera, spot or point light, effect, three-point rig.

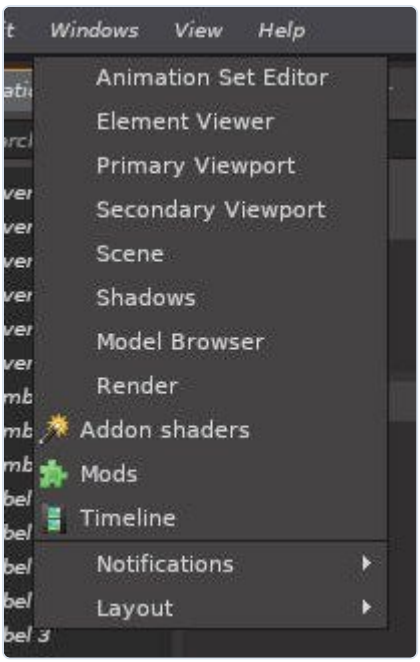
### Deselect

Clears the selection.

### Animation

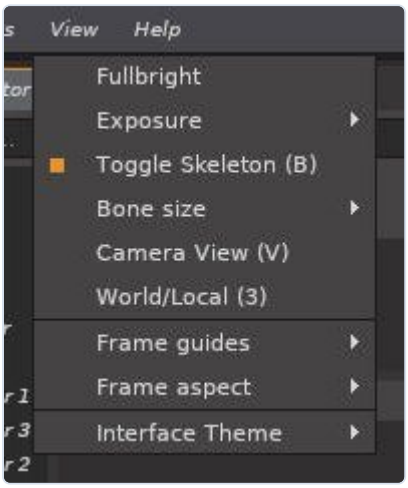
SAeditor commands (keys, import, sounds).

## Windows




Shows any window or tab: Animation Set Editor, Element Viewer, both viewports, Scene, Shadows, Model Browser, Render, Addon shaders, Timeline. **Notifications** sets up the pop-up messages; **Layout** is described on the workspace page.

## View




### Fullbright

Shows the scene without lighting — handy when building in the dark.

### Exposure

Fixed brightness of the viewports or *Auto (engine)*.

### Toggle Skeleton (B)

Bone markers on characters.

### Bone size

Tiny, Small, Medium, Large markers.

### Camera View (V)

Look through the selected scene camera.

### World/Local (3)

Gizmo axes in world or object space.

### Frame guides

Rule of thirds, center cross, safe areas, golden ratio.

### Frame aspect

The shape of the picture (Free, 2:3 poster, A-series and others).

### Interface Theme

See Themes.

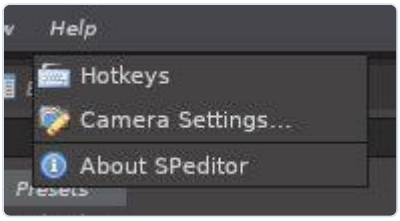
### Interface settings...

Accent, font, corners, animations, sounds, notifications (more).

### IK: keep the feet planted

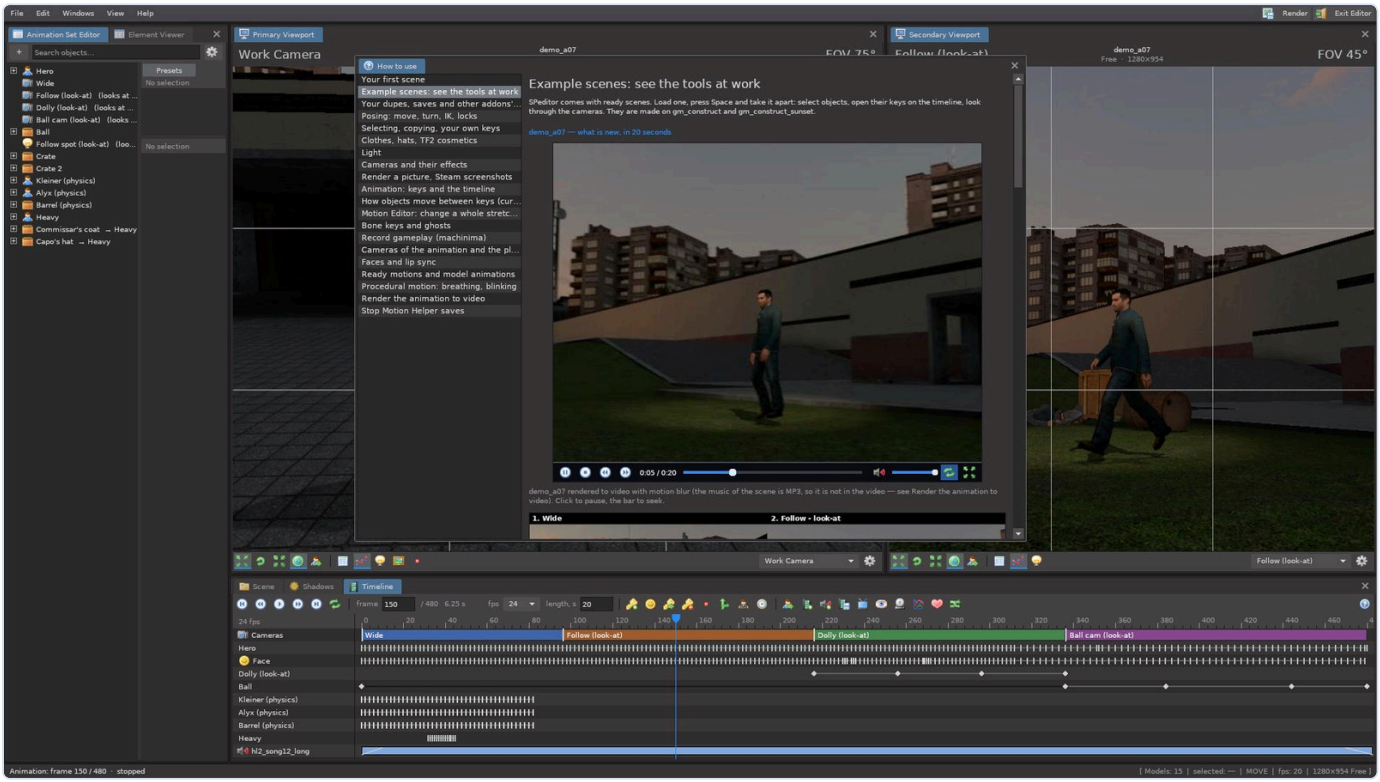
Moving the pelvis with IK on keeps the feet in place (more).

## Help

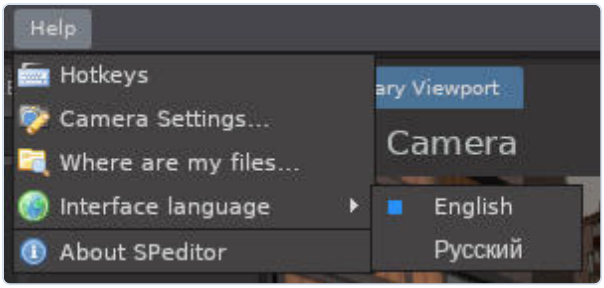


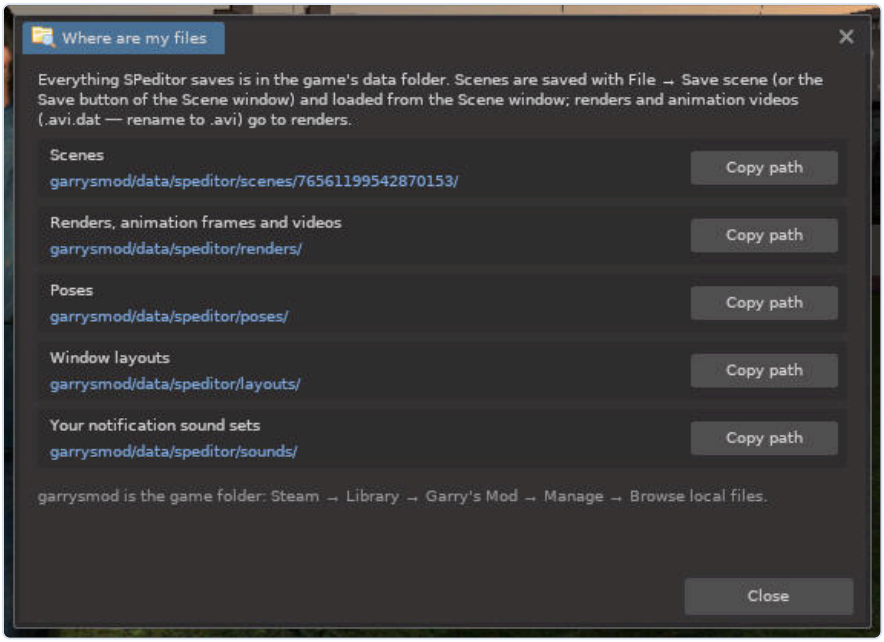

**Hotkeys** lists every key (also those of mods) with a *Change keys...* button, **Camera Settings...** sets the work camera speed and FOV, **About SPeditor** shows the version, the news and the credits.

### How to use




**Help → How to use.** Short illustrated guides inside the game: what a tool does, the steps, the result, live demonstrations (how an object moves between keys, bone keys, camera cuts) and a button that opens the tool. The **?** buttons of the tool windows open their guide. Since a0.7 the demo video plays in a player: pause, stop, seeking, volume, repeat, full screen.



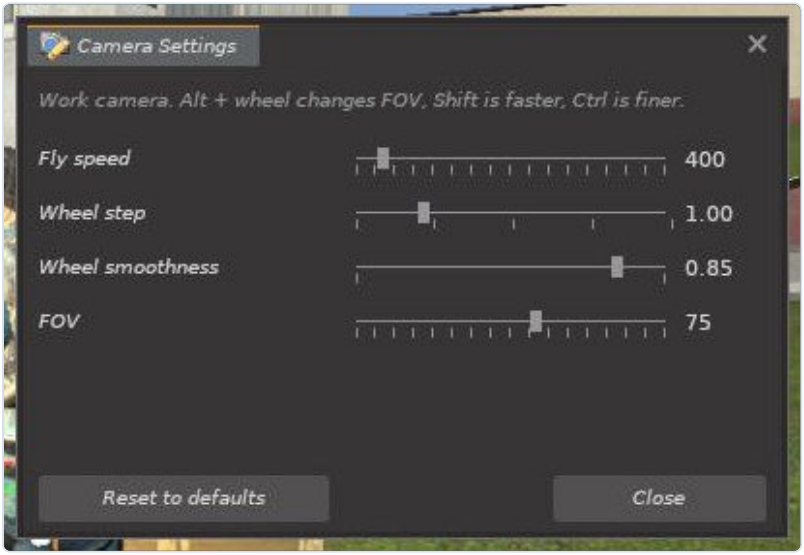

**Where are my files...** shows the folders of scenes, renders and videos, poses, layouts and notification sounds, each with a *Copy path* button; the message after saving a scene also says where the file went. **Interface language** switches the language at once, without reopening the editor: English, Russian and the installed language packs; *Write a translation template* starts a new one.

# Viewports and the work camera

You look at the scene through the **work camera** — a free camera that is not part of the scene and never appears in renders.

## Moving around

| INPUT | ACTION |
|---|---|
| `RMB` + `W` `A` `S` `D` | Fly. `Q` / `E` go down / up, `Shift` is faster, `Ctrl` slower. Moving the mouse with the right button held turns the camera. |
| `Alt` + `LMB` | Orbit around the selection. |
| Wheel | Move forward / back (dolly). |
| `Alt` + Wheel | Change the field of view (zoom lens). |
| `F` | Focus on the selected object or bone. |
| `V` | Look through the selected scene camera; `V` again goes back. |




▸ **Help → Camera Settings...** sets the **fly speed**, the **wheel step** and its **smoothness**, and the **FOV** of the work camera. **Reset to defaults** brings back 400 / 1.00 / 0.85 / 75°.

## The viewport header and toolbar

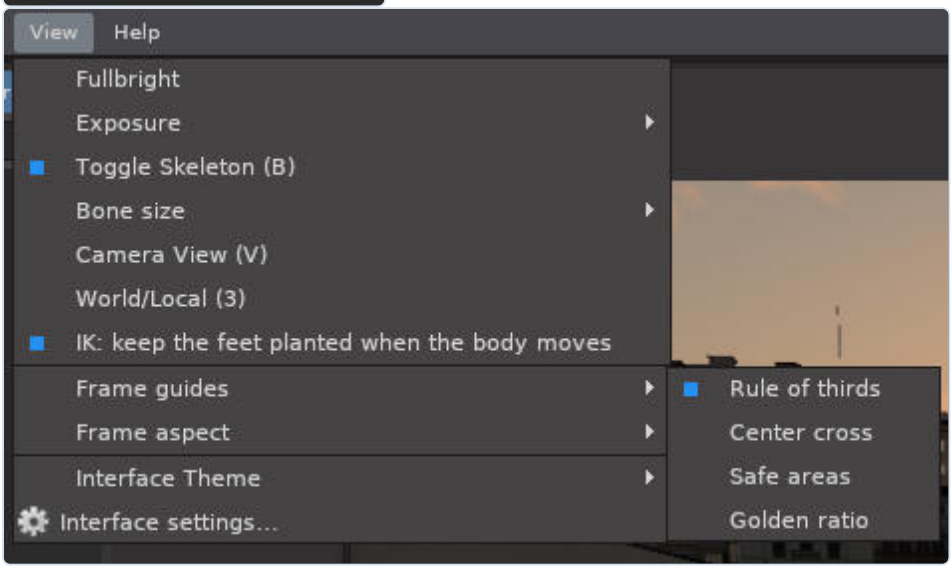
The header shows the camera on the left (*Work Camera* or the scene camera's name), the scene name, the frame aspect and size in the middle, and the field of view on the right. The dark bars

around the picture are the **frame gate**: only what is inside them gets into the render.

The toolbar under the picture holds the tool buttons (move, rotate, scale, world/local axes, IK), view toggles (frame guides, skeleton, fullbright), the camera list and the gear menu.




The **gear menu** of a viewport: Focus selected, Camera Settings, Render, Exposure, Frame aspect, Hide viewport. The camera list chooses what the viewport shows; in the Secondary Viewport **Auto** shows the selected scene camera, and the work view while you look through a camera in the Primary one.



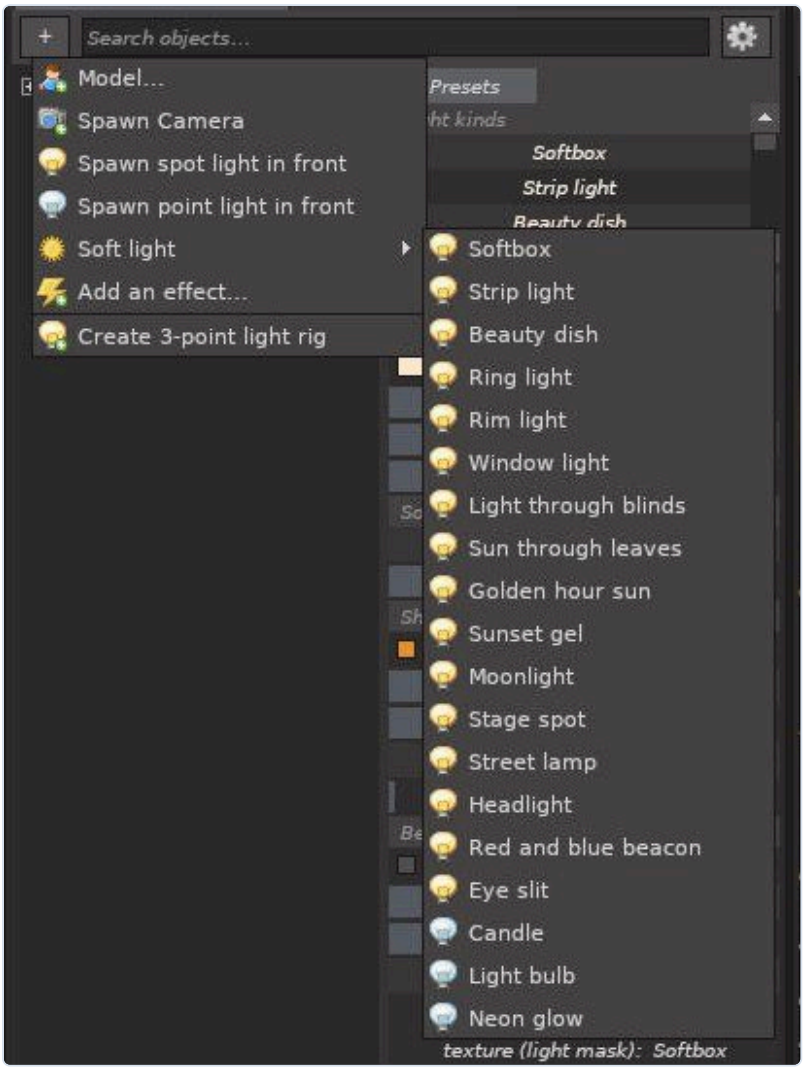

**Frame guides** help with composition: rule of thirds, a center cross, safe areas and the golden ratio. They are drawn only in the viewport, never in the render.

> ✓ The Primary and Secondary viewports pause while the Render window is open, to give the preview all the speed. The FPS counter reads about 20 when the game window is not focused — that is the engine saving power, not SPeditor.

# Objects

A scene is made of four kinds of objects: **models** (props and characters), **lights**, **cameras** and **effects**. All of them are listed in the Animation Set Editor.

## Adding objects: the + menu




### Model...

Opens the Model Browser.

### Spawn Camera

A scene camera exactly where the work camera looks.

### Spawn spot / point light in front

A plain light in front of the camera.

### Soft light

One of 19 ready-made light kinds (see Lights).

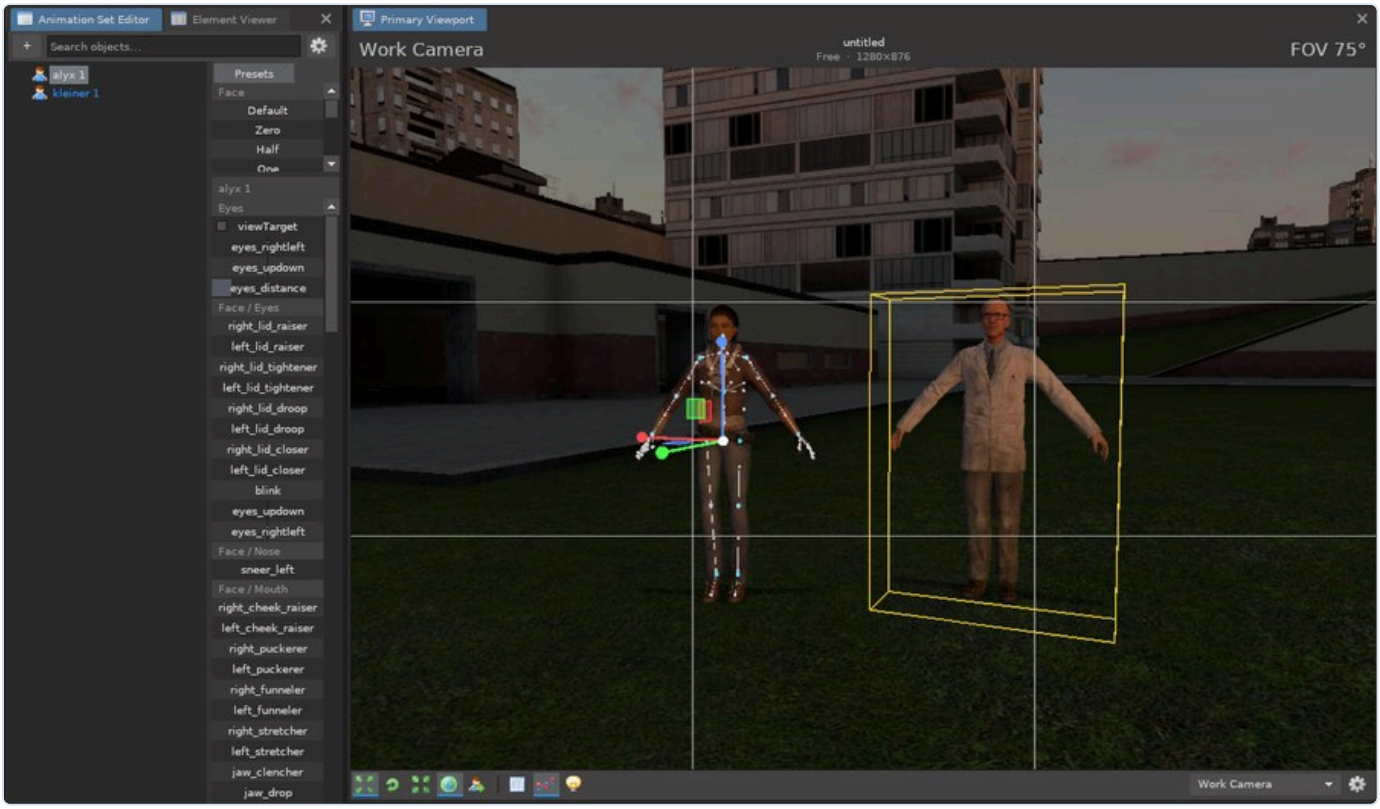
### Add an effect...

Opens the Effect Browser (see Effects).

### Create 3-point light rig

Key, fill and rim lights around the selected model.

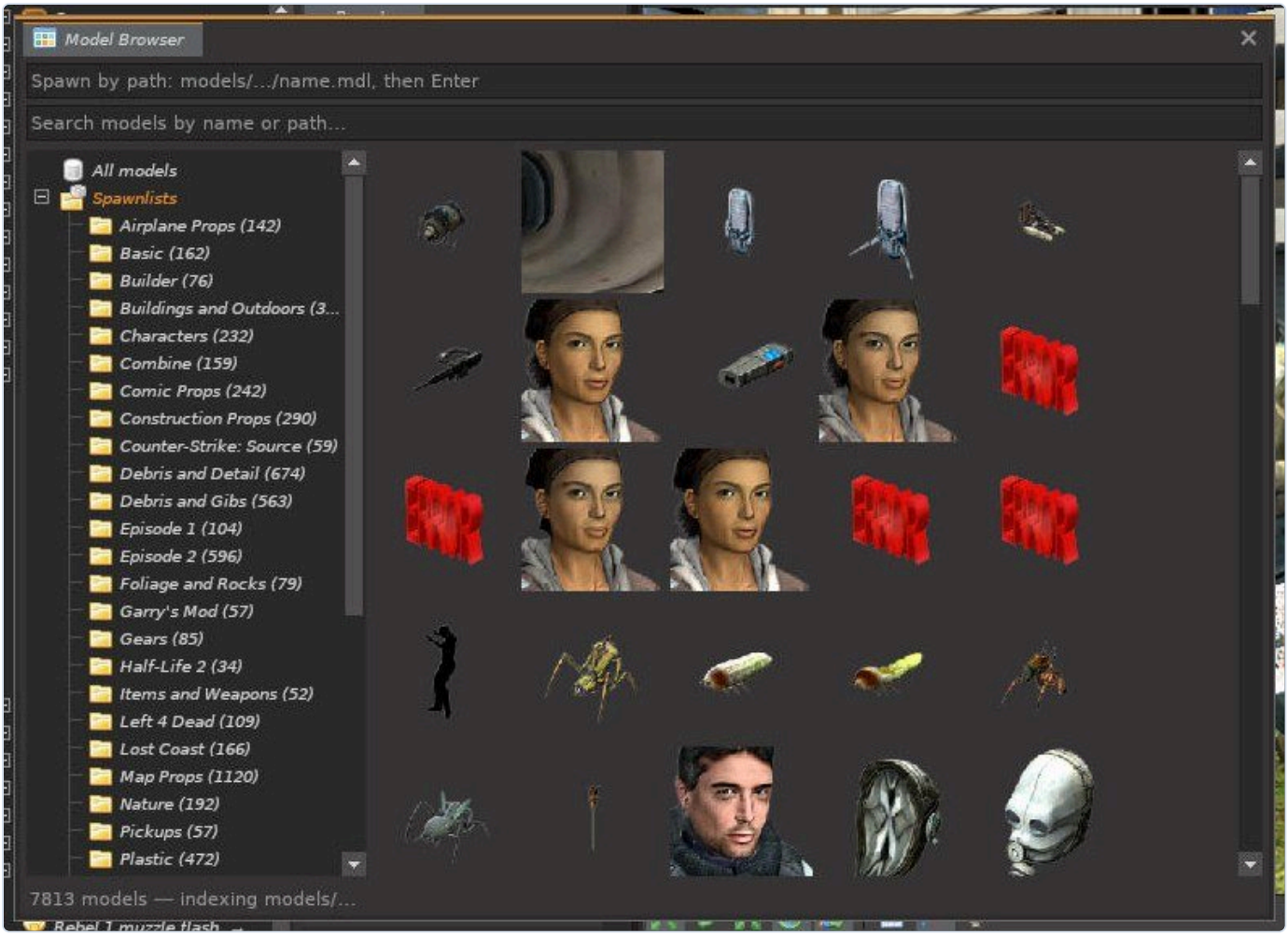
## Several objects at once




**Two characters selected.** The first one has the gizmo, the other a frame; both are highlighted in the object list, and the status bar says *selected: alyx 1 (+1)*.

- `Shift` + click (or `Ctrl` + click) adds an object to the selection in the viewport or in the object list; the same on a selected one takes it out. The gizmo then moves, turns and scales them together; `Del` and `Ctrl` + `D` work on all of them.
- With the cursor over the object list: `Ctrl` + `A` selects everything except locked and hidden objects; `Ctrl` + `C` copies the selected objects, `Ctrl` + `V` pastes copies a little aside. Paste again and the next copies go one more step further; one `Ctrl` + `Z` takes a whole paste back. The same commands are in the ▸ **Edit** menu.
- A duplicate (`Ctrl` + `D`) of several objects is one undo step too, and the copies become the new selection.

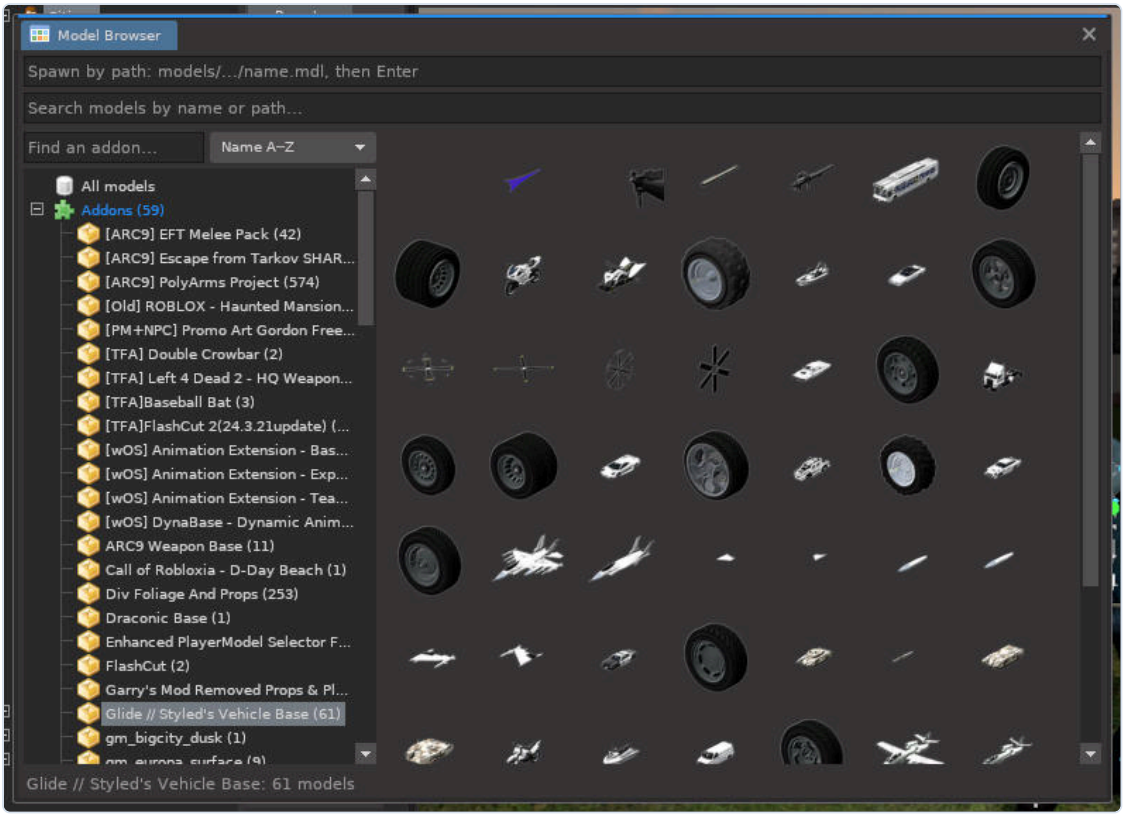
# The Model Browser




**Model Browser.** Folders on the left: *All models* and the game's spawnlists; icons on the right. Click an icon to add the model to the scene.

- **Search** by part of the name or the path, e.g. `alyx`, `combine`, `props_c17`.
- **Spawn by path**: type a full path such as `models/player/group01/male_07.mdl` and press `Enter` — handy for models from addons that are not in any spawnlist.
- Any model works: props, ragdolls, player models and NPC models from the game and from the Workshop. Models with a physics skeleton become posable **ragdolls**; the others are **props** that can still have bones.
- The status line shows how many models are indexed; the list of *All models* grows while indexing runs in the background.
- Legacy addons — folders in `garrysmod/addons`, not from the Workshop — are listed with the Workshop ones.

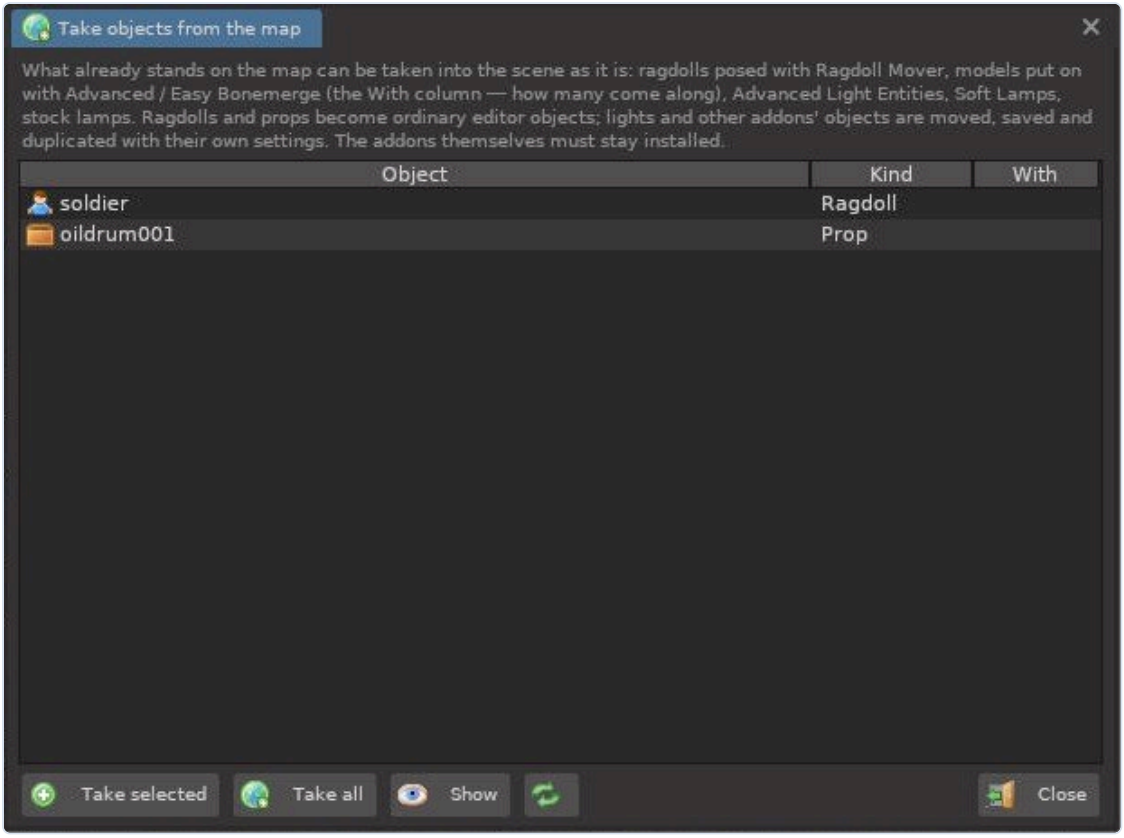
## Addons and Games




**Addons** lists every Workshop subscription that has models, with their number. *Find an addon...* filters by name; the list can be sorted by name (A–Z, Z–A), newest or oldest subscriptions, recently updated, or most models. **Games** shows the models of mounted games.

The icons are the same as in the spawn menu. If one is broken, right-click it → *Render the icon again*.

# Taking objects from the map




▸ **File → Take objects from the map** lists what stands on the map but is not in the scene yet:

- ragdolls posed with **Ragdoll Mover** or the physgun, and props;
- models put on with **Advanced Bonemerge** or **Easy Bonemerge** — they come along with their model (the *With* column counts them);
- lamps of **Advanced Light Entities**, **Soft Lamps** and the stock lamp tool.

*Take selected*, *Take all*; *Show* points the camera at the line. Ragdolls and props become ordinary editor objects. Lights and other addons' objects are moved, saved in the scene and duplicated with their own settings — the addons that made them must stay installed.

NEW **Your dupes and saves** come in the same way: spawn the dupe (or load the save) as usual, open the editor and take what it placed. An empty scene shows two buttons right in the object tree — *Model...* and *Take objects from the map*. Settings that other tools keep on an object (paint, glow and similar duplicator data) are saved with the scene.

# The object tree



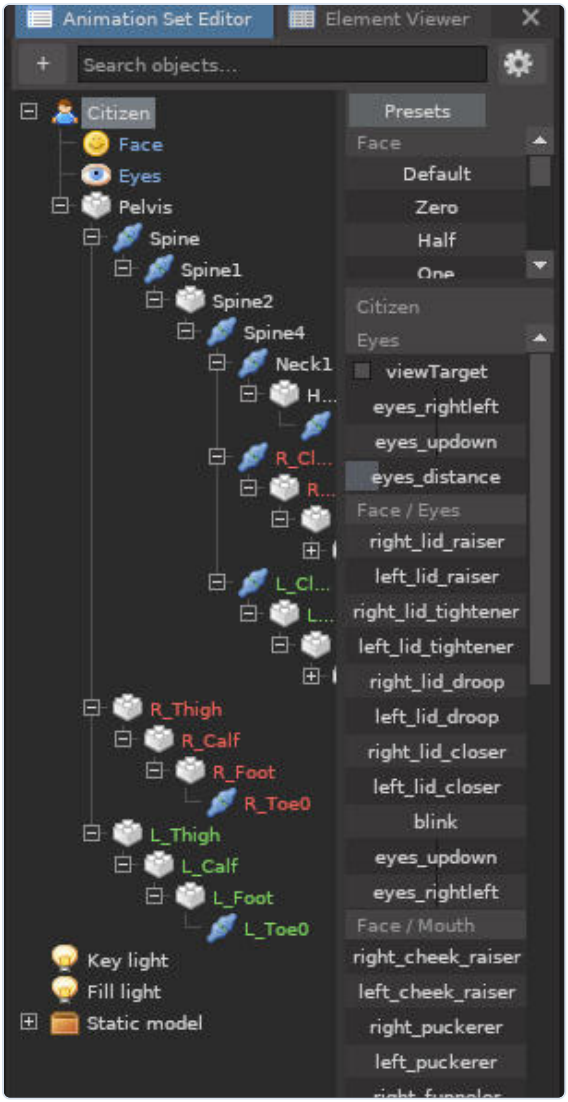

The left column of the Animation Set Editor lists every object of the scene. Icons tell the kind: a person for characters, a box for props, a bulb for lights, a camera, a lightning bolt for effects. An attached object shows its parent after an arrow: *Rebel 1 weapon → Rebel 1*.

Expand a character to see its **control groups**: Face, Eyes, Body, Head, RightArm, LeftArm, RightLeg, LeftLeg, RightFingers, LeftFingers. Clicking a group shows only its controls in the right column.

The right column holds the controls of the selection: face presets and flex sliders, eye target, bone rows (*Bone* - *pos*, - *rot*, - *scale*), or the settings of a light or camera.

The search field filters the tree. The gear button holds tree options (expand / collapse all).

### The skeleton view




Cog → *Tree: skeleton (as in Ragdoll Mover)* shows each character's bones in their real hierarchy instead of control groups. Physical bones and bones without physics have different icons; the right side is red, the left side green. *Skeleton shows* adds bones without a physics object and bonemerged models (under their model). *Tree: objects only* hides both.

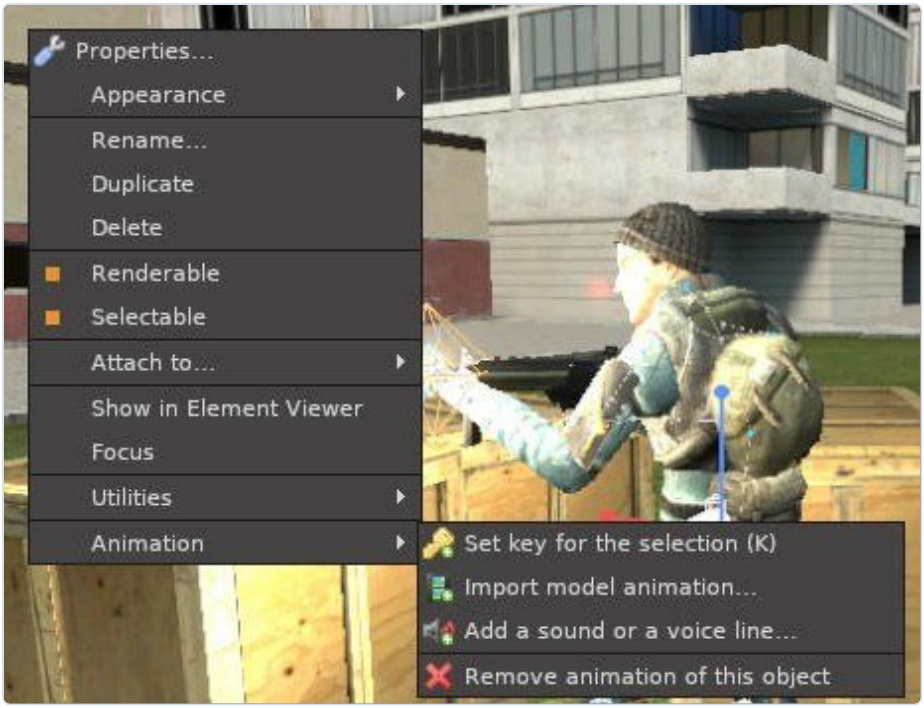
Click a bone to select it; the bone under the mouse is marked in the viewports. Right-click a bone to lock it to another bone (Posing).

## Selecting

- `LMB` on an object in a viewport or in the tree selects it. `LMB` on a bone marker selects that bone.
- `Ctrl` + `LMB` adds an object to the selection or removes it. Several selected objects are moved, rotated, scaled, duplicated and deleted together.
- Click on empty space or use ▸ **Edit → Deselect** to clear the selection.

## The object menu

A short `RMB` click on an object (without moving the mouse) in the viewport or in the tree opens its menu.




### Properties...

Colour, textures, size, shadow (below).

### Appearance

Quick colour, remove / restore textures, cast shadow, reset.

### Rename... / Duplicate / Delete

As named.

### Renderable

Off: the object is hidden in viewports and renders.

### Selectable

Off: the object is locked and cannot be clicked by accident.

### Attach to...

See Attaching and wearing.

### Show in Element Viewer / Focus

As named.

### Utilities

Drop to floor, 3-point rig, reset pose, copy / paste pose.

### Set Skin / Set Body

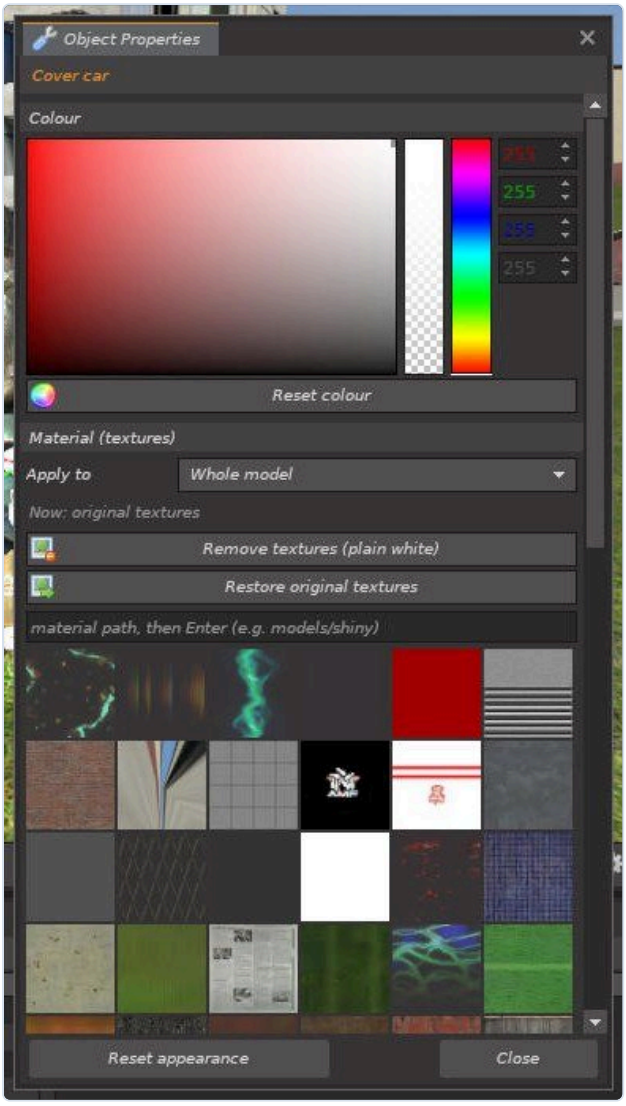
Skins and bodygroups of the model.

### Animation

SAeditor: set a key, import an animation, add a sound.

Cameras and lights also have **Aim camera at object...**; cameras have **Look through this camera** and **Match camera to work view**; effects have **Play from the start** and **Choose another effect...**.

## Object Properties




### Colour

Tint and opacity (the fourth field is alpha: below 255 the object becomes see-through). *Reset colour* returns white.

### Material

Replace textures for the *Whole model* or one material slot. Pick a material from the grid or type a path such as `models/shiny`. *Remove textures* makes the model plain white so it takes the object colour; *Restore original textures* undoes it.

### Size

Props only: overall scale and stretching in length (X), width (Y) and height (Z). The Scale tool `4` does the same with the mouse.

### Shadow

*Cast shadow* off removes the object from shadow maps — useful for an object in front of a lamp.

### Reset appearance

Everything back to the original.

# Element Viewer




The Element Viewer tab (next to the Animation Set Editor) shows every property of the selection as numbers: name, model, position x/y/z, pitch/yaw/roll, scale, skin, renderable, selectable, and for lights and cameras all their settings with SFM names (*intensity, fieldOfView, castsShadows, shadowFilterSize…*). Type a value for exact placement.

> ✓ Any slider in SPeditor: `Shift` + drag is fine, `Ctrl` + drag is coarse; right-click a slider for *Reset to default* and *Enter value…*.

# Posing

Everything is posed with one gizmo: whole objects and single bones, including fingers.

## The gizmo



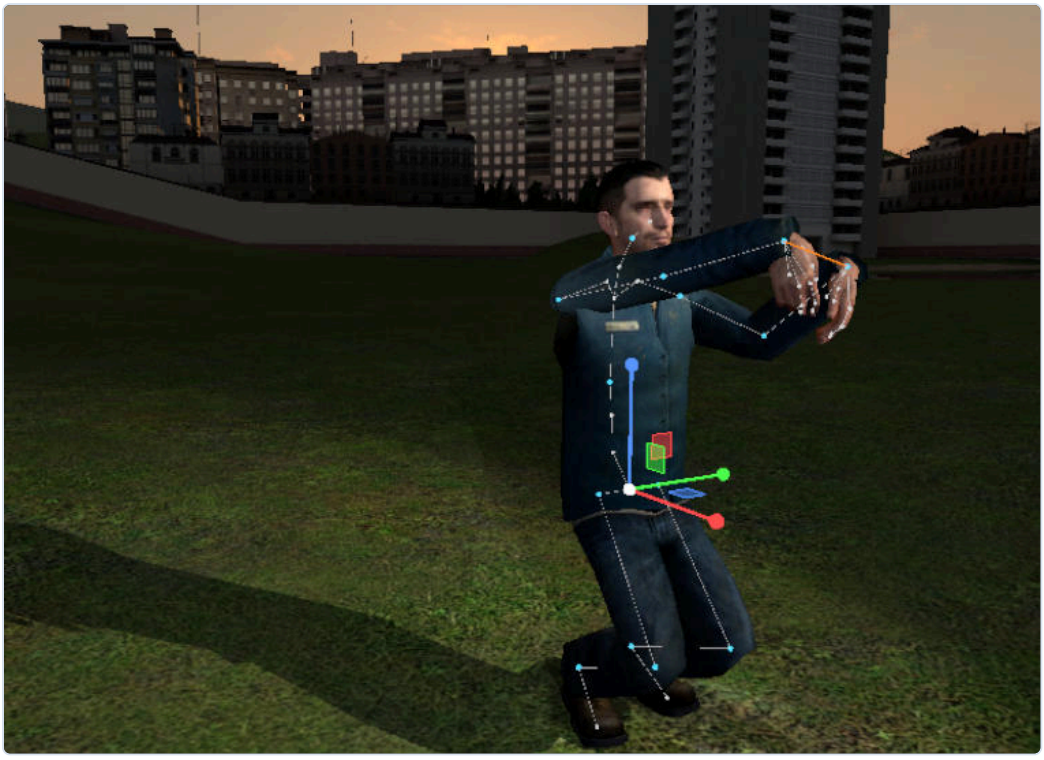
Move 1



Rotate 2

| KEY | TOOL |
| --- | --- |
| 1 | **Move.** Drag an arrow to move along an axis, the centre to move in the screen plane. |
| 2 | **Rotate.** Drag a coloured ring to turn around its axis. The white outer ring turns around the line of sight (as you see it on the screen); drag anywhere inside the rings for a **trackball** — the bone rolls after the mouse in every direction at once. A click there without dragging still selects the bone under the cursor. |
| 4 | **Scale.** The centre scales evenly, an axis stretches along it. Works on props and on bones of any model. |
| 3 | **World / local axes.** Local axes follow the object's own rotation. |
| I | **Inverse kinematics.** Move a hand or a foot and the whole arm or leg follows. |

### Plane handles



The Move gizmo has three small coloured squares between its axes. Drag one to move in two axes at once: the square lying flat moves along the floor, the upright ones along a wall — the third axis does not change.

## Bones



**The left hand selected with IK on.** The small white cubes are bone markers.

Click a bone marker to select the bone, or click a row *Bone - pos / - rot / - scale* in the Animation Set Editor: it selects the bone and the matching tool at once.

Bones with a physics body (pelvis, spine, head, upper arms, forearms, hands, thighs, calves, feet) move the ragdoll itself. Bones without one (clavicles, fingers, toes, face bones) are moved with bone offsets. Both are saved in poses and animated by SAeditor.

With ▸ **View → Toggle Skeleton (B)** the markers are hidden; ▸ **View → Bone size** makes them larger or smaller.

A click on the body of the *selected* character picks the body part under the cursor — no need to hit the small markers.

## Scaling bones

Select any bone and press `4` (or click its - *scale* row): a big head, long arms, a giant hand. In a0.3 the Scale tool worked only on bones without physics (fingers) and silently acted as Move on a hand or the head; a0.4 scales every bone.

## IK keeps the feet planted

With IK on, moving or turning the pelvis keeps both feet where they stand and bends the knees: a character crouches, leans or shifts its weight in one drag. Switch it off with ▸ **View → IK: keep the feet planted when the body moves.** NEW Since a0.7 IK is on from the start, and the editor remembers your choice.

## Stretching bones NEW

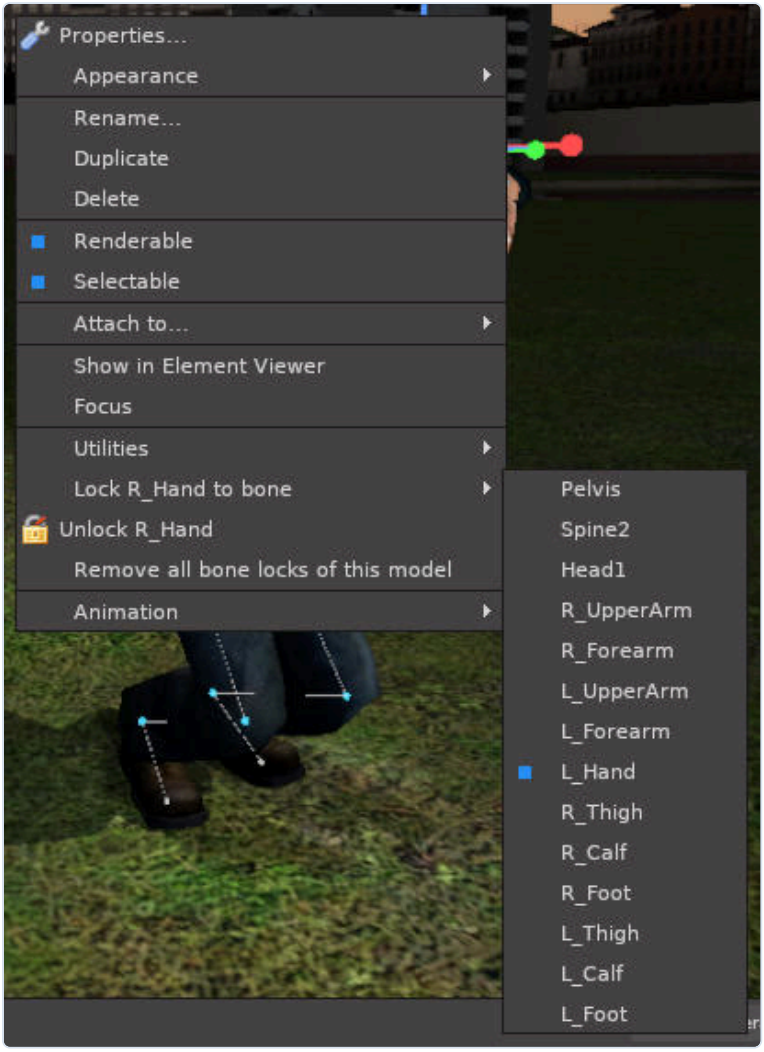


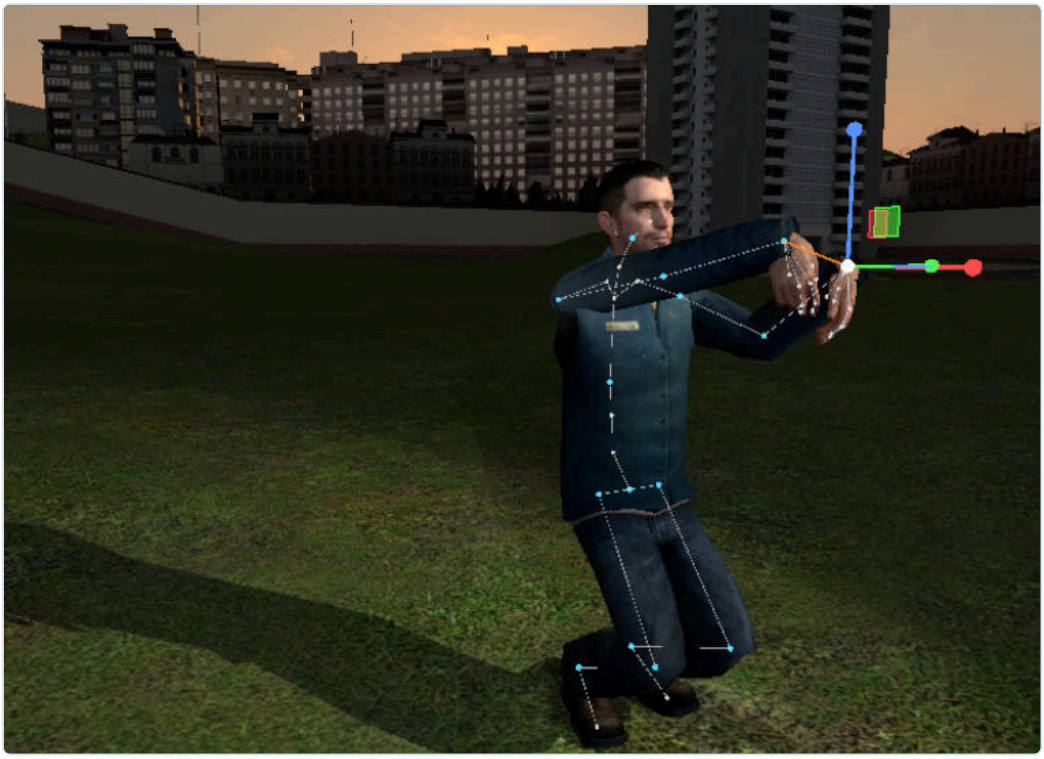
**Stretch on:** the head moved up, the neck grew longer.

Press `T` or the *Stretch* button of the viewport bar. While it is on, the Move tool takes a bone *away* from its parent instead of swinging the limb: a longer neck, arm or leg, a cartoon pose. It

works on ragdolls too, on physical bones as well. Press T again to go back to ordinary posing; the stretch stays and is keyed in SAeditor like any bone edit.

## Locking a bone to another bone

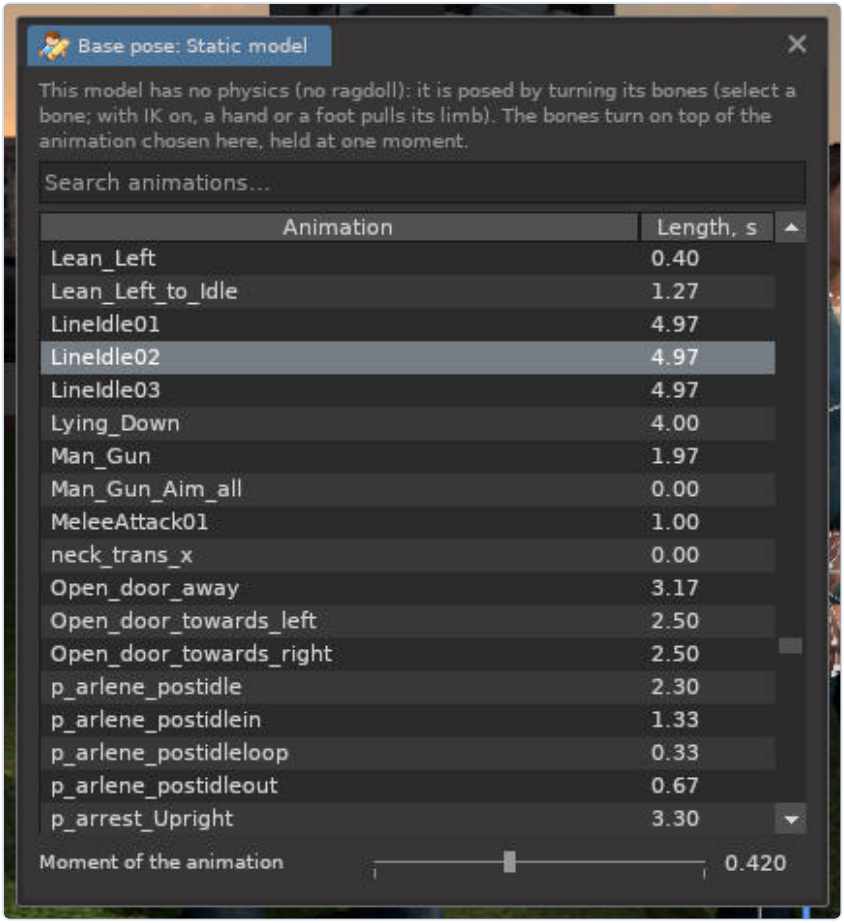




**The right hand locked to the left one:** move the left hand and both go together.

Select a physical bone, for example the right hand, right-click → *Lock R_Hand to bone* and pick the bone it should hold on to — the left hand. From then on it follows that bone; with IK on it pulls its whole arm. Two hands on a rifle, a sword or a steering wheel stay together while you pose. *Unlock* and *Remove all bone locks of this model* are in the same menu (in the skeleton tree: right-click a bone). Locks are saved with the scene.

# Models without physics




**Base pose:** the animation and the moment of it under your bone turns.



**IK on a model without physics:** the hand pulls the arm.

Some models have bones but no physics skeleton — for example characters ported from SFM as props. They now stand still and are posed by turning their bones: select a bone and rotate it, or

turn on IK and drag a hand or a foot. Object menu → *Base pose* chooses an animation of the model and its moment; your bone turns are added on top. Model animations can be imported onto such models too.

## Face and eyes

- **Face presets** at the top: *Default, Zero, Half, One* set all flexes at once.
- **Flex sliders** are grouped as Brows, Eyes, Nose, Mouth, Other (e.g. *smile, jaw_drop, blink*). Models without face flexes do not show the section.
- **Eyes:** tick *viewTarget* to aim the eyes; *eyes_rightleft*, *eyes_updown* and *eyes_distance* set where they look.
- **Skins and bodygroups**: object menu → ▸ **Set Skin** / ▸ **Set Body**.

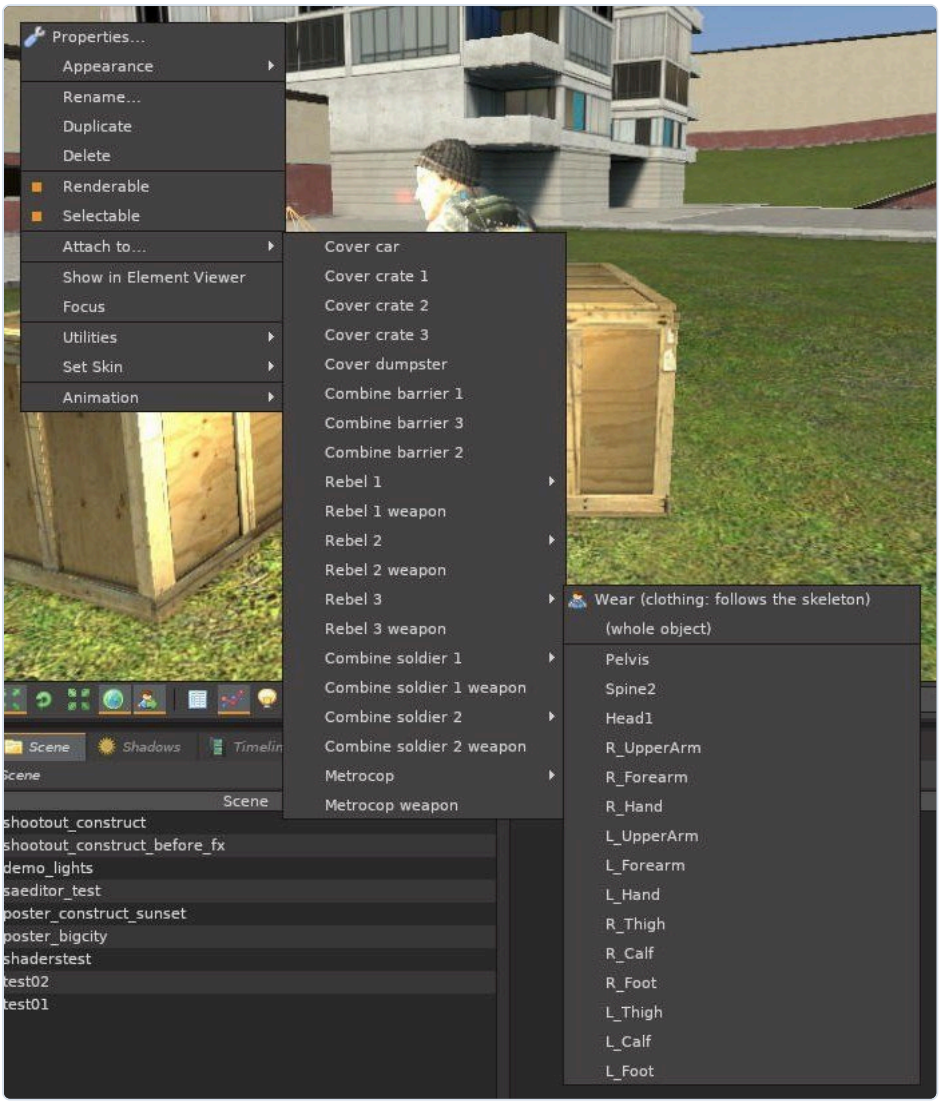
## Poses: copy, paste, library

- `Ctrl` + `C` / `Ctrl` + `V` copy the pose of the selected character and paste it onto another. Between different models with the same skeleton the pose is carried over by bone names, keeping each model's proportions.
- The **Pose library** in the Scene tab: *Save pose*, *Load pose*, *Delete*. The list shows the pose name and the model it was made on.
- ▸ **Utilities** → **Reset pose of selected** returns bones, face and eyes to the original.

> **!** Addons that change ragdolls (active ragdoll systems, Fedhoria-like mods) can fight SPeditor and make limbs impossible to pose. If limbs jump back or fall, disable such addons while you work in the editor.

# Attaching and wearing

Objects can follow other objects: a weapon in a hand, a hat on a head, a muzzle flash on a gun, or clothing that bends with the body.




**Object menu → Attach to... → a character.** At the top *Wear*, then *(whole object)*, then every physics bone.

## Three ways to attach

| CHOICE | RESULT | USE FOR |
|---|---|---|
| **R_Hand**, **Head1**... | Follows that bone rigidly, keeping its offset. | Weapons, props in hands, hats, helmets as separate props. |
| **(whole object)** | Follows the object's root (pelvis of a character). | A rider on a vehicle, a light on a car. |

| | | |
|---|---|---|
| **Wear** | Every bone of the worn model follows the bone with the same name on the wearer (bone merge). | Clothing, armour, hair, accessories made for the same skeleton (e.g. ValveBiped). |

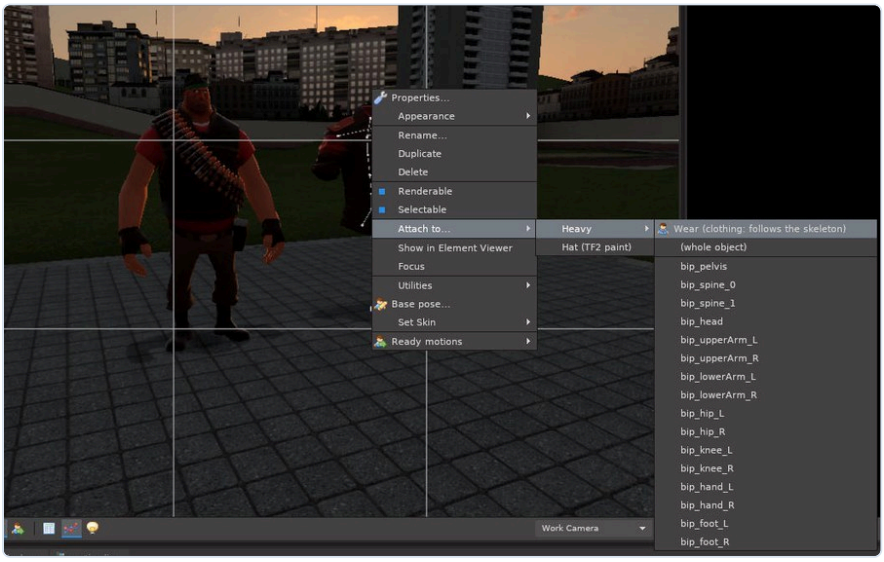
## Putting clothes on a character

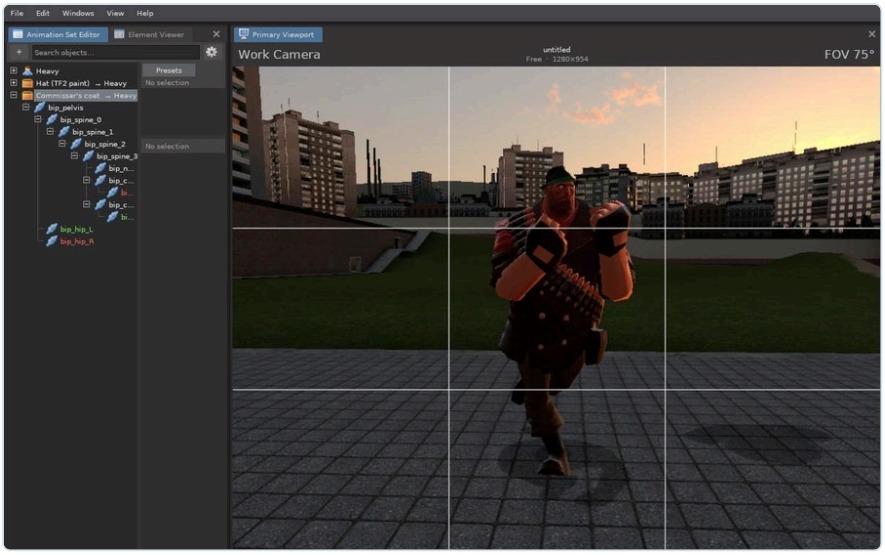
1. Add the clothing model (from a clothing addon) and the character.
2. Right-click the clothing → ▸ **Attach to...** → the character → ▸ **Wear (clothing: follows the skeleton)**.
3. Pose the character: the clothing bends with every limb. Skin, bodygroups and colour of the clothing still work.
4. To take it off: object menu → ▸ **Detach**.

> **i** Before a0.4 clothing could only be attached rigidly to one bone, so it did not follow the arms and legs. *Wear* arrives in a0.4; it is saved with the scene, survives duplicating and can be undone.

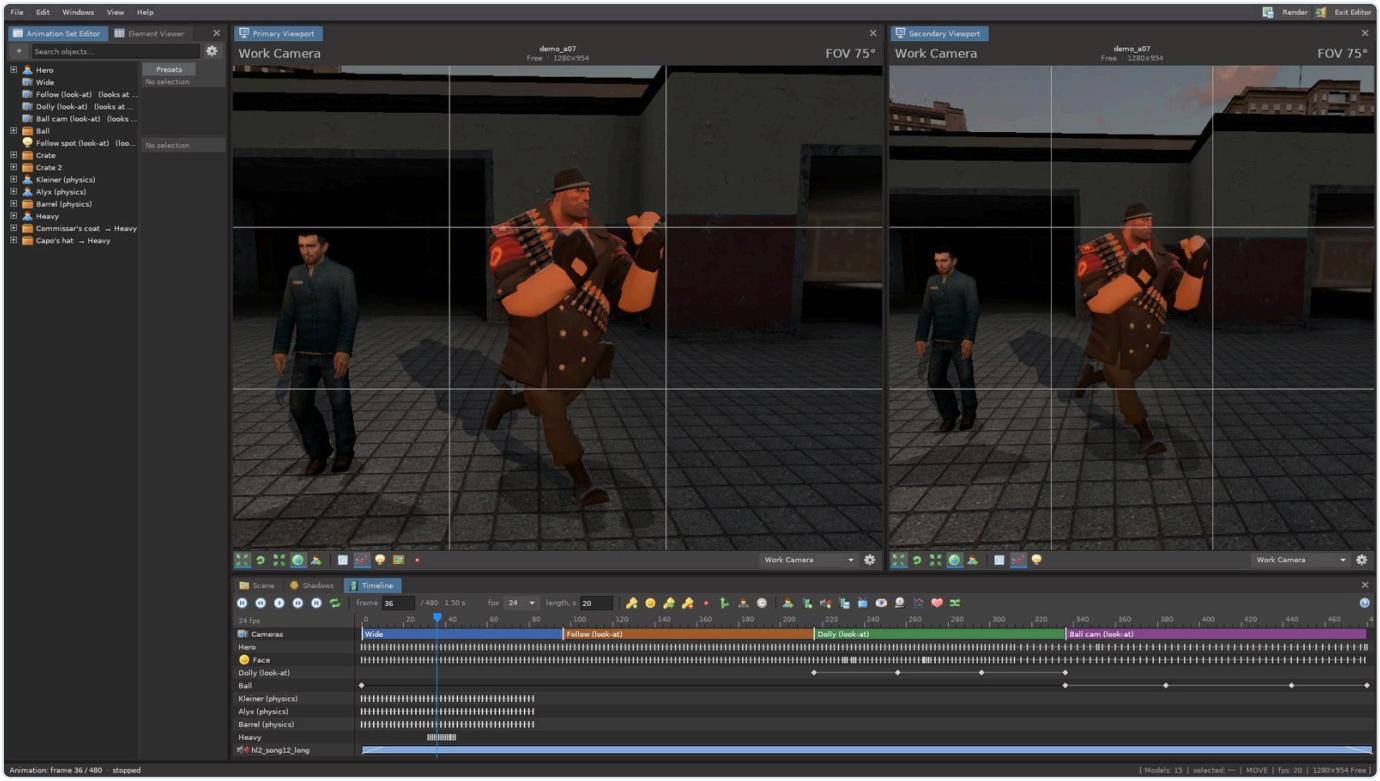
> **!** Wearing works only when the bone names match. A model made for another skeleton (a custom rig) stays at the wearer's root. You can still click and select the worn object: its invisible copy stays where it was attached.

## TF2 cosmetics and any skeleton NEW






**A TF2 coat worn by Heavy.** Its own bones (the tails) are in the object tree and can be posed.



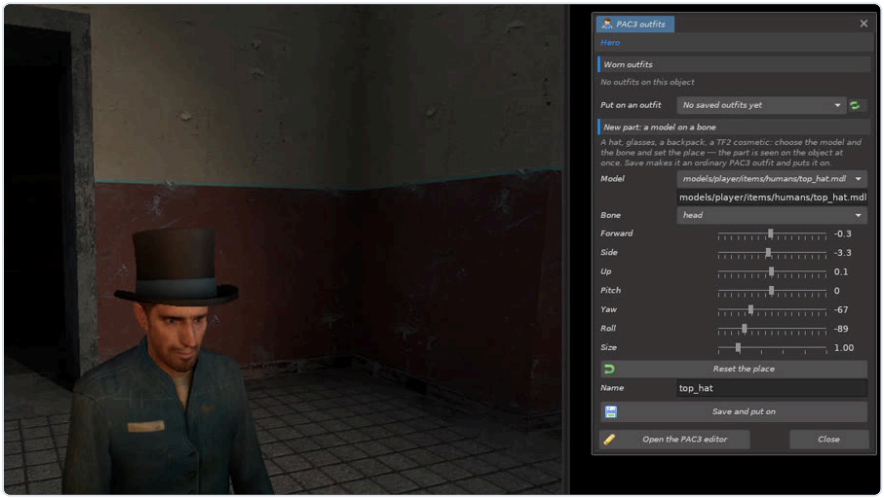

- *Wear* is offered for any model with a skeleton, also models without physics — TF2 cosmetics on TF2 characters, hair, capes.
- The own bones of worn clothes (coat tails, hair, straps) can be selected and posed.
- TF2 paint and the crit glow of the **Hat Painter & Crit Glow Tools** show on worn cosmetics and are saved with the scene.
- Characters whose models report wrong physics bones (the Enhanced TF2 MvM bots) now pose, attach and import their own animations normally.

# PAC3 outfits NEW

With **PAC3** installed (from the Workshop), characters and props of the scene can wear PAC3 outfits. They follow the bones, are saved with the scene and are seen by other players on a server. Without PAC3 the outfits stay in the scene file but are not shown.

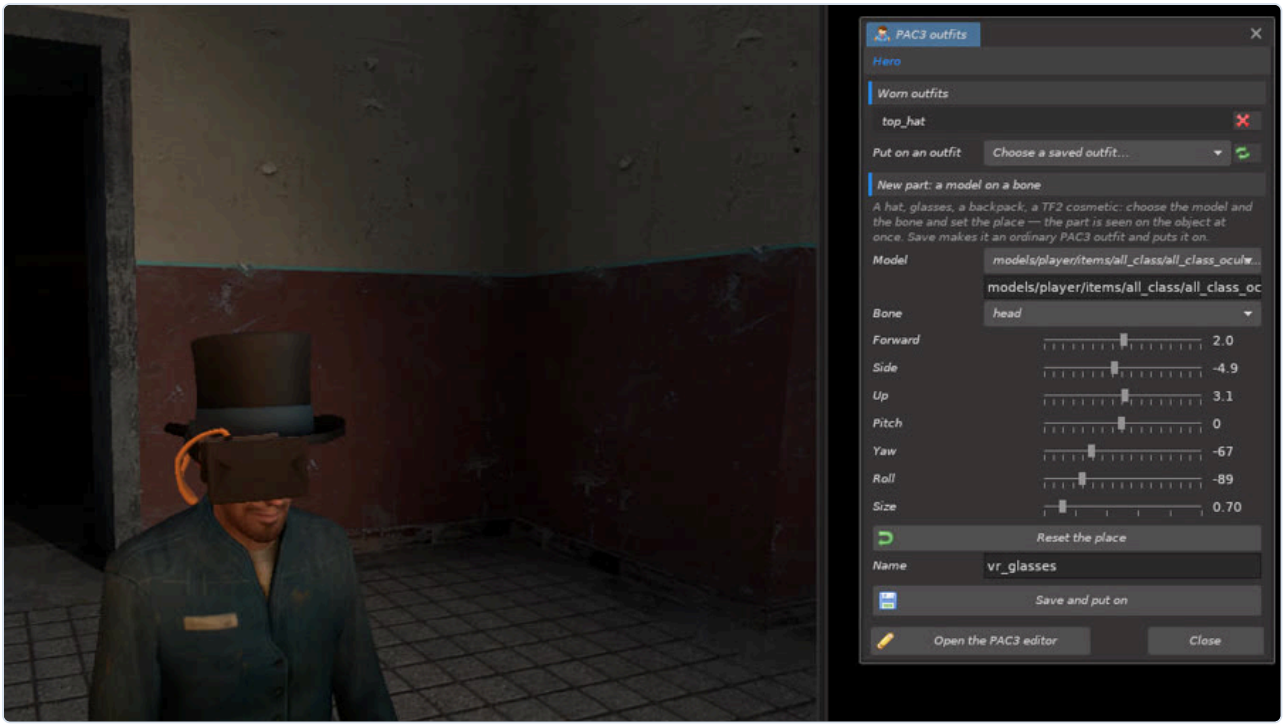



**Right click → PAC3 outfits:** put on a saved outfit, take one or all off, open the PAC3 window or PAC3's own editor.




**The PAC3 window:** a model on a bone with sliders, without PAC3's editor.

1. Right-click a character → ▸ **PAC3 outfits** → ▸ **PAC3 window: outfits and a model on a bone...**
2. Under *New part: a model on a bone* type the model path (a hat, glasses, a backpack, a TF2 cosmetic) and press Enter, pick the bone.
3. Set the place with *Forward*, *Side*, *Up*, *Pitch*, *Yaw*, *Roll* and *Size* — the part is seen on the character at once.
4. Give it a name and press *Save and put on*: it becomes an ordinary PAC3 outfit in `garrysmod/data/pac3/`, which you can put on other objects or open in PAC3's editor.




**A second part:** glasses on the head bone over the saved top hat.

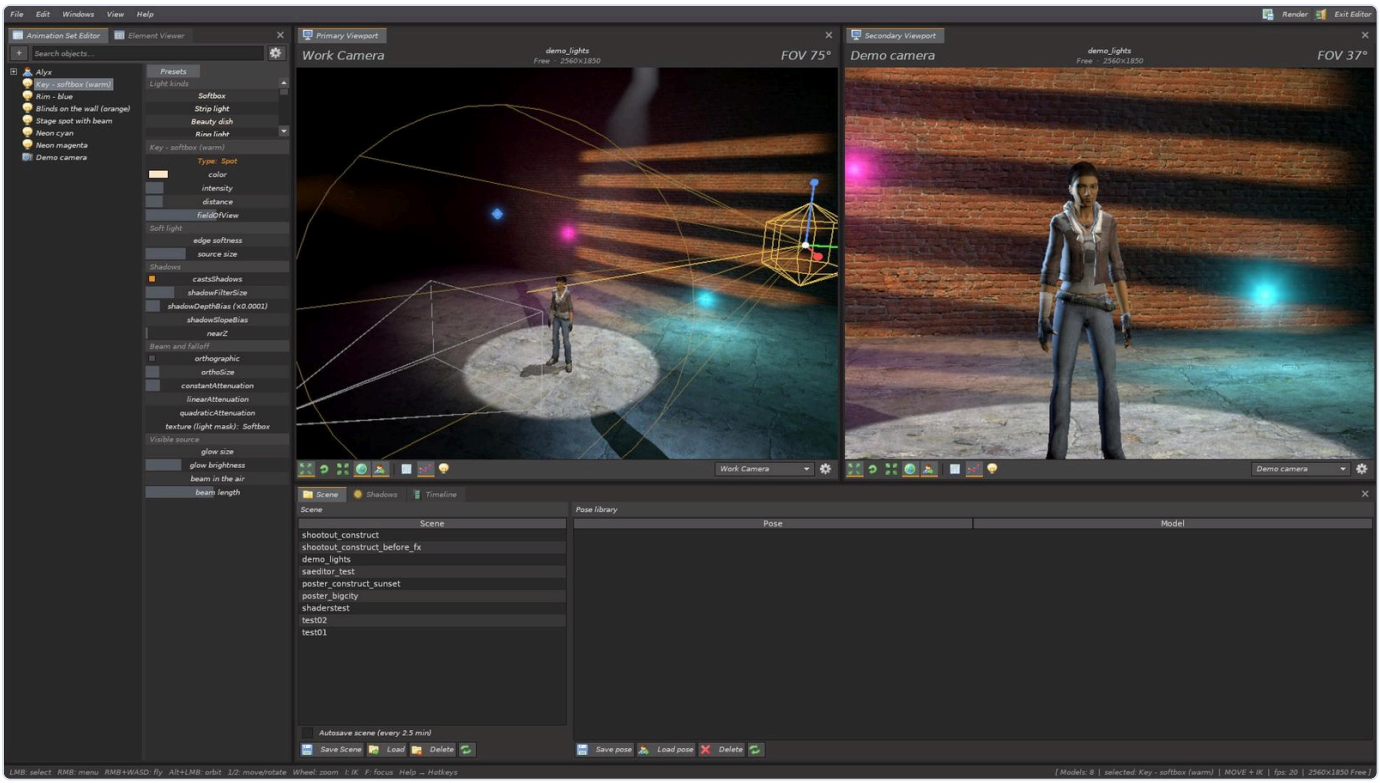
> Outfits made in PAC3's own editor work too: save them there and they appear under *Put on an outfit*. A character keeps its SPeditor pose while it wears an outfit, unless the outfit itself moves bones.

## Rules

- Move an attached object by hand and the new offset is kept.
- Chains are allowed (a scope on a gun in a hand); an object cannot be attached to its own child.
- When the parent is deleted, the object stays where it was.
- Attachments are saved in scenes and followed in SAeditor animations.

# Lights

Light makes a poster. SPeditor has point lights and spot lights; both cast shadows, and spot lights have a soft edge, penumbra, a glow and a visible beam.



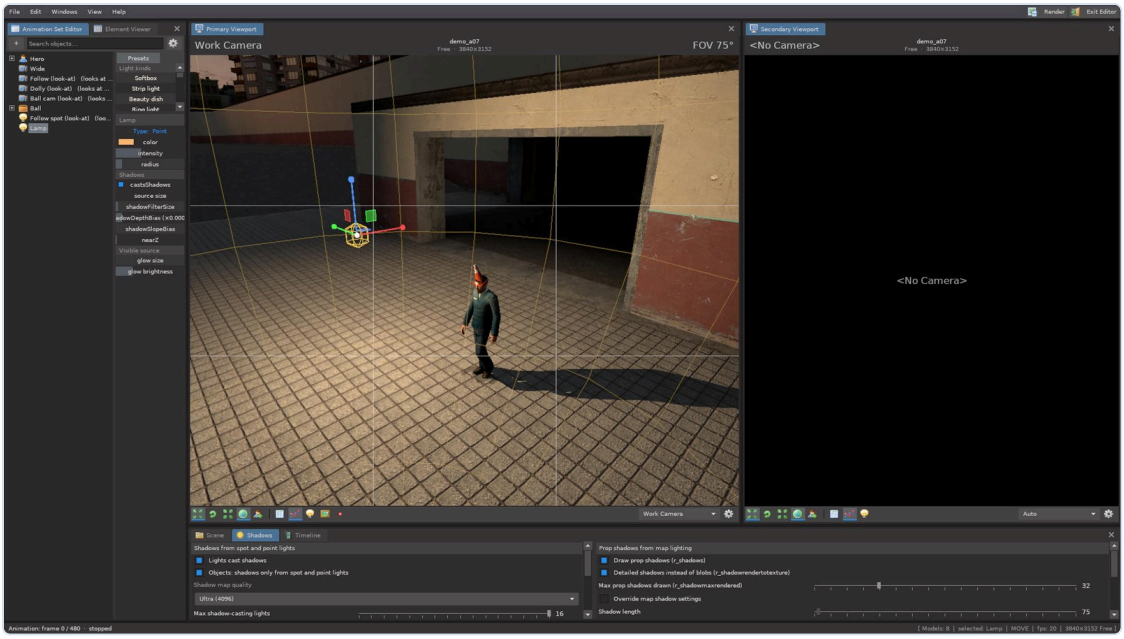

**Six lights:** a warm softbox as the key light, a blue rim light, orange “light through blinds” on the wall, a stage spot with a beam, two neon glows. The selected softbox shows its cone and its controls in the Animation Set Editor.

## Point and spot

| | POINT LIGHT | SPOT LIGHT |
|---|---|---|
| Shines | in all directions | in a cone or through a mask |
| Shadows | no (an engine limit) | yes, soft, adjustable |
| Good for | fill light, neon, fire, bulbs, muzzle flashes | key and rim light, sun, windows, lamps |

Switch a light between the two kinds with its *Type* control in the Animation Set Editor.

### Shadows of a point light NEW




A point light has a *Shadows* section in its panel. With shadows on it shines as six spot lights looking along the sides of a cube, each with its own shadow map, so it lights and shadows all around. The game has 8 shadow maps for all lights together (a spot light takes one, a point light six), so only the point light nearest to the camera gets shadows; the others shine without them.

## Light kinds

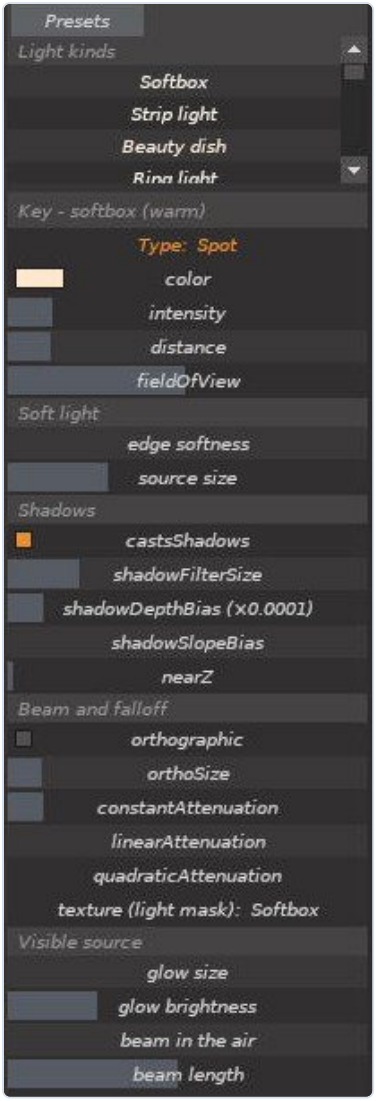
**+** → ▸ **Soft light** creates a light already set up for a purpose. With a light selected, the *Light kinds* list at the top of the right column turns that light into another kind.

| KIND | WHAT IT IS |
|---|---|
| Softbox | Large soft key light, the studio standard. |
| Strip light | A narrow tall softbox for edges and reflections. |
| Beauty dish | Round, contrasty portrait light. |
| Ring light | Ring around the lens: flat face light, round catchlights. |
| Rim light | Back light that outlines the silhouette. |
| Window light | Daylight through a window frame. |
| Light through blinds | Stripes of light, the noir look. |
| Sun through leaves | Dappled sunlight. |

| | |
|---|---|
| Golden hour sun | Low warm sun with long shadows. |
| Sunset gel | A coloured gradient of sunset light. |
| Moonlight | Cold dim parallel light. |
| Stage spot | Hard round spot with a visible beam. |
| Street lamp | Warm cone from above. |
| Headlight | Car headlight with a glow. |
| Red and blue beacon | Police lights. |
| Eye slit | A narrow strip of light across the eyes. |
| Candle · Light bulb · Neon glow | Small point sources with a visible glow. |

**Presets** (colour only): Soft white, Warm lamp, Cool moonlight, Neon cyan, Neon magenta, Fire.
**Create 3-point light rig** places key, fill and rim lights around the selected model.

# Light settings




### color · intensity

Colour (click the swatch for a palette and a mixer) and brightness.

### distance · fieldOfView

How far the light reaches; the cone angle of a spot.

### edge softness

0 keeps the mask as it is, 1 fades the light from the centre to the edge.

### source size

Size of the lamp: a larger source gives softer shadows that blur with distance (penumbra). Calculated in the render and its preview.

### castsShadows · shadowFilterSize

Shadows on/off and the blur of their edge.

### shadowDepthBias · shadowSlopeBias

Raise if lit surfaces get stripes; lower if the shadow detaches from the object.

### nearZ

The light starts at this distance from the source.

### orthographic · orthoSize

A parallel square beam like sunlight: shadows keep their size at any distance.

### constant / linear / quadratic attenuation

Falloff with distance. Quadratic is like a real lamp; constant does not fade at all (good for far subjects).

### texture (light mask)

The shape of the beam: soft masks of SPeditor (softbox, strip, round, ring, window, blinds, bars, slit, leaves, sunset...), Lamp tool textures, other game textures or any texture by path.

### glow size · glow brightness

A visible glow around the source, like a headlight or a beacon.

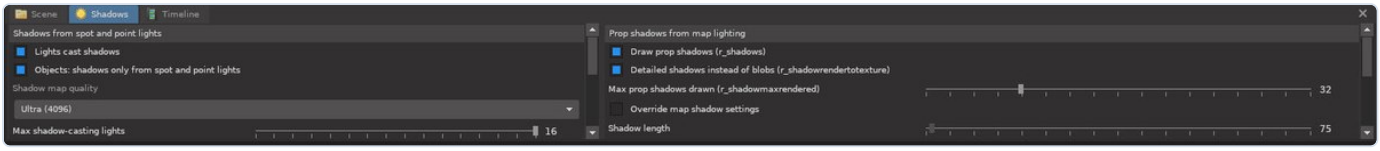
### beam in the air · beam length

A visible cone of light, as in dust or fog.

Every shadow-casting spot light costs FPS. The Shadows tab limits how many of them cast shadows at once. Penumbra (*source size*) is only calculated in the render; the viewports show a sharp shadow.

# Shadows

The Shadows tab of the bottom window controls two different things: shadows of the editor's spot and point lights and the simple prop shadows the map's own lighting casts.




## Shadows from spot and point lights

### Lights cast shadows

The master switch.

### Objects: shadows only from spot and point lights NEW

On by default: while the editor is open, props and characters of the scene drop no grey blob from the map's sun, only the real shadows of your lights.

### Shadow map quality

Low, Medium, High, Ultra. The engine reads it at start: after a change you may need `vid_restart` or the launch option `+r_flashlightdepthres 2048`.

### Max shadow-casting lights

Only the N lights nearest to the camera cast shadows. The line below shows how many cast now.

### Shadow draw distance

A light whose lit area is farther than this casts no shadow (0 = unlimited). Saves FPS in large scenes.

### Softness multiplier

Multiplies the edge blur of every light.

### Slope bias (global)

Removes stripes on surfaces lit at a shallow angle; each light can override it.

## Prop shadows from map lighting

These are the soft blobs or silhouettes that props and ragdolls drop from the map's sun. *Draw prop shadows*, *Detailed shadows instead of blobs* and *Max prop shadows drawn* are local engine settings. Length, colour and direction belong to the map: tick *Override map shadow settings* to

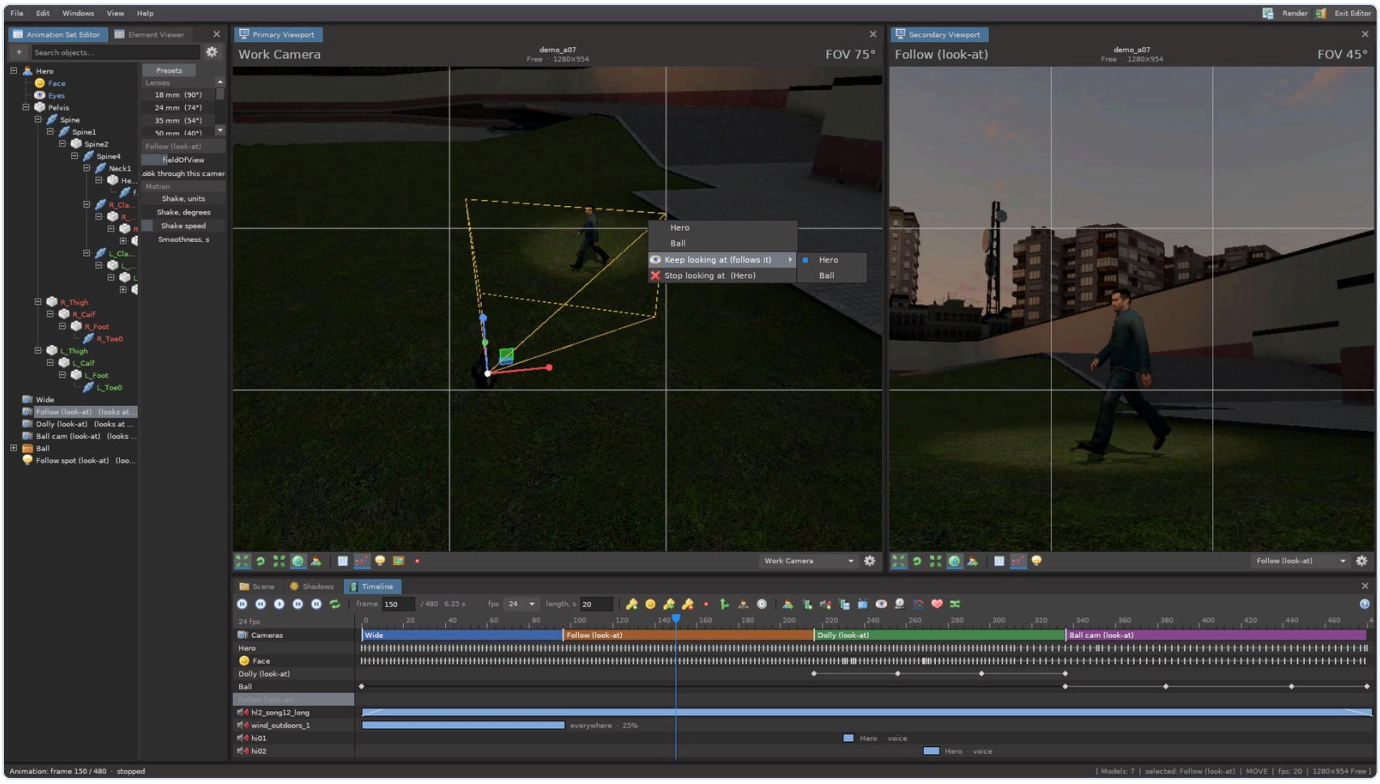
change them (this affects every player on a server; untick to restore). *Disable prop shadows on the map* removes them entirely.

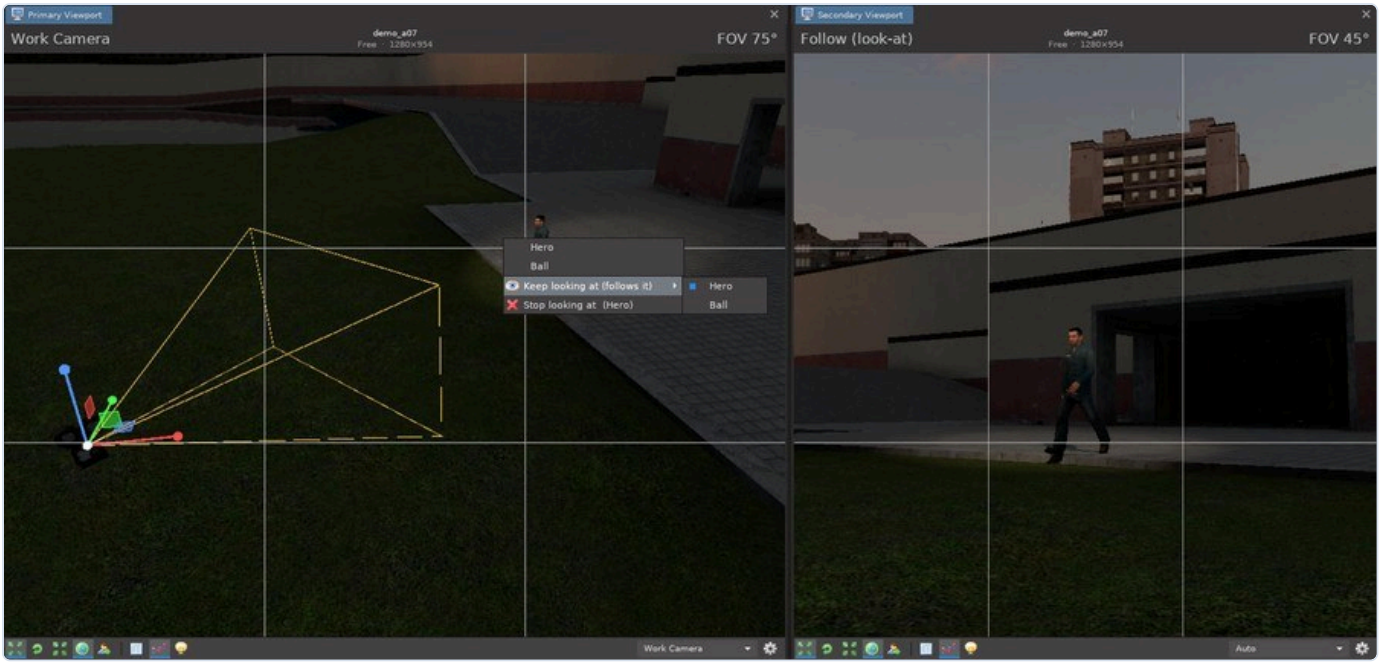
# Cameras

Scene cameras are objects of the scene. A render is always taken through a scene camera (or through the Primary Viewport), and camera effects belong to scene cameras.

## Creating and aiming

- **+** → ▸ **Spawn Camera** creates a camera exactly at the work camera's position and angle.
- **Match camera to work view** (object menu) moves an existing camera to where you look now.
- **Aim camera at object...** turns it towards any object.
- NEW **Aim at...** → **Keep looking at** makes the camera (or a light) keep turning to the object — the head of a character, the middle of a prop — while the animation moves either of them. Its place stays yours (keys, flying, attaching); the object tree shows *looks at ...* next to it. **Stop looking at** in the same menu. It is saved with the scene.
- Move and rotate it with the gizmo like any object.



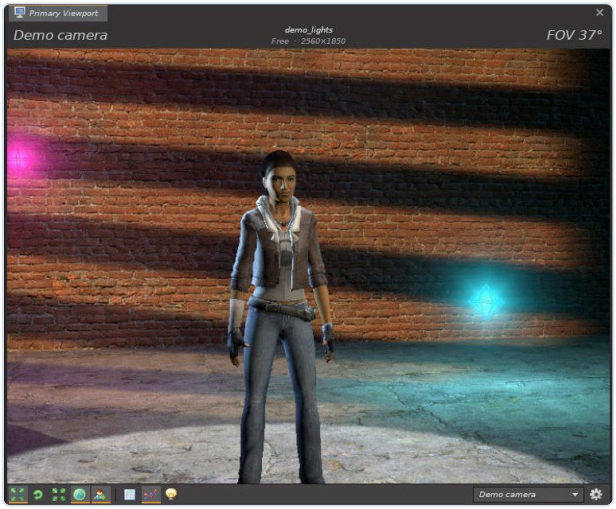



**Aim at… → Keep looking at → Hero:** the Secondary Viewport looks through the camera that follows him.

## Recording a camera NEW

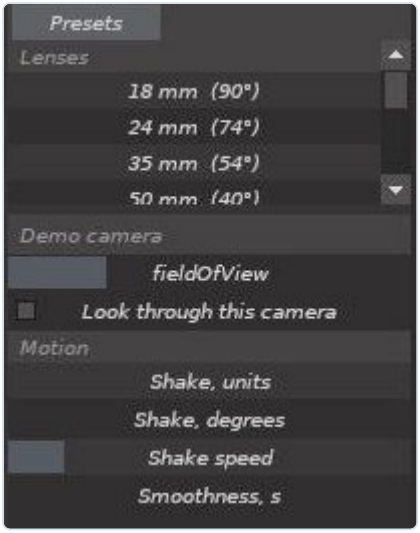
A camera can record its own moves into SAeditor keys: right-click it → *Record this camera's moves (keys)*, or press the red dot on the Primary Viewport bar while you look through it. From then on every move, turn and zoom sets a key at the current frame, even with Auto key off — play the animation and fly the camera, or step through frames and frame each shot. ● *REC* blinks beside its name in the viewport header. The same menu item stops it; the choice is saved with the scene.

## Looking through a camera




Select a camera and press `V`, choose it in the viewport's camera list, or tick *Look through this camera*. The header then shows the camera name. Flying with `RMB` + `WASD` now moves the scene camera itself — the fastest way to frame a shot. `V` goes back to the work camera.

# Camera settings




### Lenses

Ready focal lengths: 18 mm (90°), 24 mm (74°), 35 mm (54°), 50 mm (40°) and longer.

### fieldOfView

Any angle.

### Shake, units / degrees

How far the camera is thrown about and turned. A small value gives a handheld camera.

### Shake speed

Shakes per second: 1–2 handheld, 8+ an explosion or an engine.

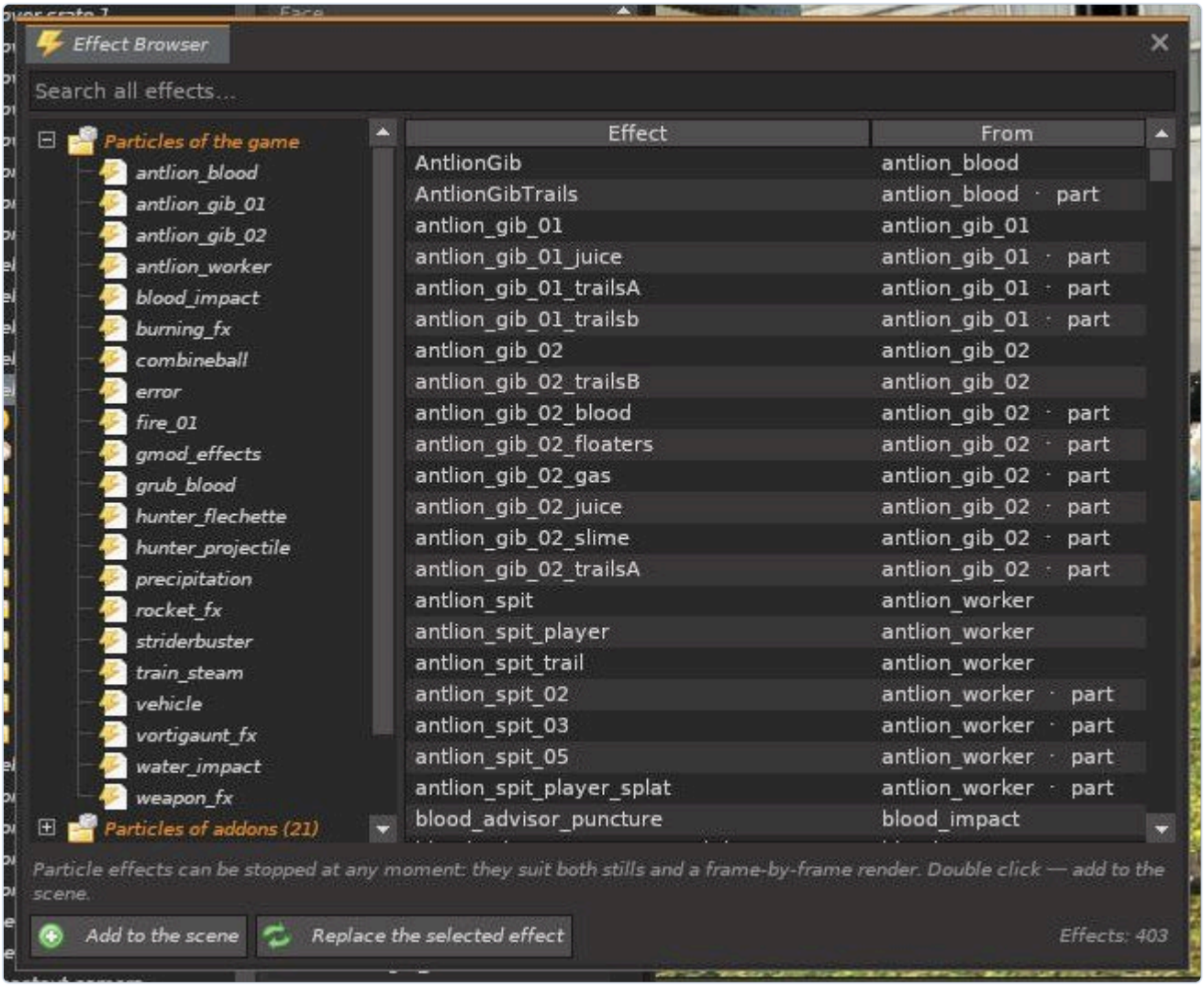
### Smoothness, s

In an animation the camera floats through its keys instead of stopping at each one.

A camera's own **depth of field** (focus distance, aperture, quality) and **exposure**, as in SFM, are set in the Render window under *Camera effects → Camera (as in SFM)*. See Rendering.

# Effects

Explosions, fire, smoke, blood, sparks and muzzle flashes are objects too. A particle effect can be stopped at any moment and stays frozen in the viewport and in the render.




**Effect Browser** (**+** → ▸ **Add an effect...**). Groups on the left, effects and the file they come from on the right. Double-click adds the effect to the scene.

## Kinds of effects

| GROUP | CAN BE FROZEN | NOTES |
|---|---|---|
| Particles of the game / of addons | ✓ | The best choice for stills and frame-by-frame video. Includes every particle file of installed Workshop addons. |
| | | |

| | | |
|---|---|---|
| Emitter tool effects | — | Play by themselves; a still render starts them and waits for the chosen moment. |
| Engine effects · Lua effects | — | Some need a weapon or a character and show nothing alone. |

## Effect settings

### Moment, s

How long after its start the effect is shown. 0 means it has not started yet. After a move or a change of the moment the effect is replayed up to that moment, which takes as long as the moment itself.

### Size

Scale of the effect (not every effect uses it).

### Length

Distance to the second point: where a tracer or a beam goes. 0 means a one-point effect.

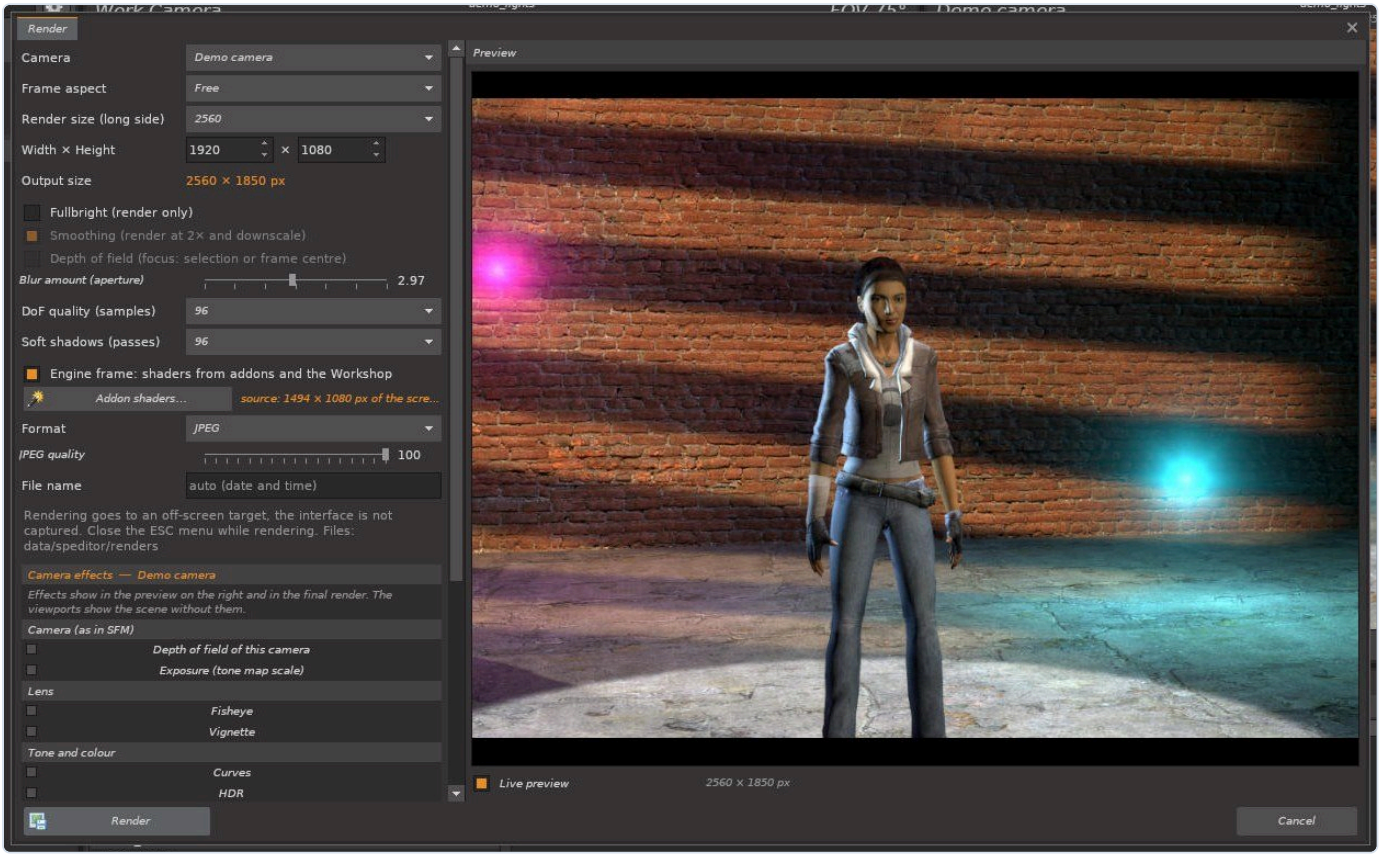
Effects can be attached to an object or a bone (a flash on a muzzle), duplicated, hidden, aimed and saved with the scene. In SAeditor, **Play the effect from this frame** keys the moment for you.

> An effect advances one game frame at a time. Render animations with the game window focused: an unfocused game runs at about 20 FPS, and effects shorter than a frame (a muzzle flash lasts ~0.03 s) may not show.

# Rendering

The Render window turns the scene into a picture. It renders off-screen, so the interface never gets into the file, and shows a live preview of exactly what you will get.

Open it with the ▸ **Render** button at the top right, ▸ **File → Render...** or the viewport gear menu.




**The Render window for a still scene.** Settings on the left, the preview on the right. The preview grows with the window.

NEW The window has two tabs: **Picture** (one still of the current frame) and **Animation** (the video, see Video). A scene with animation opens on the Animation tab. The look of the frame — size, smoothing, depth of field, SSAO, shaders, camera effects — is shared by both.

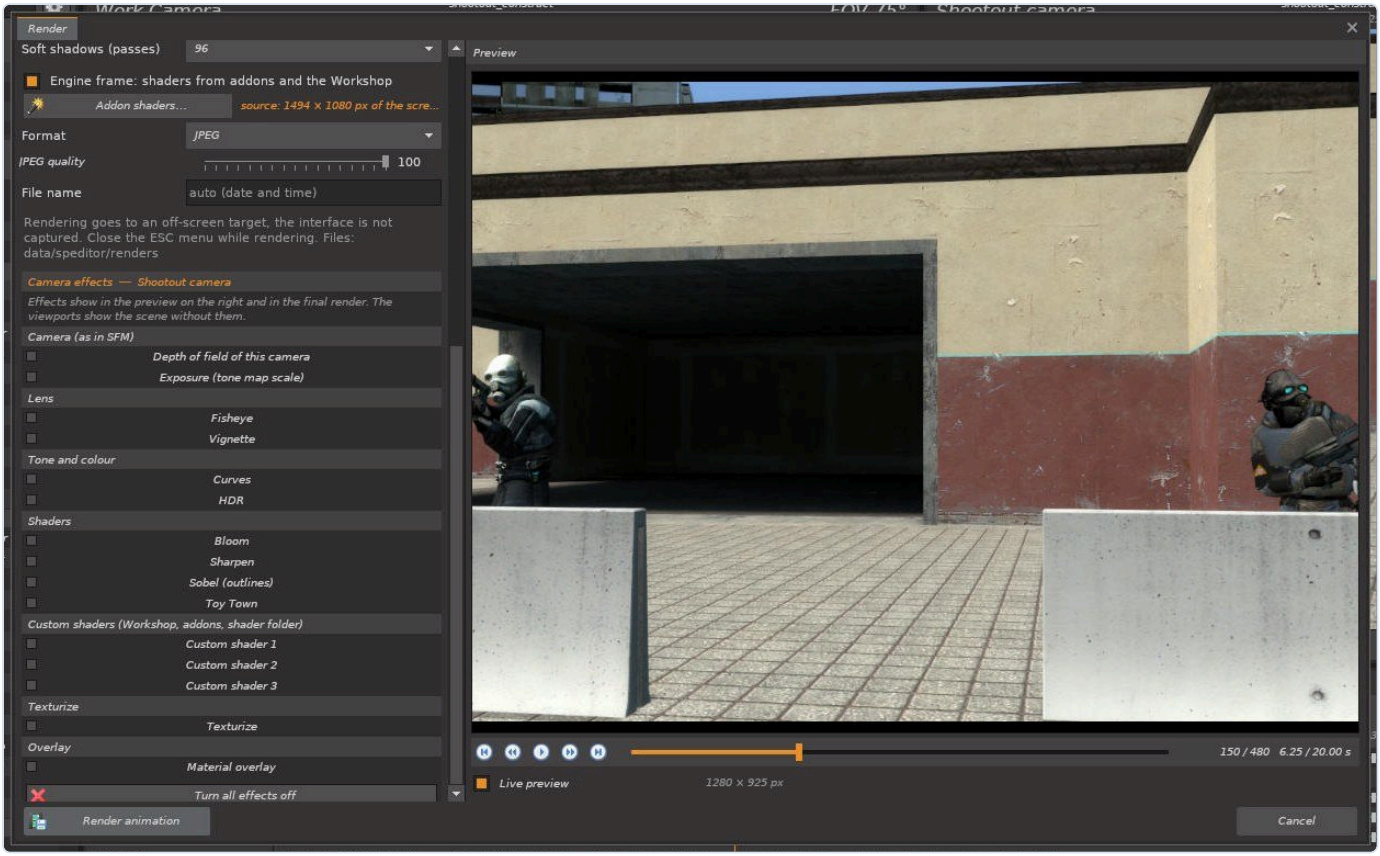
## Settings

| SETTING | MEANING |
| --- | --- |
| Camera | A scene camera, or *Primary Viewport source* (whatever the Primary Viewport shows). |

| | |
|---|---|
| Frame aspect | Free (the viewport's shape), 2:3 poster, A-series paper and other ratios. |
| Render size (long side) | The long side of the picture, up to 3840 px. *Custom size* sets width × height exactly. *Output size* shows the result. |
| Fullbright (render only) | Render without lighting. |
| Smoothing | Renders at 2× and scales down: clean edges. |
| Depth of field | Focus on the selection or the frame centre; *Blur amount* (aperture) and *DoF quality* (passes). |
| Ambient occlusion (SSAO) NEW | Soft contact shadows where things meet — corners, folds of clothes, under feet and objects — as in SFM. *SSAO radius* (units) is how far around a point the shadow reaches, *SSAO strength* how dark it gets. The editor's own effect, no other addon needed; in the preview, the picture, the video, the viewport Render look and the Engine frame. Saved with the scene. |
| Soft shadows (passes) | How many times the scene is rendered for lights with a *source size*. More passes, smoother penumbra, longer render. |
| Engine frame | Lets the game's own renderer draw the picture so that post-processing addons and Workshop shaders get in. See Shaders. |
| Format · JPEG quality | PNG (lossless), JPEG or **Steam screenshot**: the PNG is saved and the picture also goes into your Steam screenshots — the game shows it full screen for a moment and takes a screenshot. Publish it from ▸ **Steam → View → Screenshots → Garry's Mod → Upload**. The Steam copy has the size of your screen. |
| File name | Empty means date and time. Files go to `data/speditor/renders/`. |

> **!** Close the ESC menu while rendering. A large render with depth of field and soft shadows can take a while — the preview shows the progress (“DoF preview: 12 of 96 samples”).

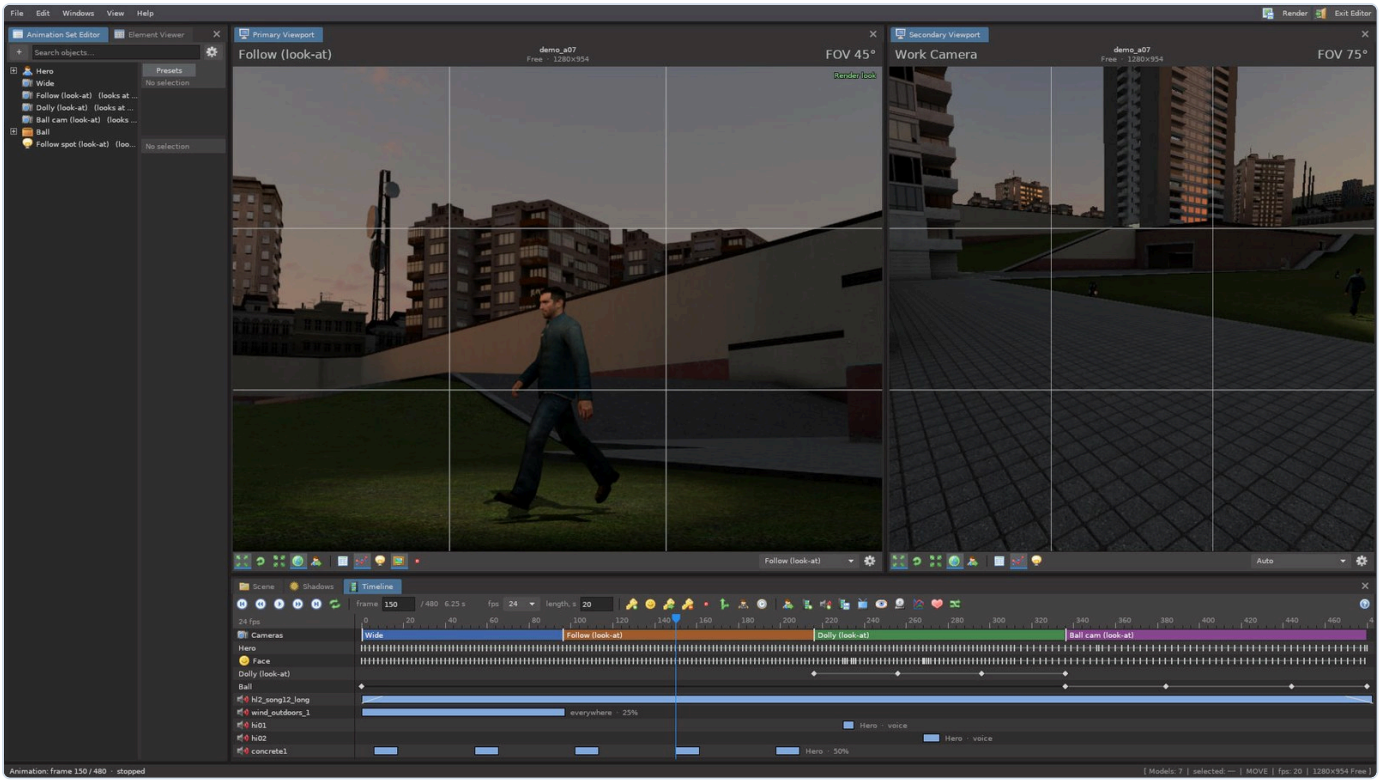
# Camera effects




**Camera effects** at the bottom of the settings. They belong to the chosen scene camera and are saved with it.

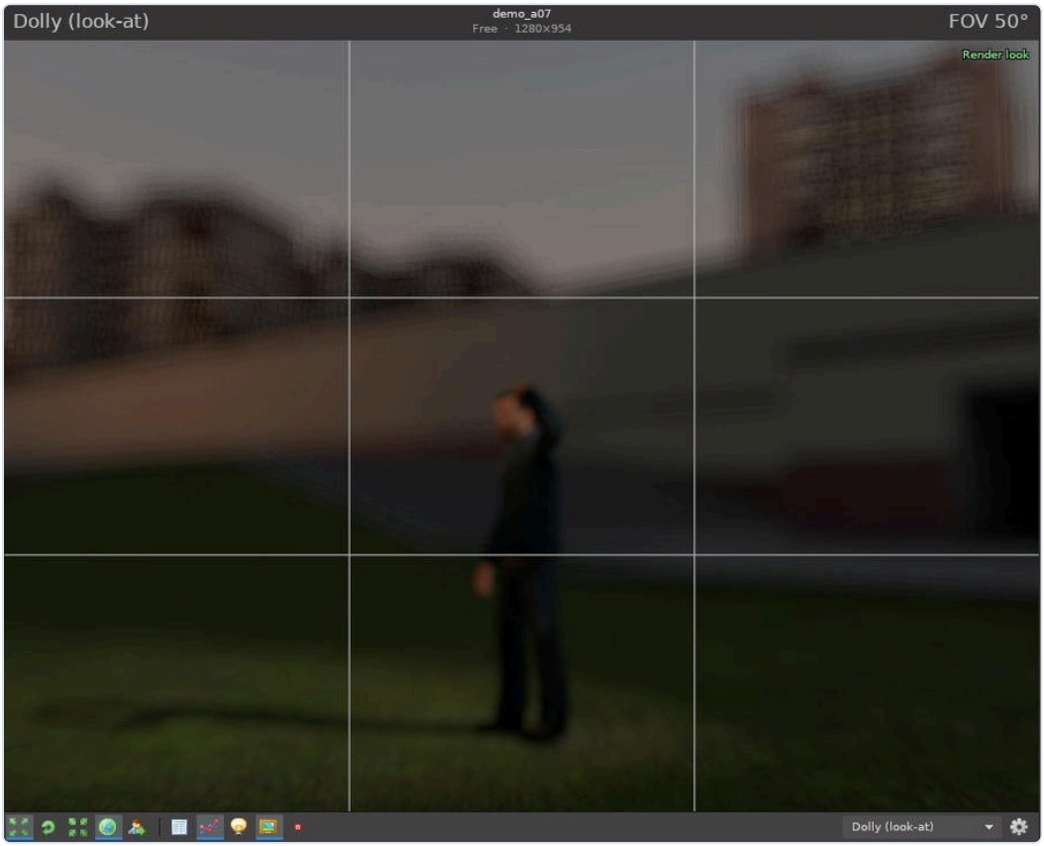
| SECTION | EFFECTS |
|---|---|
| Camera (as in SFM) | Depth of field of this camera (focus distance, aperture, quality) · Exposure (tone map scale) |
| Lens | Fisheye (strength) · Vignette (strength, clear radius) |
| Tone and colour | Curves for RGB and each channel (click to add a point, drag to move, right-click to remove) · HDR |
| Shaders | Bloom · Sharpen · Sobel (outlines) · Toy Town |
| Custom shaders | Up to three shader materials from the Workshop, addons or the shader folder |
| Texturize · Overlay | A texture pattern over the picture · any material over the picture |

The effects show in the preview and in the final render; the viewports stay clean. The work camera has no effects — choose a scene camera. *Turn all effects off* resets them.




**SSAO on:** soft shadows under the walking character and in the corners of the walls.

## Render look in the viewport NEW




The photo button of the Primary Viewport bar, as in SFM: stop on a frame and the viewport shows the finished picture — depth of field, soft shadows, smoothing, SSAO and the camera

effects (bloom, colour, vignette…). While you fly, drag or play it shows a quicker draft of the same, so motion and effects are seen together. The gizmo and the skeleton are hidden in it: turn it off to pose. Motion blur is seen only in the video.

> ✓ Things far from the focus are blurred on purpose while the Render look is on — that is the depth of field.

## Animated scenes



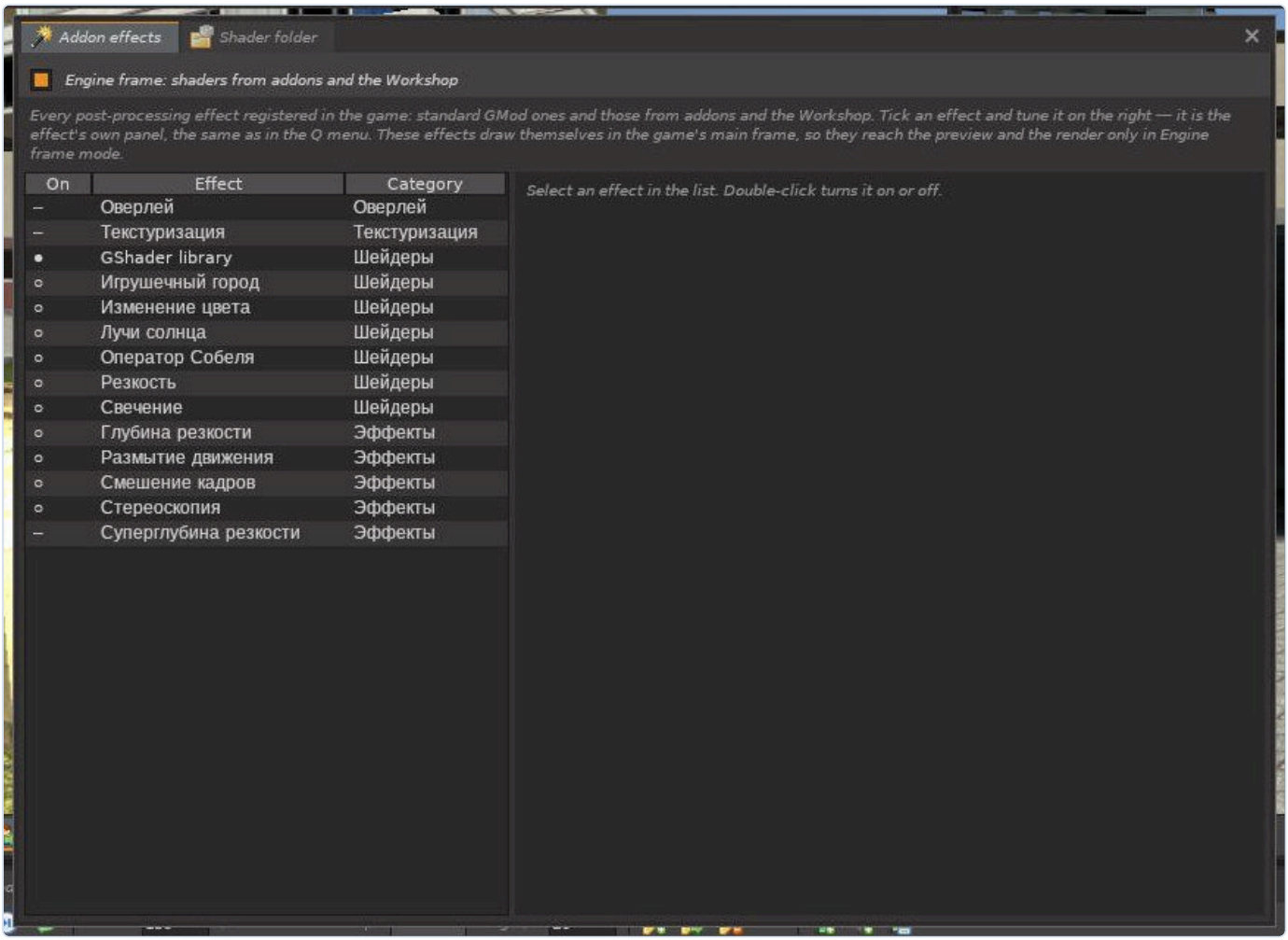

**The Animation tab:** frames, video settings, the cameras of the video with the list of shots, and a player under the preview.

See Animation → Video for the video file, the frame rate, motion blur and the shots.

> ✓ `speditor_render` in the console renders with the current settings — bind it to a key for quick test shots.

# Workshop shaders

Post-processing addons and shader packs from the Workshop can get into your renders in two ways: through **Engine frame** or as **custom shaders**.




**Windows → Addon shaders.** Every post-processing effect registered in the game, with its category. Double-click turns an effect on or off; its own settings panel appears on the right.

## Engine frame

Tick *Engine frame* in the Render window. The picture is then drawn by the game's renderer through the chosen camera with every enabled effect — standard GMod effects, Workshop shaders and shaders built on GShader Library — and taken from the screen.

> Engine frame limits: the picture is no larger than the screen, and 2× smoothing, penumbra and the editor's depth of field are not applied. The Render window shows the source size, e.g. “source: 1494 × 1080 px of the screen”.

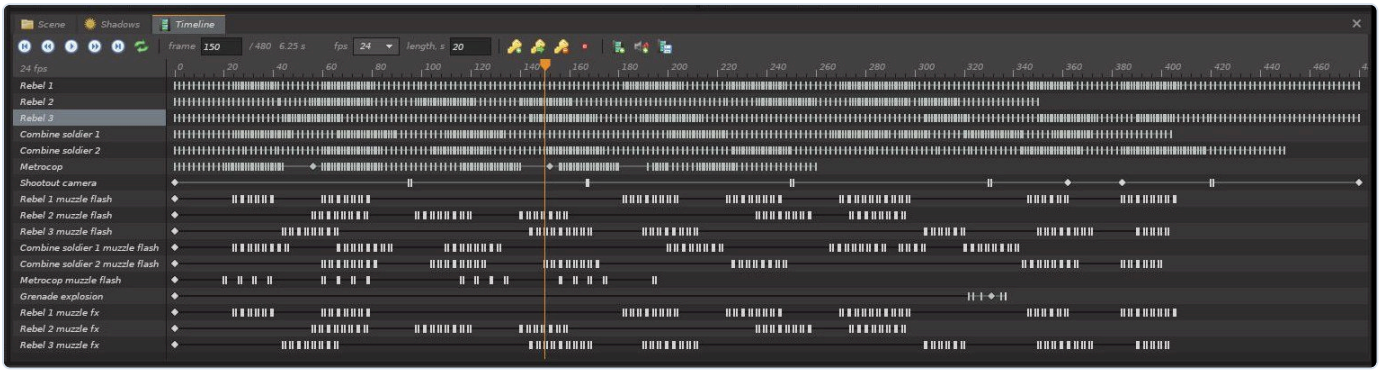
## The shader folder

The *Shader folder* tab of the same window: put a downloaded shader pack (with `materials/` and `shaders/fxc/`, or loose `.vmt` and `.vcs` files) into `garrysmod/addons/speditor/shaders/` (create the folder if it is missing) and press *Rescan*. The pack is mounted without restarting the game, and its shaders appear in *Camera effects* → *Custom shader*. Lua from a pack is not run: addons with Lua are installed the usual way and work through Engine frame.

> On Linux the folder name and the files inside must be lowercase, otherwise the game cannot read them.

# Animation: keys and the timeline

SAeditor is the animation mod that comes with SPeditor. It adds the **Timeline** tab to the bottom window. You set keys — full snapshots of an object at a frame — and SAeditor calculates the motion between them.




**The timeline of the example scene.** One row per animated object. Diamonds are keys; dense marks are baked animations (a key every frame or two). The orange line is the playhead.

## Your first animation

1. Select an object, leave the playhead on frame 0 and press `K`. A key appears in the object's row.
2. Move the playhead (click the ruler or press `.` several times), change the object — move it, change the pose, the light colour, the camera angle — and press `K` again.
3. Press `Space` to play. Between the keys everything is interpolated.

A key remembers everything about the object: position, physics pose, bone offsets (scale too), face flexes, colour and opacity, light settings, camera FOV and shake, the moment of an effect, and skin and bodygroups (they switch at the key instead of blending).

## The toolbar

| CONTROL | ACTION |
| --- | --- |
| Transport buttons | First frame (`Home`), previous key, play / pause (`Space`), next key, last frame (`End`), loop playback. |
| frame | Current frame; type a number to jump. Next to it: total frames and time. |

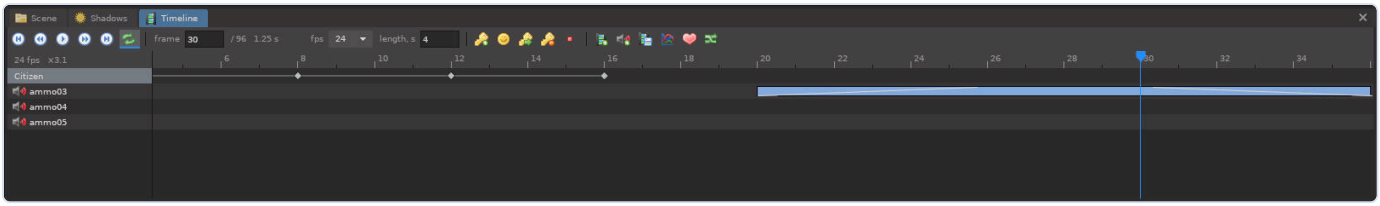
| | |
|---|---|
| fps · length, s | Frame rate (12, 15, 24, 30, 60) and length in seconds — up to an hour (it was 3600 frames). |
| Key buttons | Set key for the selection (`K`), face key (the smile, `Shift` + `K`), set key for every object, delete key, **Auto key** (the red dot: every edit sets a key at the current frame; the status bar shows AUTO KEY). |
| Right buttons | Import model animation, add a sound, render animation; curve editor, procedural motion (the heart), Stop Motion Helper (the arrows); ready motion (the walking figure), bone keys (the branch), ghosts (the grey figure), record gameplay (the webcam), the player (the TV) and walking around the scene (the eye). |

## Working with keys

- `LMB` on a key selects it and moves the playhead there; drag to move it; `Ctrl` + drag copies it. `Ctrl` + click picks several keys, `Shift` + drag over empty space picks keys in a box — then they move, change their motion and are deleted together.
- `RMB` on a key: *Motion to the next key* — **Continuous** (a spline through the keys, the kind of every new key), **Smooth** (stops at the key), **Linear**, **Speed up** (ease in), **Slow down** (ease out), **Hold** (jump: the value holds until the next key) and in *More... With a swing*, *Springy*, *Bouncing*, *Gentle*, *Sharp* (see them move); *Copy key*; *Delete key*.
- `RMB` on an empty place of a row: *Set key here*, *Paste key here*, *Remove animation of this object*.
- `LMB` on the ruler or empty space moves the playhead; drag to scrub. The wheel scrolls the list of objects; `Ctrl` + wheel zooms the timeline, `Shift` + wheel scrolls it in time.

| KEY | ACTION |
|---|---|
| `Space` | Play / pause |
| `K` | Set a key for the selection |
| `Shift` + `K` | Set a face key for the selection |
| `Ctrl` / `Shift` + wheel | Zoom the timeline / scroll it in time |
| `,` / `.` | Previous / next frame |
| `Home` / `End` | First / last frame |

# Long animations and zoom



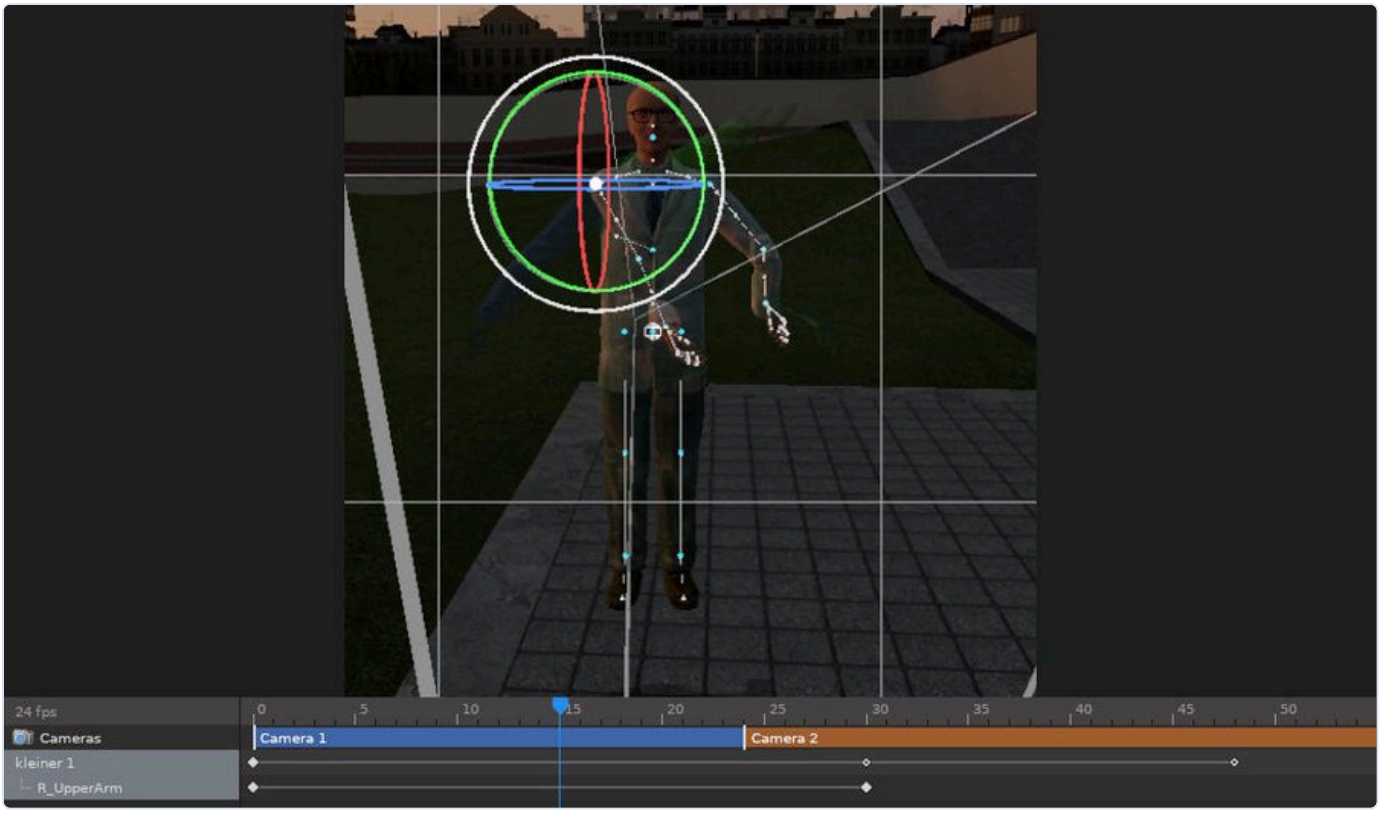

**The timeline zoomed in three times** (×3.1 next to the frame rate): frames 5–36 fill the width, so close keys can be picked and moved one by one.

An animation can be up to an hour long, and an object can hold up to 100 000 keys — a 12-minute animation baked with a key on every frame fits. `Ctrl` + wheel zooms the timeline around the mouse, `Shift` + wheel scrolls it in time; while the animation plays, the view follows the playhead. Zoom all the way out to see the whole animation again.

# Face keys

`Shift`+`K` (or the smile button, or the menu of a track → *Separate face keys*) gives the face a track of its own, right under the object. A blink or a line of speech then has its own rhythm over a slow body motion. With Auto key on, face edits key the face track.

# Bone keys and ghosts




**Ghosts on:** the arm at the key before the playhead is drawn see-through in blue, at the key after it in green. Below the character — the row of the selected bone with its own keys.

Normally a key holds the whole character. With **Bone keys** on (the branch button of the timeline, ▸ **Edit → Animation → Bone keys**) a key set while a bone is selected holds only that bone — every bone moves between its own keys, as in SFM: an arm can wave twice while the head turns once. With Auto key, dragging a bone keys just the bones the drag moved. Keys of some bones are small hollow diamonds on the character's row.

**Ghosts** (the grey figure button) show the selected character at its previous and next keys, so you see where the motion comes from and goes to without scrubbing.

## Camera effects in keys

A camera key also holds its effects — depth of field, exposure — so the focus can travel from one actor to another. With Auto key on, flying a camera you look through (`V`) or changing it keys it.

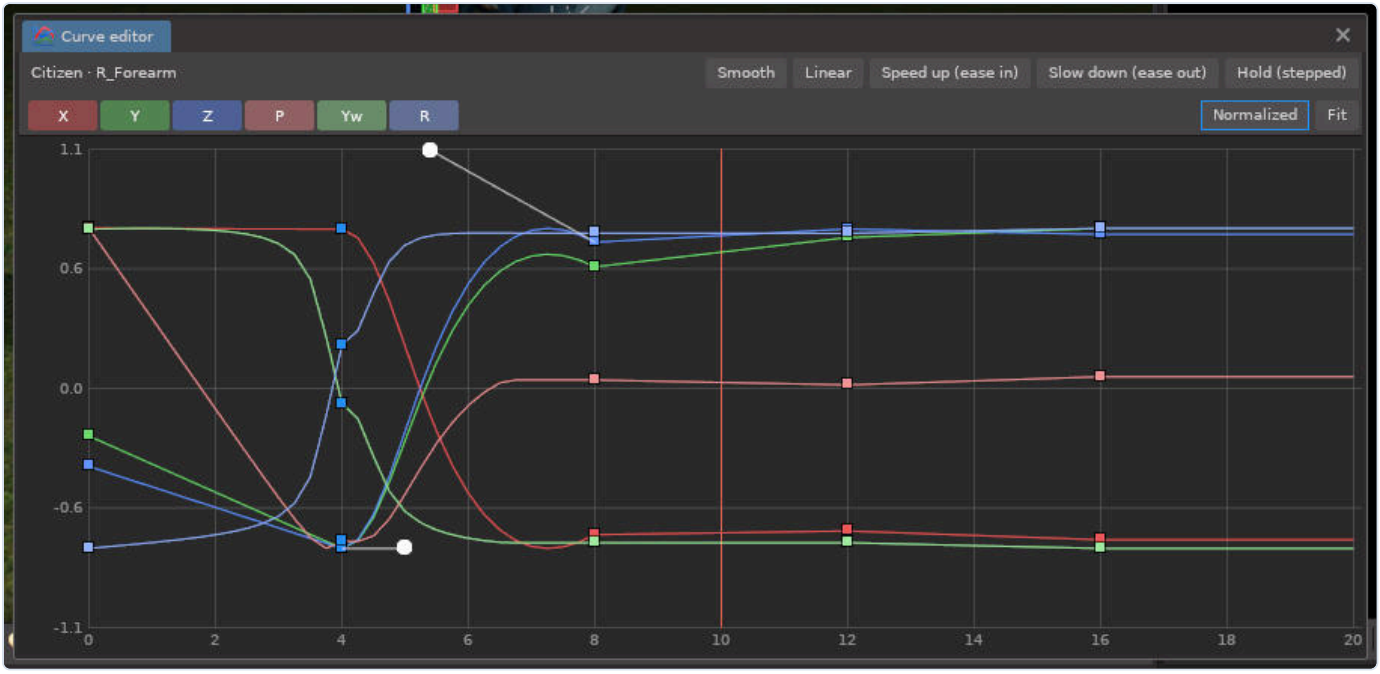
> Changed a pose and it jumped back when the playhead moved? The frame had no key: without Auto key an edit between keys is not kept. Press `K` (or turn Auto key on) before moving the playhead — the editor reminds you with a hint.

The animation is stored with the scene: ▸ **File → Save Scene...** saves it too. Physics poses are set on the server, so playback is smooth in singleplayer and on a listen server; on a busy dedicated server it depends on the network.

# Curves and procedural motion

Two tools work on top of the keys: the curve editor shapes the motion between them, procedural motion adds life that needs no keys at all.

## The curve editor




**Curves of the right forearm.** One line per value: X, Y, Z of the place and P, Yw, R of the turn. Squares are keys; the white circles are the handles of the selected key; the red line is the playhead.

Open it with the curve button of the timeline or ▸ **Animation → Curve editor**. It shows the selected object, or the selected bone of a character, over time.

- The coloured buttons show and hide the channels.
- **Normalized** draws every channel in its own range, so degrees and units fit together; off, the real values are drawn. **Fit** shows all the keys.
- Drag a key up or down to change its value, left or right to move it in time.
- *Smooth*, *Linear*, *Speed up*, *Slow down*, *Hold* set the motion after the selected key.
- Drag a round handle to bend the motion between two keys yourself — it may overshoot and come back. Turns are drawn continuously past ±180°, so a spin does not jump.

## Kinds of motion

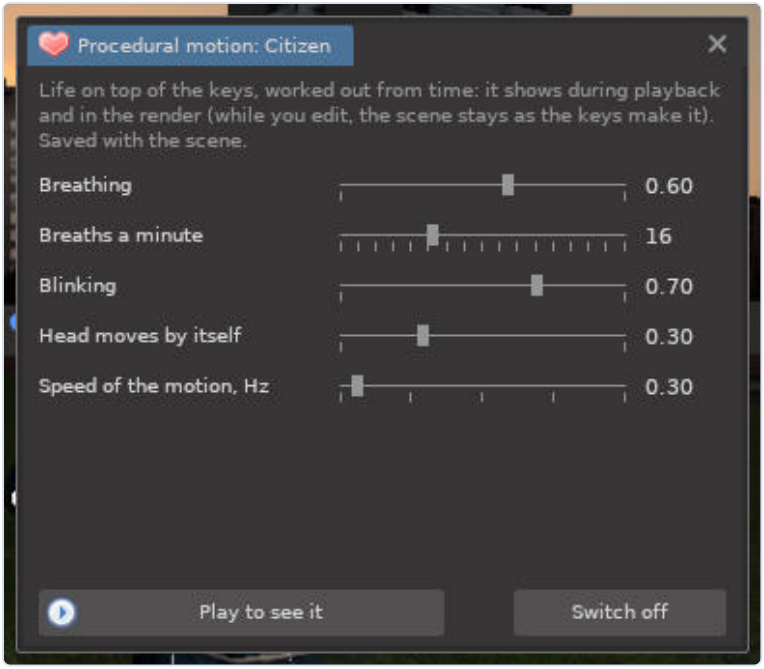



**A recorded hand with Continuous keys:** the lines pass through the points without breaks. The strip at the bottom shows how the selected key moves — in words and with a moving ball.

- **Continuous** (spline) — every new key: the motion flows through the keys without stopping, as in SFM and Blender.
- **Smooth** stops at every key; **Linear** — machines, camera pans; **Speed up** — falling, throwing; **Slow down** — arriving, landing; **Hold** — cuts and switches.
- *More...*: **With a swing** (an anticipation and an overshoot), **Springy**, **Bouncing** (a ball hitting the floor), **Gentle**, **Sharp**.

▸ **Help → How to use → How objects move between keys** shows every kind moving side by side.

# Procedural motion

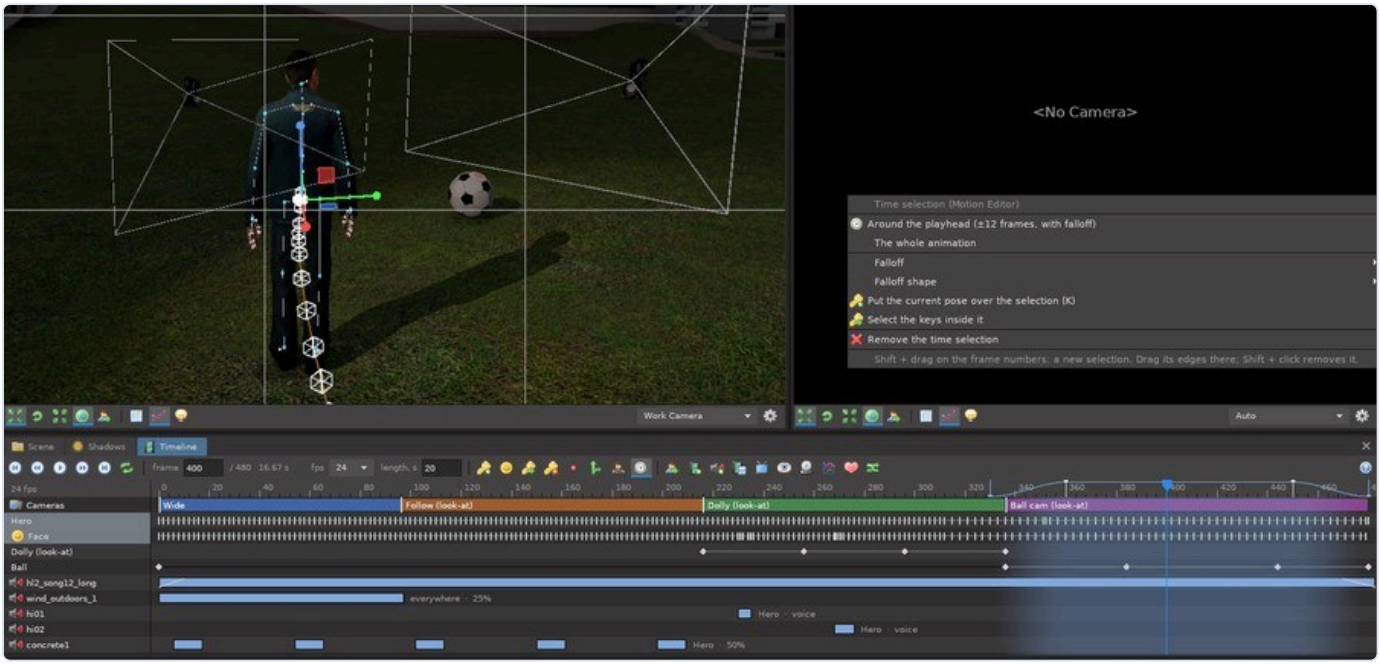



The heart button of the timeline, or right-click an object → ▸ **Animation → Procedural motion...** Characters get *Breathing*, *Breaths a minute*, *Blinking* and a *Head that moves by itself*; props, lights and effects get a slow *Drift*. *Speed of the motion* sets how fast it wanders.

It is worked out from time: it shows during playback and in the render, but never while you edit, so keys never capture it. *Play to see it* previews it, *Switch off* removes it. It is saved with the scene.

# Motion Editor: an edit over a stretch of time NEW

A key changes one moment. The Motion Editor, as in SFM, changes a whole stretch at once: lift an arm a little higher during the whole wave, push a walk a step aside, make a head look down for two seconds — and keep the motion smooth at both ends.




**A time selection** on the timeline (the tinted stretch at the right) and its menu (the clock button).

## The time selection

- `Shift` + drag on the frame numbers of the timeline makes a time selection. Its middle is tinted; the curve drawn over the numbers shows how much an edit counts at every frame.
- It has four edges. Between the inner two an edit counts in full; from each inner edge to the outer one it fades out — the **falloff**. Drag the edges on the frame numbers; edges standing together part when dragged (outwards — a falloff, inwards — a narrower middle).
- The clock button of the timeline (and a right click on the frame numbers) opens its menu: from the selected keys, around the playhead, the whole animation; the falloff in frames (none, ¼, ½, 1 or 2 seconds); its shape — *Smooth* or *Linear*; select the keys inside; remove it.
- `Shift` + click on the frame numbers removes it. The status bar shows it while it is there.

## Editing

1. Put the playhead inside the selection.

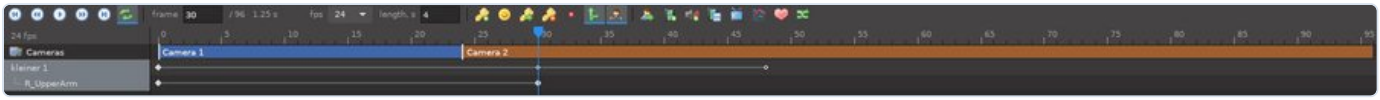
2. Change the object: move it, pose bones, change the face, a light's colour, a camera's field of view.

3. Press K (or let *Auto key* do it). The difference between the object now and what its animation gave at that frame is added to every key in the selection — in full in the middle, less and less over the falloff. Keys are added at the edges first, so nothing outside the selection moves.

> It works for bone keys and the face track too, and one undo takes the whole edit back. A key set with *Bone keys* on and a bone selected stays an ordinary bone key.

# Camera cuts and the player

The **Cameras** row at the top of the timeline says which scene camera the animation is watched through, from which frame on — like shots in a film. The player and the video follow it.




**The Cameras row:** a coloured band per shot, named after its camera.

1. Add two cameras to the scene and place them.
2. Right-click the Cameras row at a frame → *Switch to Camera 1 here*. Do the same later for Camera 2.
3. Drag a cut mark to move it; right-click it to change its camera or delete it. `Ctrl` + `Z` undoes.

## The player



**The player** (the TV button of the timeline, or ▸ **File → Watch in the game → Player**): the scene and the camera at the top, the seek bar with the shots marked, play / pause, stop, the previous / next shot, repeat, render look, volume.

Keys: `Space` play / pause, `←` `→` 5 seconds, `PgUp` / `PgDn` previous / next shot, `Home` / `End`, `↑` `↓` volume, `M` mute, `R` repeat, `V` render look, `Backspace` or the cross — leave. A click on the camera name at the top picks another camera; a click on the picture pauses.

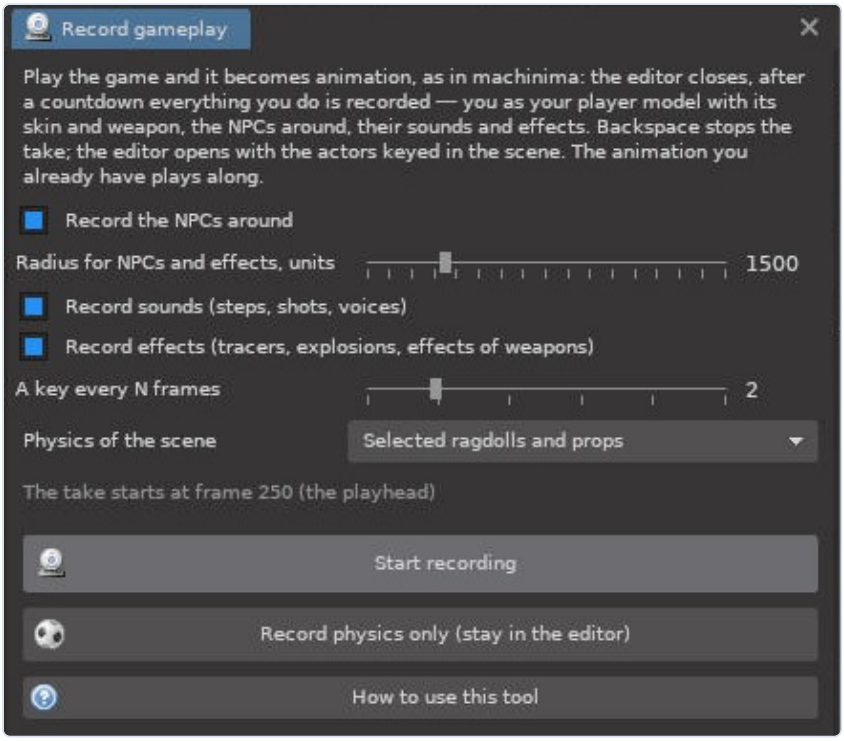
**Render look** (the button beside repeat) shows the video as the Render window will make it — its frame proportions with black bars, camera effects, depth of field, soft shadows, the engine frame — before you render anything. While it plays, depth of field and soft shadows are lighter; on pause the frame is full.

## Walking around the scene

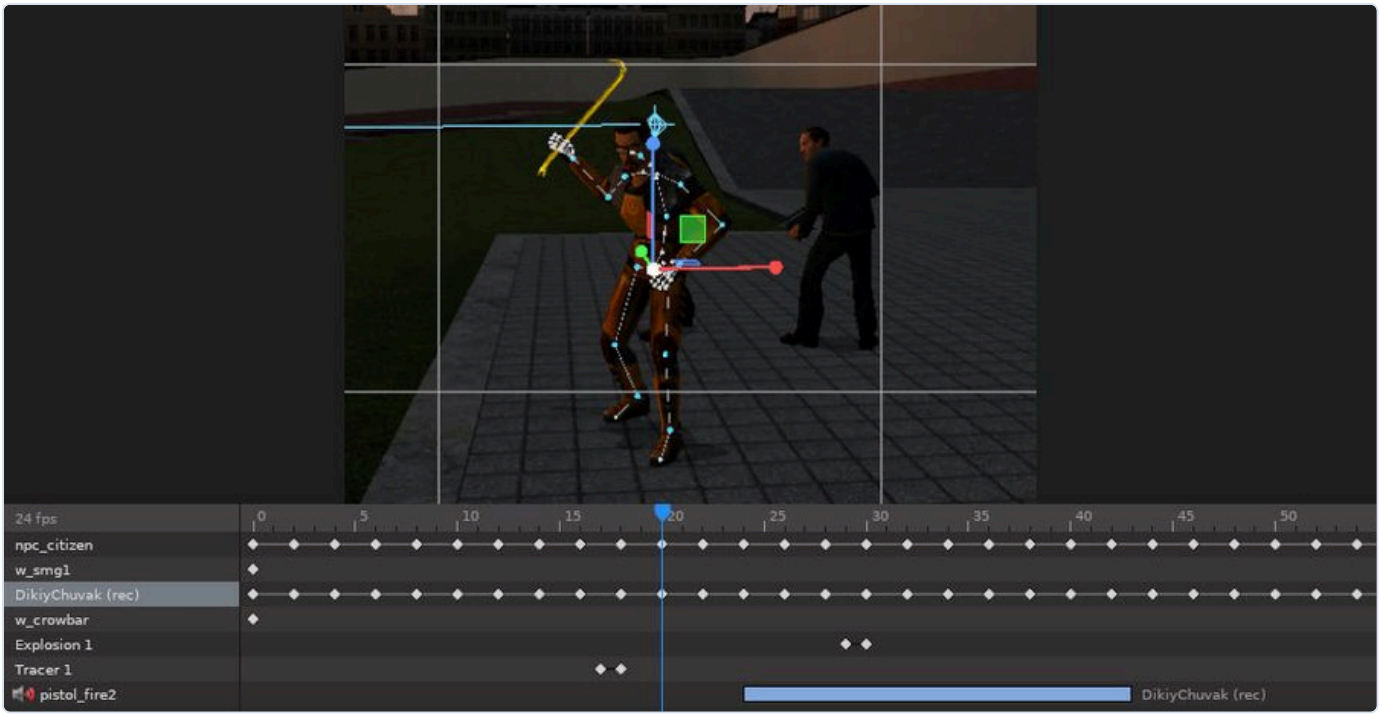
The eye button (▸ **File → Watch in the game → Walk around the scene**, console `saeditor_show eyes`): the animation plays while you walk around it yourself, with your HUD and weapon. `Backspace` stops it.

# Record gameplay (machinima)

Instead of posing every frame you can simply play: walk, run, shoot, let the NPCs around do their thing. The take becomes ordinary keys, sounds and effects that you edit like any animation.




1. Put the playhead where the take should start. Open ▸ **Record gameplay** (the webcam button of the timeline or ▸ **File → Record gameplay...**) and choose what to record: the NPCs within a radius, sounds, effects, how often a key is taken.
2. Start: the editor closes, a countdown runs, then *REC* blinks in the corner. Play. The animation you already have plays along, so you can act against it.
3. `Backspace` stops the take and the editor opens.




**A recorded take:** the player with a crowbar and a citizen with an SMG, keyed every 2 frames; the tracer and the explosion are effect objects, the shot is a sound from the player.

You become a ragdoll of your player model (skin, bodygroups, player colour) with the weapon in your hand, every NPC gets its weapon, rockets and grenades fly as models, sounds go onto the timeline. Then edit it like any animation: delete keys, fix a pose (with bone keys — one arm), add cameras and cuts, record another take on top.

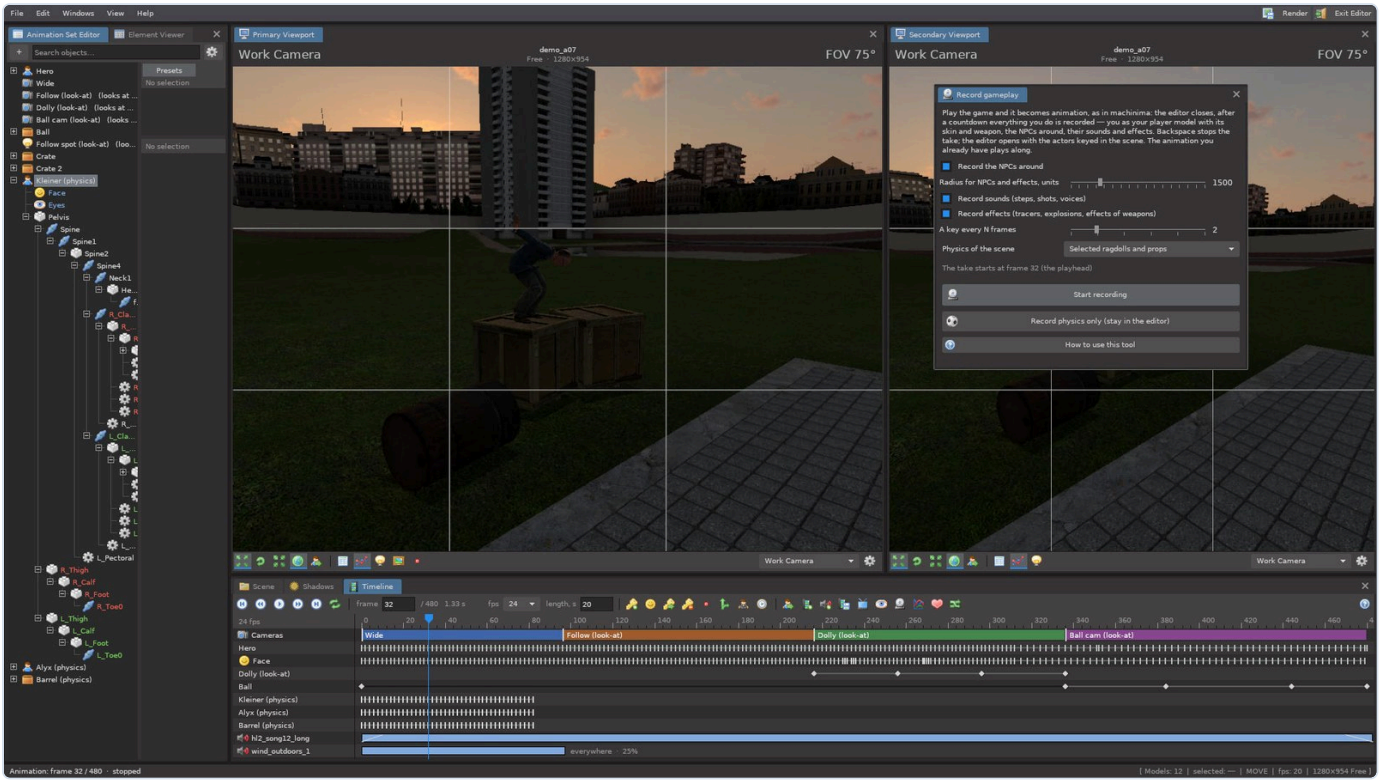
## Your view as a camera NEW

*Record my view as a camera (first person)*, on by default: what you saw while playing — where you looked, how you turned, the zoom — becomes the scene camera *<your name> view (rec)*, keyed along the take. If the animation has no camera cuts yet, it becomes the first one, so the player and the video go through your eyes at once. Your own body, what you hold and PAC3 outfits are not drawn in that camera, as in the game; from other cameras they are seen as usual. The first-person weapon (the viewmodel) is not recorded.

## Doors, lifts and buttons NEW

Doors, lifts, trains, buttons and other moving parts within the radius are recorded too — of the map and, since a0.7.1, those spawned by door tools and addons. A door you opened during the take opens in the animation at the same moment. *Remove all animation*, a cleanup and loading a scene put them back where the take found them. Map doors are saved with the scene for its map; spawned ones are found again only while they exist.

# Physics of ragdolls and props NEW

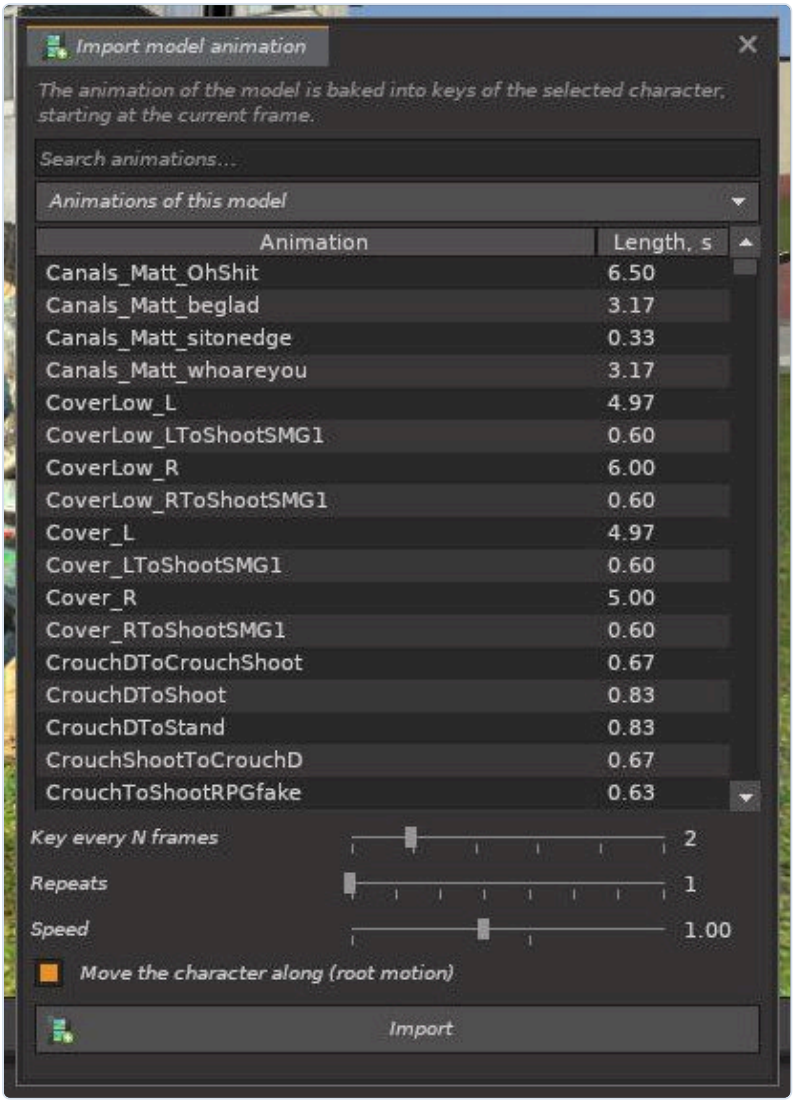



- **Record physics only (stay in the editor):** select ragdolls or props (nothing selected — all of them), put the playhead where the fall starts and press it. They are let go and fall, roll and collide; *Stop* (or `Backspace`) writes their keys from that frame on.
- **In a game take:** *Physics of the scene* → *Selected / All ragdolls and props*. They are unfrozen for the take — throw them with the physics gun, shoot them — and get their own keys next to the recorded player and NPCs.
- The keys go onto the objects themselves (their old keys inside the take are replaced). Locked and attached objects are not moved.

> ! Not recorded: fingers (the hands keep their rest grip), the faces of the actors, effects the engine draws by itself (bullet holes, the muzzle flashes of HL2 weapons). An NPC whose body the game removes on death is hidden after its last frame; ragdolls of dead NPCs that stay are followed.

# Importing model animations

A character can get a ready-made animation — walking, running, gestures, dances — baked into keys. Right-click a character → **▸ Animation → Import model animation...**




### Source list

*Animations of this model*, *Player animations (male / female / zombie)* or *Animations of another model...* (type a path). Any model with the same skeleton works: a citizen can dance like a player model or use an animation pack made for players.

### Search

Filter by name; the list shows each animation's length.

### Key every N frames

1 is the most exact; 2–3 is lighter for long animations.

### Repeats

Loop a walk cycle several times.

#### Speed

Play faster or slower.

#### Move the character along

Root motion: a walk moves the character forward instead of walking in place.

The animation starts at the current frame. If it is longer than the timeline, the timeline is lengthened. Select several characters first to give all of them the same animation — they move in step.

### Fingers

In a0.3 only bones with a physics body were imported, so fists, pointing and gripping hands stayed open. Since a0.4 the import also brings fingers, toes and clavicles as bone offsets: a fist animation clenches the fist.

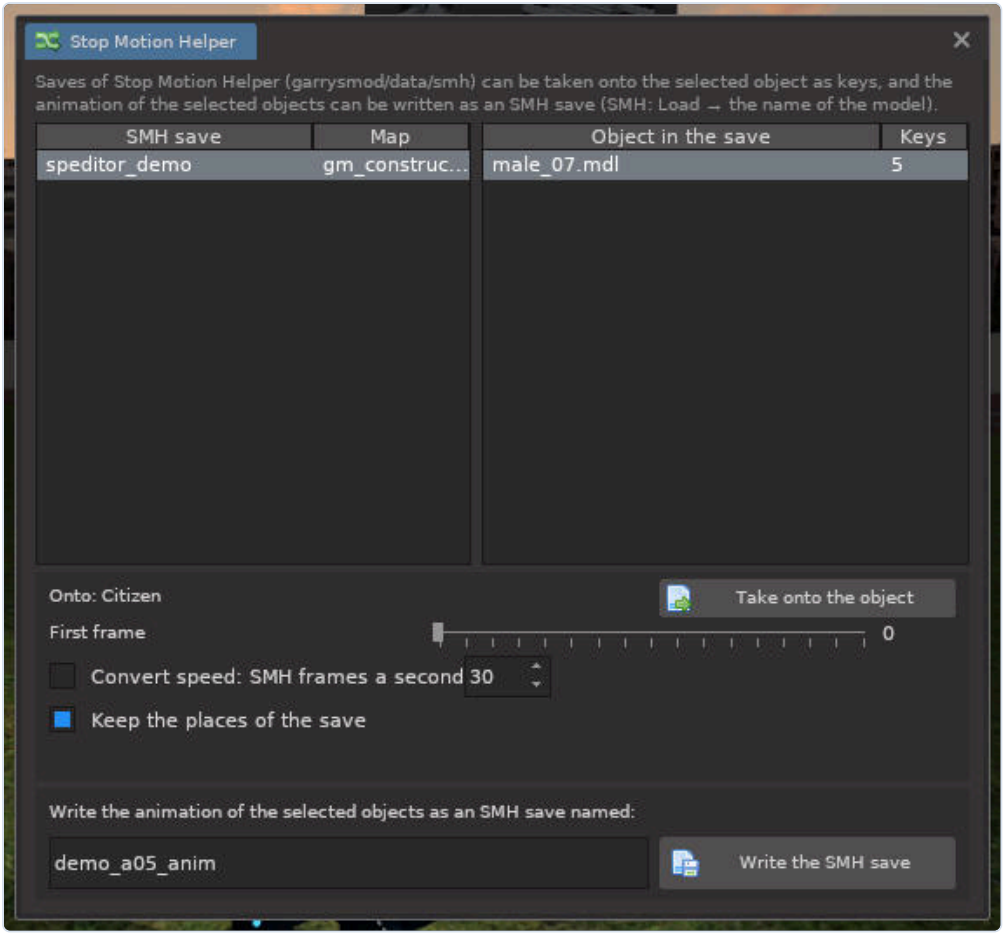
The import no longer breaks the neck and shoulders: the top of the spine, the neck and the clavicles keep the ragdoll's own shape, so the head and the arms stay joined to the body. Re-import animations made in a0.4 if a neck looked stretched. Models without physics can receive imported animations too (Models without physics).

> ! “This model has another skeleton” means the source model's bones do not match the character. Choose a source made for the same skeleton (most humans in GMod use ValveBiped).

## Ready motions and animation packs

- **Ready motion** (the walking figure button): a motion for the selected character, or a camera move — dolly, orbit, pan — for a camera, placed at the playhead.
- ▸ **Source → Animation pack of an addon…** in the import window: pick an addon (Workshop or a folder in `garrysmod/addons`), then a model of it — animation packs are models that hold only animations. Its animations are offered when the skeletons share bone names.
- The Stop Motion Helper import also finds SMH saves inside addons.

# Stop Motion Helper




Saves of Stop Motion Helper ( `garrysmod/data/smh/` ) become SAeditor keys, and back. Open the window with the arrows button of the timeline or ▸ **File → Stop Motion Helper**.

### Take onto the object

Choose a save on the left and an object of it on the right; its keys go onto the object selected in the scene.

### First frame

Where the keys start on the timeline.

### Convert speed

Rescales SMH frames a second to the timeline's frame rate; off, frame numbers are taken as they are.

### Keep the places of the save

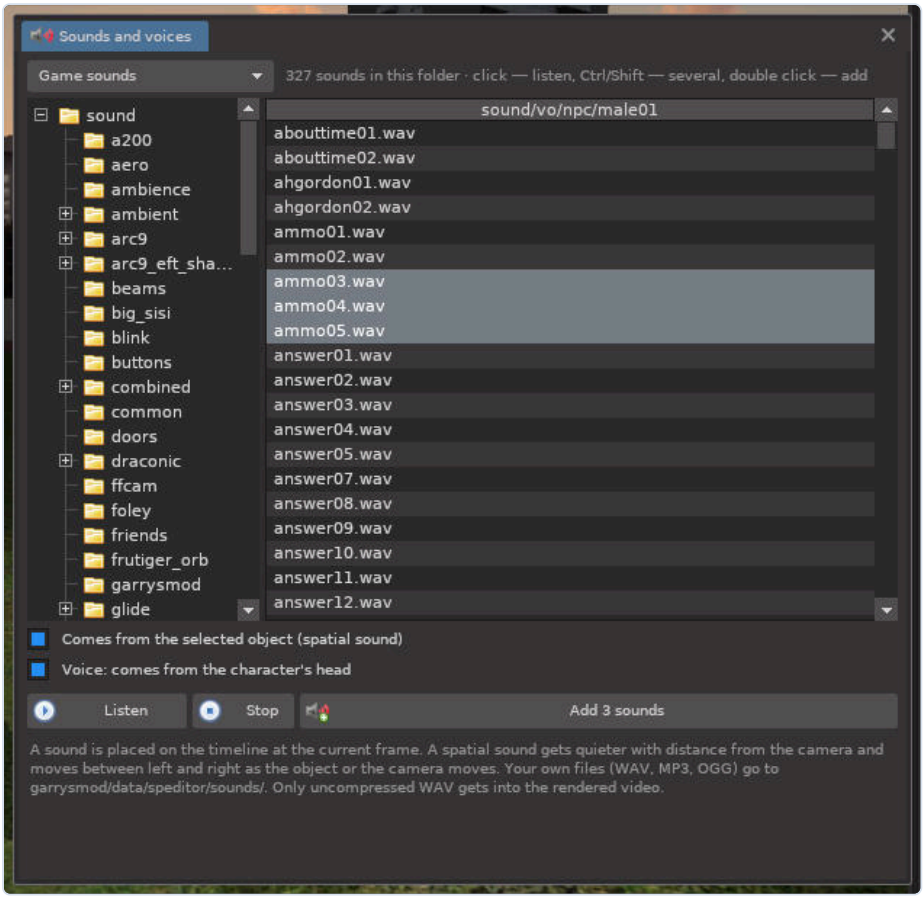
The object moves where it stood in SMH; off, the motion is moved to the object's current place.

### Write the SMH save

Writes the animation of the selected objects as an SMH save; load it in SMH with *Load* (matched by the model).

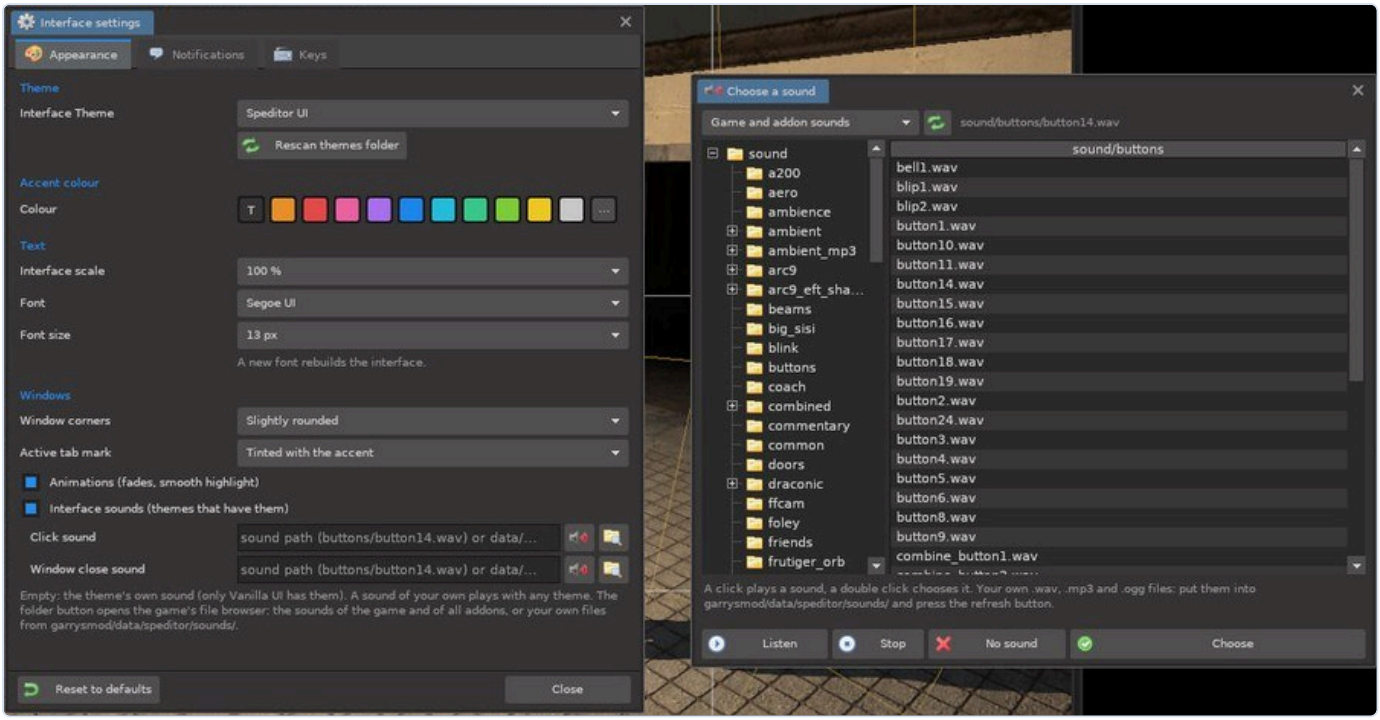
# Sounds and voices

Sounds are placed on the timeline and played with the animation. A sound can come from an object (spatial), from a character's head (voice) or be heard everywhere (music, a narrator).




Open it with ▸ **Animation → Add a sound or a voice line...** or the speaker button of the timeline.

- **Game sounds** browses the `sound/` folders of the game and addons; **My files** shows `garrysmod/data/speditor/sounds/` (WAV, MP3, OGG).
- **Listen** / **Stop** preview a file; a click on a file also plays it.
- **Several at once:** `Ctrl` / `Shift`-click files, then *Add 3 sounds* — they are placed one after another from the current frame.
- **Comes from the selected object**: quieter with distance from the camera, moves between left and right.
- **Voice**: comes from the character's head (ticked by itself for files in `sound/vo/`).
- **Add at the current frame** or double-click a file.




**Choose a sound** (a0.7): the game's file browser for interface sounds, notification sounds and *Replace the sound…* — a click plays, a double click chooses.

# Sounds on the timeline

Every sound is a blue bar under the objects; the same file from the same source shares one row (a burst of shots is one row). Drag a bar to move it in time. Right-click it for:

### Listen · Comes from · Voice

Preview; change the source object or make it heard everywhere; toggle voice.

### Volume · Speed and pitch · Distance of full volume

10–200 %, 50–200 %, 50–2400 units.

### Fade in · Fade out

0.1–4 s; drawn as a slope at the end of the bar, heard in playback and in the video.

### Start at the current frame · Duplicate · Remove the sound

As named.

### Replace the sound… NEW

Picks another file in the sound browser; the clip keeps its place, source, volume and fades.

## Cutting and trimming

- **Drag the left or right edge** of a bar (the cursor becomes ↔) to cut off the start or the end of the sound. A trimmed side is marked with a light stripe.

- Menu: *Cut in two at the playhead*, *Trim the start to the playhead*, *Trim the end to the playhead*, *Restore the whole sound*.
- Trimming works in playback and in the video's sound track; a few milliseconds of fade at each cut prevent clicks.

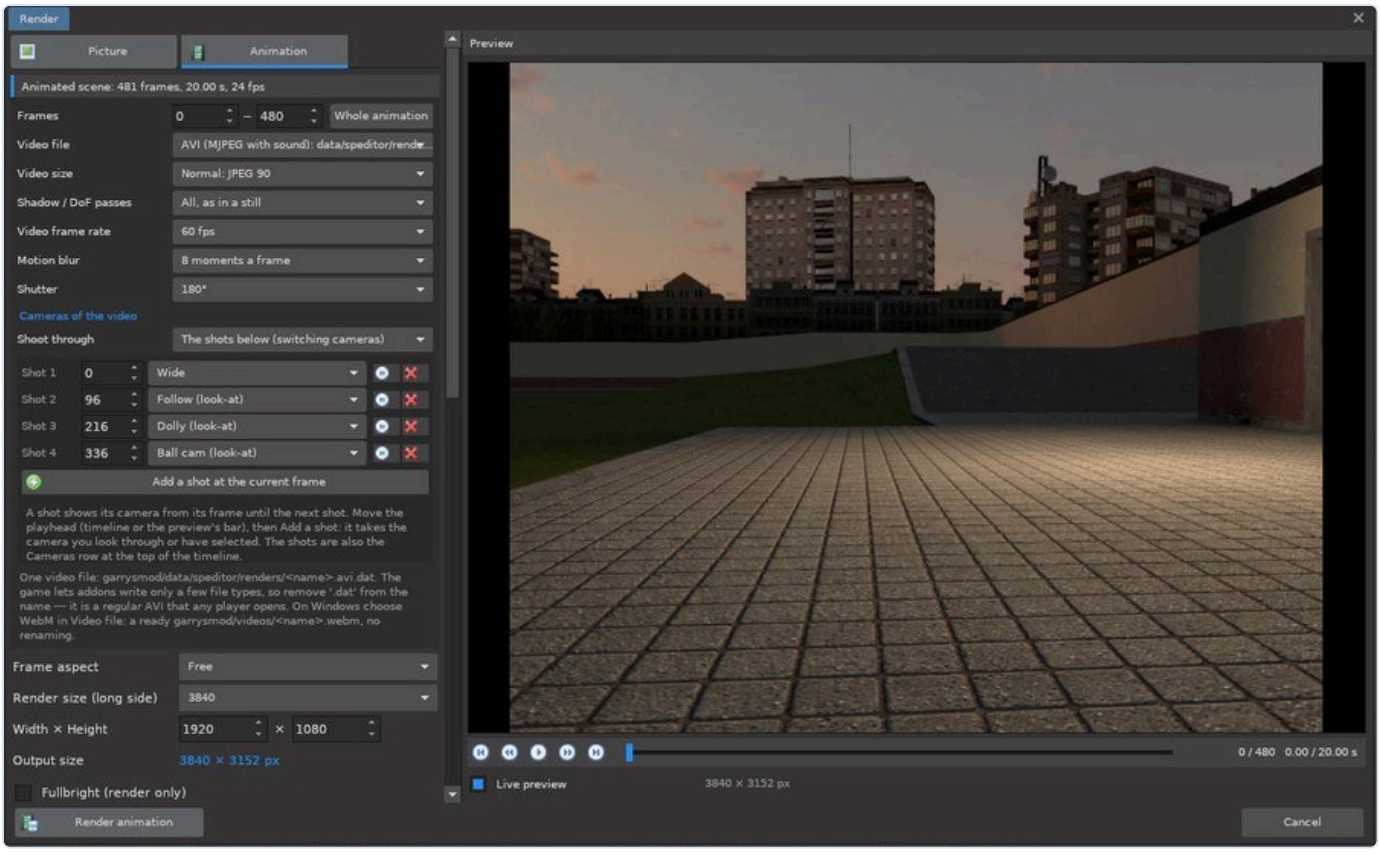
## Lip sync

Right-click a voice sound → *Lip sync* → the character. The mouth follows the phonemes written in the WAV (as Faceposer and Source Filmmaker write them; voices of Valve games have them), otherwise the loudness of the sound (MP3 and OGG too). The keys go to the character's Face row; the rest of the face is kept. Too strong or too weak: *Lip sync* → *How wide the mouth opens*, then run it again.

> Only uncompressed WAV gets into the rendered video. MP3 and OGG play in the editor but are left out of the video (the render tells you which ones). Convert them to WAV first.

# Rendering an animation to video

A scene with animation renders on the *Animation* tab of the Render window; the *Picture* tab renders one still of the current frame instead.




**The Animation tab** with the four shots of demo_a07.

The video is one AVI file `renders/<name>.avi.dat` with sound (MJPEG). Remove `.dat` from the name — it is a regular AVI any player opens; a video over 1.9 GB continues in a second file.

### Frames NEW

The whole animation or a part (from – to). The player under the preview shows any moment of the video before you render it.

### Video file NEW

AVI, or on Windows **WebM**: the game's own video writer makes a ready `garrysmod/videos/<name>.webm` (no renaming, smaller). It is at most the size of the screen, its width rounded down to a multiple of 16; the game writes no sound into it, so the sound goes next to it as `data/speditor/renders/<name>.wav` — put both into a video editor. On Linux the writer makes no file, so there it is AVI only.

### Cameras of the video NEW

*The shots below* — a list of shots right in the window: *Add a shot at the current frame* takes the camera you look through or have selected; each shot can get another camera or start frame, the arrow jumps to

it. The shots are the same as the Cameras row of the timeline. Or shoot the whole video through one camera or the Primary Viewport.

The window follows the loaded scene: after another scene is loaded the frames, the cameras and the file name (if its field is empty) are that scene's.

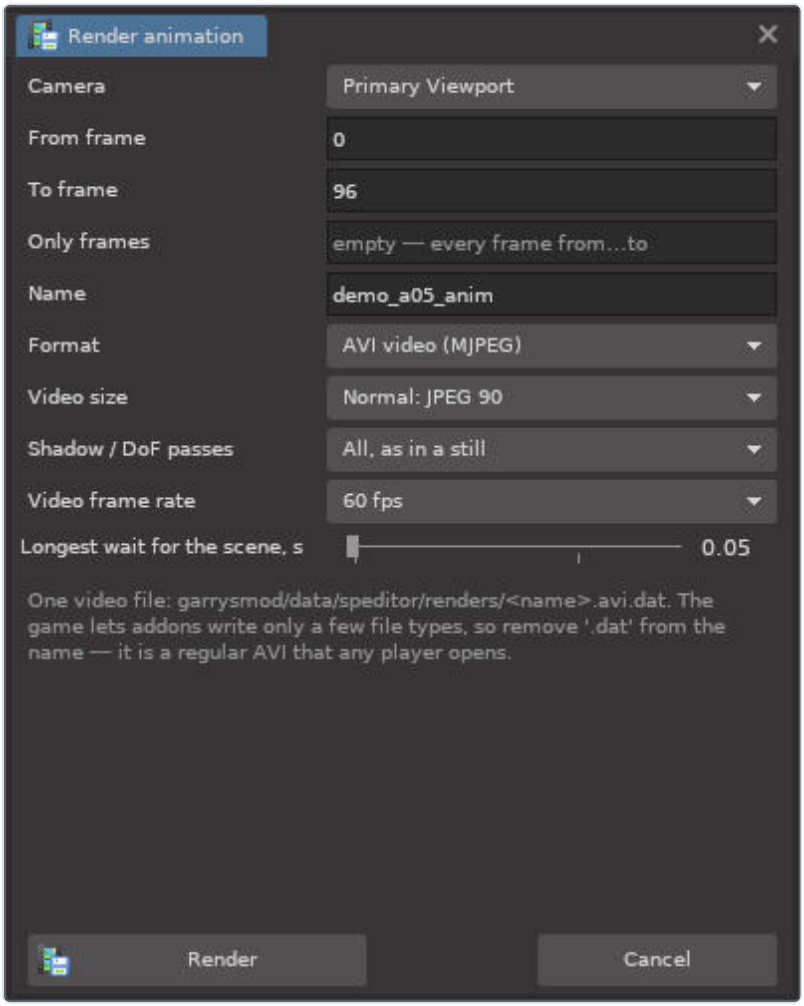
### Video size

Five JPEG levels: Largest (100), Normal (90), smaller by 40 % (70), by 65 % (50), by 80 % (30). Frames that did not change are not written twice.

### Shadow / DoF passes

All (as in a still), Half or Quarter — fewer passes render several times faster.

## Video frame rate




*Video frame rate* (in the Render window and in ▸ **File → Render animation...**): *As the animation* or 24, 25, 30, 48, 50, 60, 120 fps. Above the animation's own rate the frames in between are posed between the keys, as in SFM: a 24 fps animation comes out as a smooth 60 fps video of the same length. The sound track follows.

More frames take longer to render: 60 fps is two and a half times the frames of 24.

## Motion blur NEW

As a film camera sees it: every frame of the video becomes the mean of several moments around it (4, 8 or 16), so fast motion leaves a trail and a pan stays smooth. *Shutter* says how long the shutter stays open in a frame: **180°** is the usual film look, 360° the longest blur, 90° a sharper, staccato motion. The render takes as many times longer as there are moments.

Size, smoothing, depth of field and camera effects come from the same Render window. A progress line appears on the timeline; *Stop the animation render* cancels it.

## Making an MP4

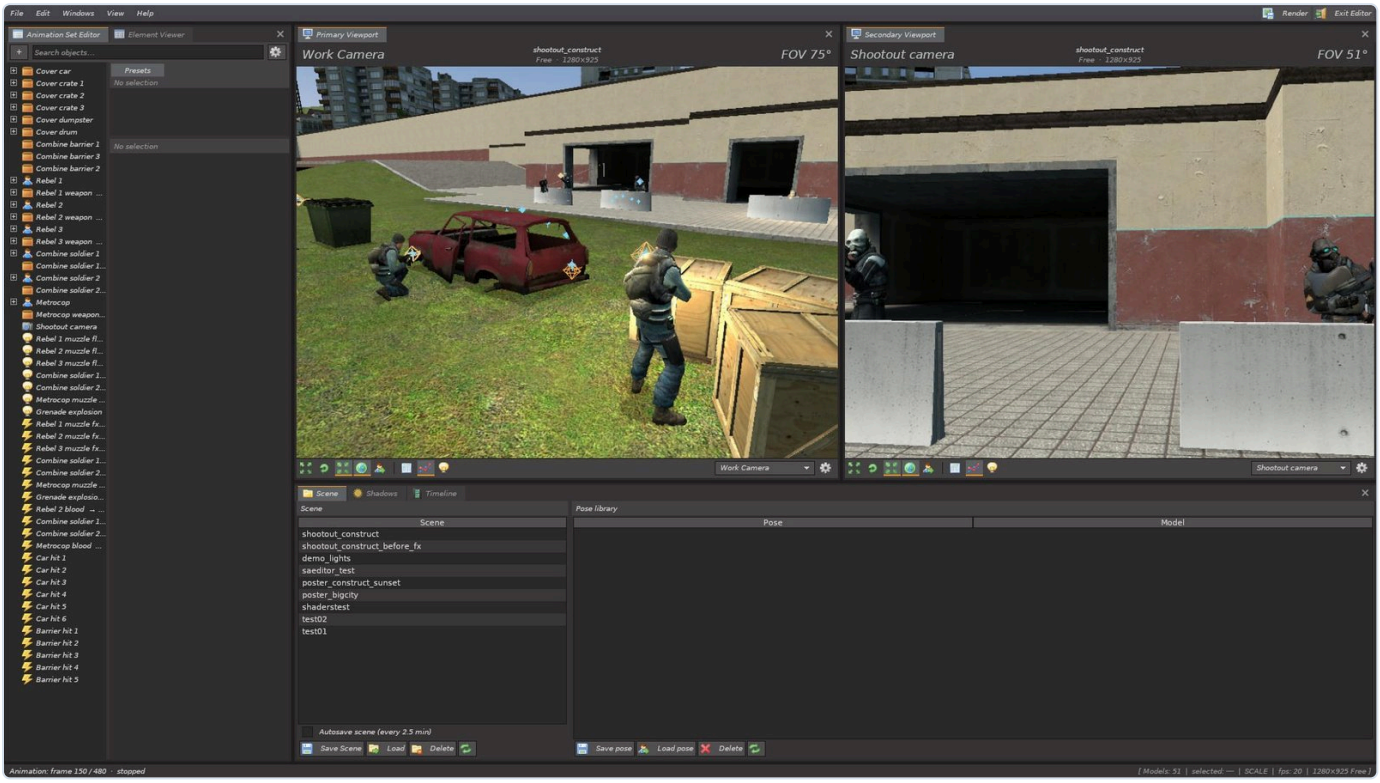
Garry's Mod cannot write MP4 itself. After the render the console prints a ready ffmpeg command. For an AVI:

```
ffmpeg -i myclip.avi -c:v libx264 -crf 18 -pix_fmt yuv420p -c:a aac myclip.mp4
```

> **!** Keep the game window focused while an animation renders: unfocused, the game runs at about 20 FPS, which slows the render and can skip very short effects.

# Scenes, poses and undo

The Scene tab of the bottom window keeps your work between sessions.




**Bottom window, Scene tab:** the scene list on the left, the pose library on the right.

## Scenes

- **Save Scene** asks for a name. **Load** replaces the current scene (asks first). **Delete** removes the file. The refresh button rereads the list.
- A scene keeps: every object with its pose, face, appearance and attachments; light settings; camera settings and camera effects; the work camera; the frame settings (aspect, size, exposure, depth of field); the SAeditor animation and sounds; what cameras and lights keep looking at.
- The map is recorded too. Loading on another map warns you: coordinates would put objects in random places.
- **Autosave scene (every 2.5 min)**: `autosave` is the newest copy, `autosave_2` ... are older ones.
- `shootout_construct` is an example that comes with the addon; it cannot be deleted.

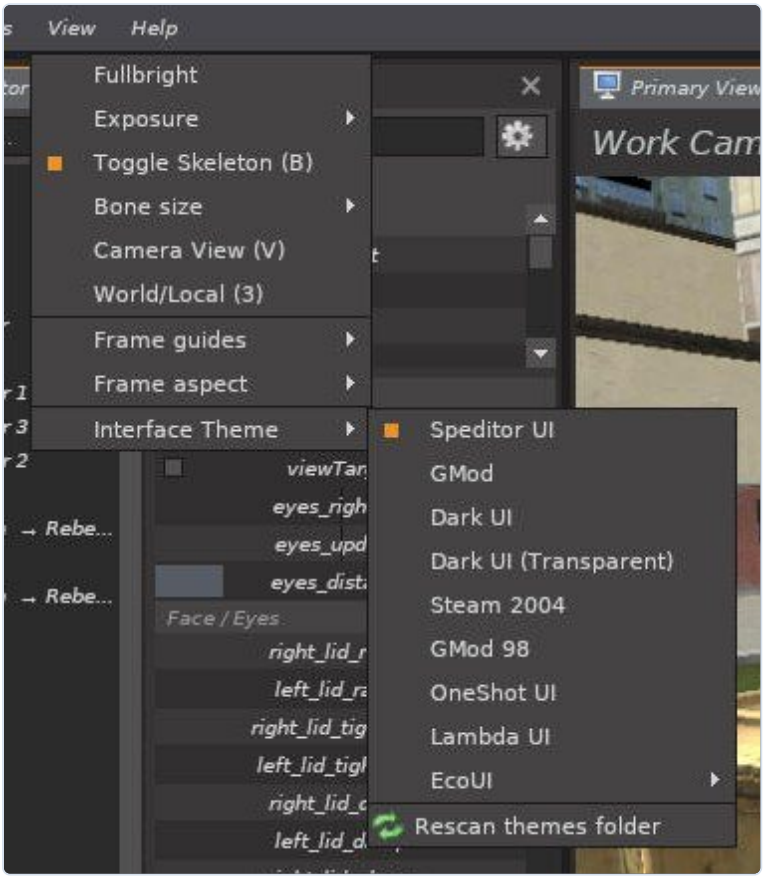
## Undo and redo

`Ctrl` + `Z` / `Ctrl` + `Y` work for every edit: moving, posing, face, lights, cameras, appearance, attaching, keys and sounds of SAeditor, even deleting objects (a deleted object waits in a trash until the history forgets it).

# Themes and notifications

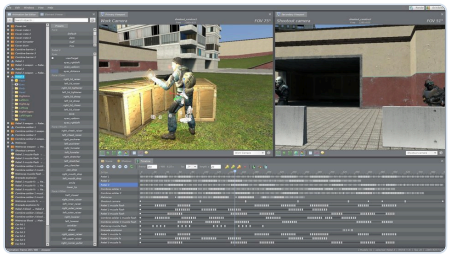
## Interface themes

▸ **View → Interface Theme** or Q menu → SPeditor → **Theme**. Changing the theme keeps open windows, the selected camera and fullbright. The editor uses only its own copies of the skins, so interface addons from the Workshop cannot break it.

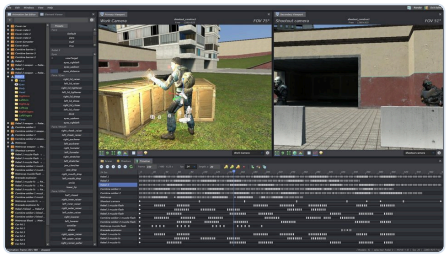





Speditor UI



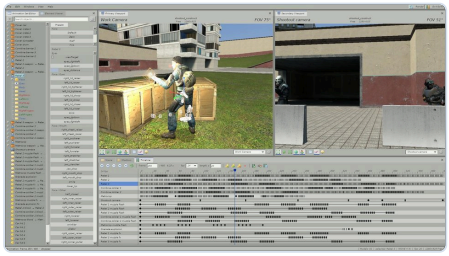
GMod



Dark UI



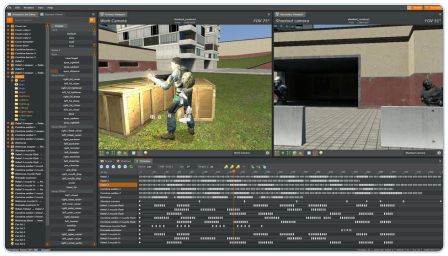
Steam 2004



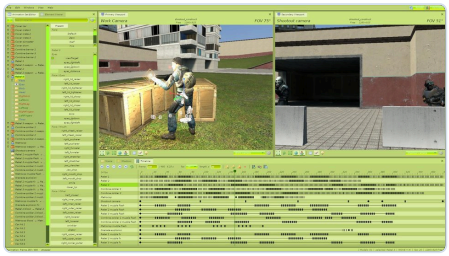
GMod 98



OneShot UI



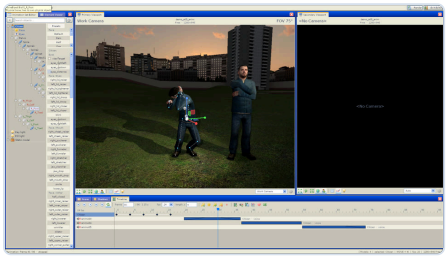
Lambda UI



EcoUI (16 colour variants)



Vanilla UI (navy and black)



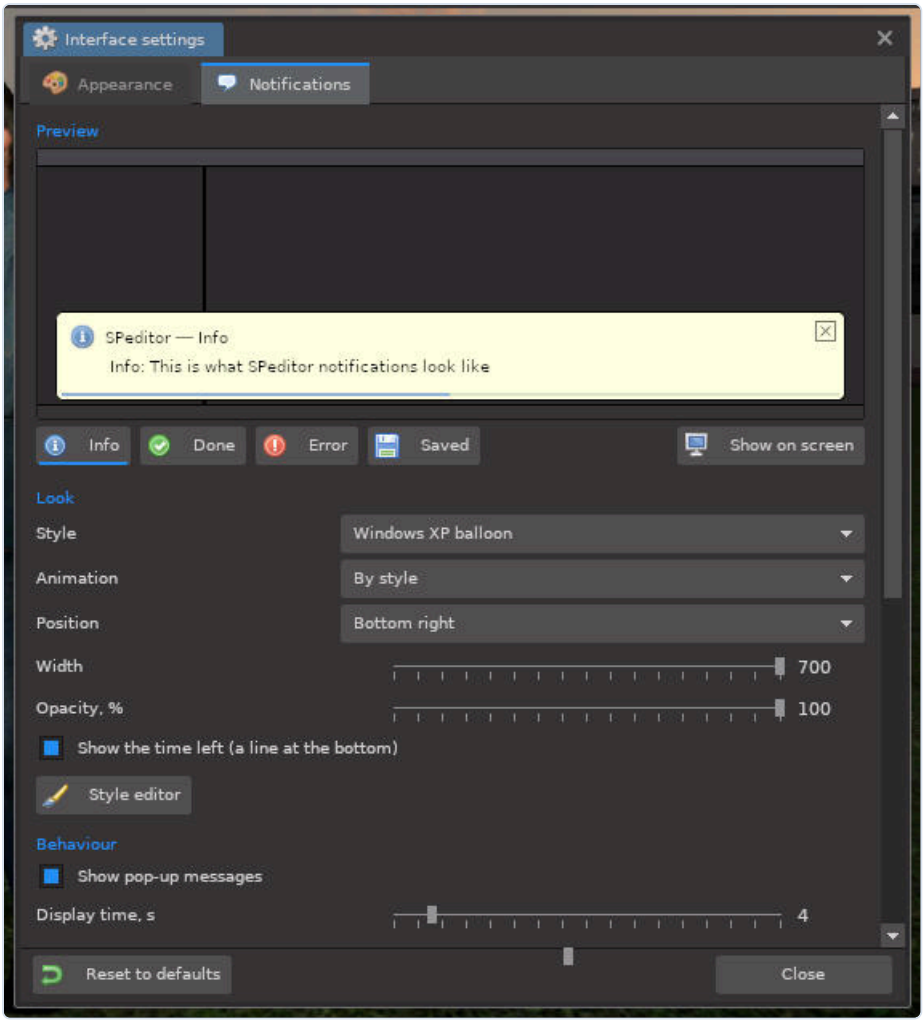
GMod XP

Vanilla UI and GMod XP also draw the editor's own windows, tabs, menu bar and status bar their own way. The accent colour, font and window corners are set in Interface settings.

There is also *Dark UI (Transparent)*. Your own Derma skin: put it as a folder into `garrysmod/addons/speditor/themes/` (create the folder if it is missing; a Lua file with `derma.DefineSkin`) and use *Rescan themes folder*.

## Notifications

Saves, autosaves, renders, errors and hints appear as cards on the screen.
▸ **Windows → Notifications** has the quick switches and the *Message log* with every message of the session; the full settings are on the **Notifications** tab of ▸ **View → Interface settings...**




### Preview

A live card on top; *Info*, *Done*, *Error*, *Saved* show each kind, *Show on screen* plays it for real.

### Style

By theme, SPeditor, Garry's Mod, Aero glass, Flat, Solid colour, Vanilla UI, Windows XP balloon, Console.

### Animation

By style, Fade, Slide in, Garry's Mod spring, Drop, Pop.

### Position · Width · Opacity

Six places on the screen; the size and transparency of the cards; how many cards at once; a line showing the time left.

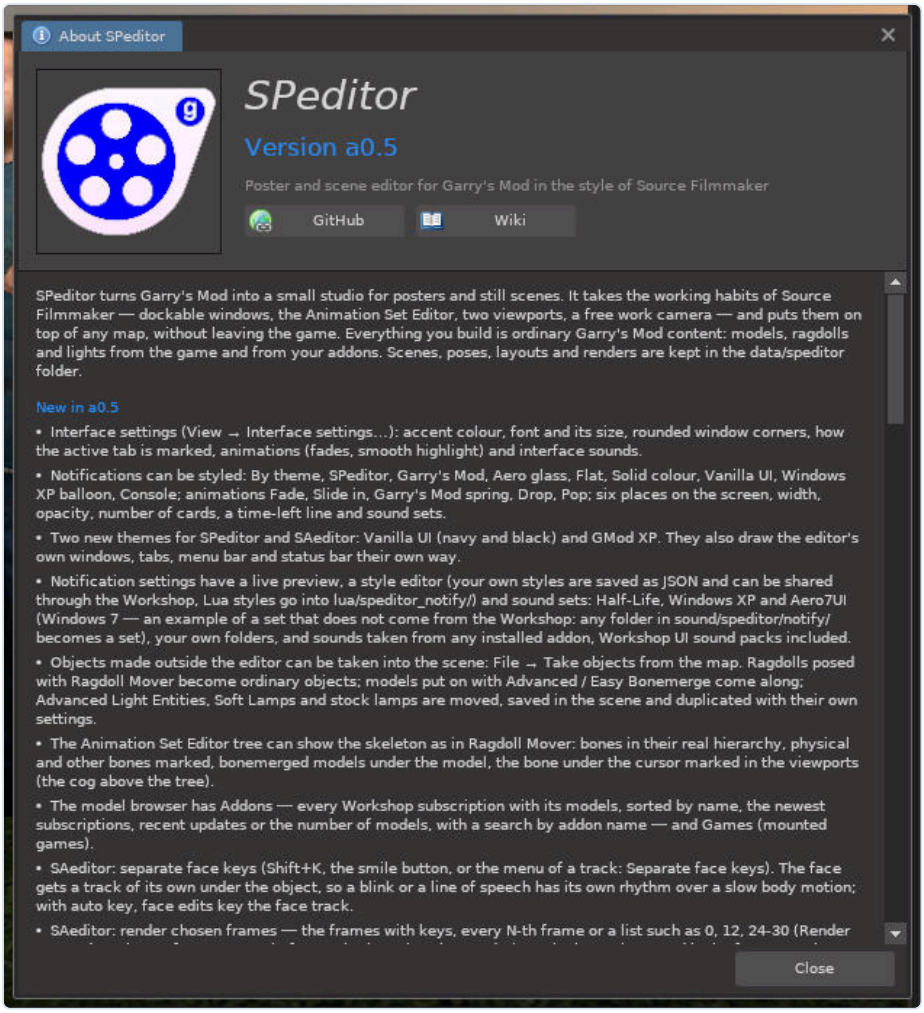
### Style editor

Your own style: colours, corners, icons. It is saved as JSON in `data/speditor/notify_styles/`; to share one in a Workshop addon put it into `data_static/speditor/notify_styles/`. Lua styles that draw everything themselves go into `lua/speditor_notify/`.

### Sounds

Sets Half-Life, Windows XP and Aero7UI (Windows 7), your own folders in `data/speditor/sounds/<set>/`, *Custom* (a file per kind) and *Sounds from installed addons* — any addon's sounds, Workshop UI sound packs included. Any folder `sound/speditor/notify/<set>/` in an addon becomes a set.

## About




▸ **Help → About SPeditor**: the version, what is new in it, the list of features and the authors of the bundled skins and sound sets (click a name to open its Workshop page, right-click to copy the link).

# Interface languages NEW

English and Russian are built in. Other languages come from the community as language packs: small addons that anyone can make and share in the Workshop.

- Switch the language: ▸ **Help → Interface language** in the editor (applied at once) or ▸ **Q menu → SPeditor → Language**. Installed packs are listed there with their authors.
- A string the pack has not translated yet is shown in English.
- On a server the server must have the pack installed too.

## Making a language pack

1. ▸ **Help → Interface language → Write a translation template**, type the language code (two Latin letters: `ja`, `de`, `pt_br`...). The file `garrysmod/data/speditor/lang/<code>.txt` holds every string of the editor and of the Help guides, in English, marked `TODO` (console: `speditor_lang_template <code>`).
2. Translate the texts on the right. Keep `%s`, `%d`, `%.1f` and `\n` as they are — SPeditor puts names and numbers there; a line that loses them is skipped. Set `name` (the language's own name) and `author`; `font` can name a font that has the letters of the language.
3. Rename the file to `<code>.lua` and put it into an addon of your own: `garrysmod/addons/<your_addon>/lua/speditor_lang/<code>.lua`. Restart the map — the language is in the menu.
4. Publish the addon to the Workshop. After an SPeditor update `speditor_lang_missing <code>` lists the new strings, and a new template keeps everything already translated.

> Until a0.6 SPeditor had a mod system. In a0.7 it is gone: SAeditor is a part of the editor, and languages are what the community adds. Scenes saved by a0.6 load as before.

# Hotkeys

The same list is in the editor: ▸ **Help → Hotkeys**, with the keys of SAeditor. Every key can be changed: ▸ **View → Interface settings → Keys** (more); below are the defaults.

| KEY | ACTION |
|---|---|
| **SELECTION AND CAMERA** | |
| `LMB` | Select object / bone |
| `Shift` / `Ctrl` + `LMB` | Add to / remove from selection |
| `LMB` on the selected character | Select the body part under the cursor |
| `RMB` (short click) | Object menu |
| `RMB` + `W` `A` `S` `D` / `Q` `E` | Fly the work camera (Q down, E up, Shift faster, Ctrl slower) |
| `Alt` + `LMB` | Orbit around the selection |
| Wheel | Move the camera in / out |
| `Alt` + Wheel | Field of view (zoom lens) |
| `F` | Focus on selection |
| `V` | Look through the selected camera / back |
| `B` | Show / hide skeleton |
| **TOOLS** | |
| `1` / `2` / `4` | Move / rotate / scale |
| `3` | World / local axes |
| `I` | Inverse kinematics |
| `Shift` + drag | Fine slider drag (Ctrl: coarse) |
| **EDITING** | |
| `Ctrl` + `Z` / `Ctrl` + `Y` | Undo / redo |
| `Ctrl` + `D` | Duplicate |

| | |
|---|---|
| `Del` | Delete |
| `Ctrl` + `C` / `Ctrl` + `V` | Copy / paste pose (in the viewport) |
| **OBJECT LIST (CURSOR OVER IT)** | |
| `Ctrl` + `A` | Select all objects (not locked or hidden ones) |
| `Ctrl` + `C` / `Ctrl` + `V` | Copy / paste objects |
| **WINDOWS** | |
| Drag tab bar | Dock: drop on a window edge / tab bar (Ctrl: keep floating) |
| Double-click tab bar | Maximize / restore window |
| Right-click tab bar | Window properties |
| **SAEDITOR** | |
| `Space` | Play / pause the animation |
| `K` | Set a key for the selection |
| `Shift` + `K` | Set a face key for the selection |
| `Ctrl` / `Shift` + wheel on the timeline | Zoom the timeline / scroll it in time |
| `,` / `.` | Previous / next frame |
| `Home` / `End` | First / last frame |
| `Ctrl` + click · `Shift` + box | Select several keys on the timeline |




**Help** → **Hotkeys** in the editor.

# Console commands and settings

Everything here can also be set in the interface; the console is useful for binds and for server owners.

## Commands

| COMMAND | ACTION |
| --- | --- |
| `speditor_toggle` | Open or close the editor |
| `speditor_render` | Render with the current settings |
| `speditor_lang_template <code>` NEW | Write a translation template to data/speditor/lang (languages) |
| `speditor_lang_missing <code>` NEW | List the strings a language has not translated yet |
| `saeditor_record` · `saeditor_record_stop` | Start / stop recording gameplay |
| `saeditor_show` [ `eyes` ] · `_stop` · `_toggle` | Play the animation in the game (the player; `eyes` — walk around the scene) |

## Settings (convars)

| VARIABLE |
| --- |
| `speditor_access` |
| `speditor_language` |

| |
|---|
| `speditor_theme` |
| `speditor_autosave` |
| `speditor_aspect` |
| `speditor_render_side` · `_custom` · `_w` · `_h` |
| `speditor_render_format` · `_quality` |
| `speditor_render_ss` · `_fullbright` · `_preview` |
| `speditor_render_engine` |
| `speditor_dof` · `_aperture` · `_samples` |
| `speditor_softlight_samples` |

`speditor_exposure`

`speditor_guides` · `speditor_bone_size`

`speditor_cam_speed` · `_wheel` · `_wheel_smooth` · `_fov`

`speditor_shadows` · `speditor_shadow_quality` · `_max` · `_dist` · `_softscale` · `_slope`

`speditor_mapshadow*`

`speditor_notify` · `_sound` · `_time`

`speditor_notify_style` · `_anim` · `_pos` · `_width` · `_opacity` · `_max` · `_bar` · `_soundset`

`speditor_ui_accent` · `_font` · `_fontsize` · `_corners` · `_tabstyle` · `_anim` · `_sounds`

`speditor_ui_scale`

`speditor_ase_tree` · `speditor_ase_nonphys` · `speditor_ase_merged`

`speditor_browser_sort`

`speditor_ragdoll_guard`

`speditor_ik` NEW

`speditor_stretch` NEW

`speditor_vp_look` NEW

`speditor_ssao` · `_radius` · `_strength` NEW

`speditor_shadow_onlylights` NEW

`speditor_ui_snd_click` · `_back` NEW

`saeditor_render_video` NEW

`saeditor_rec_phys` · `saeditor_rec_view` NEW

`saeditor_render_mode` · `_quality` · `_samples` · `_settle`

`saeditor_render_fps`

`saeditor_render_mblur` · `_shutter` NEW

`saeditor_bone_keys` · `saeditor_ghosts`

`saeditor_lip_strength`

`saeditor_rec_npcs` · `_sounds` · `_effects` · `_radius` · `_step`

`saeditor_show_camera` · `_loop` · `_look`

# Problems and answers

Collected from the questions players ask in the Workshop comments.

## The interface is huge / windows overlap on my screen

On screens narrower than ~1430 px the default layout of a0.3 broke. Fixed in a0.4. Until then: ▸ **Windows → Layout → No bottom panel** or arrange the windows yourself and *Save layout as…*.

Since a0.6 the size of the whole interface is set in ▸ **View → Interface settings → Appearance → Interface scale**: 50 % to 200 %, by default by the screen. The windows keep their places when it changes.

## Far objects look blurry in the viewport NEW

Before a0.7.1 the viewports were drawn into a picture of at most 1280 × 1024 and stretched, which blurred small far objects on 1440p / 4K screens and in an enlarged viewport — fixed. If the photo button (Render look) of the viewport bar is on, the blur is the depth of field: turn it off, or move the focus.

## The door I opened is not in the recorded animation NEW

Until a0.7.1 Record gameplay took only the doors of the map itself; doors spawned by tools and addons are recorded since a0.7.1. The door must be within the radius of the take.

## I changed a pose and it jumped back

The object is animated and the frame had no key: an edit between keys is not kept when the playhead moves. Press `K` or turn Auto key on first. See Keys and the timeline.

## Where did the JPEG / PNG frames of the animation render go?

a0.6 renders animations to AVI only. Need pictures? `ffmpeg -i clip.avi clip_%04d.png` takes the frames out of it; a single frame is *Only the current frame*.

## Clothes I attach do not move with the model

Use ▸ **Attach to… → character → Wear** (a0.4). Attaching to a bone is rigid by design: good for weapons, not for clothing. See Attaching and wearing.

## How do I resize a bone (a bigger hand or head)?

Select the bone, press `4` and drag the centre (even) or an axis (stretch). In a0.3 this works on fingers and clavicles; a0.4 allows every bone. See Posing.

## Can I pose bones without the physgun?

Yes, that is what the editor is for: click a bone marker and use the gizmo, or turn on IK (`I`) and drag a hand or foot.

## Imported animation has no finger movement

Fixed in a0.4: fingers are imported as bone offsets. Re-import the animation after updating.

## Limbs cannot be moved / snap back

Usually an addon that controls ragdolls (active ragdolls, Fedhoria and similar). Since a0.5 SPeditor holds the pose of its ragdolls while the editor is open (`speditor_ragdoll_guard 0` turns this off). If a limb still moves by itself, disable that addon while you use the editor.

## After importing an animation the neck looks stretched

Fixed in a0.5: the neck, the top of the spine and the clavicles keep the ragdoll's shape. Import the animation again.

## I posed a ragdoll with Ragdoll Mover / put on a model with Bonemerge — can the editor use it?

Yes, since a0.5: ▸ **File → Take objects from the map**. See Objects.

## A model has bones but is not a ragdoll — can I pose it?

Yes, since a0.5: it stands still and its bones turn; IK works on hands and feet; *Base pose* picks its animation. See Models without physics.

## How long can an animation be? Can the video be 60 fps?

Since a0.5 an animation can be up to an hour long, and the video has its own frame rate up to 120 fps: a 24 fps animation renders as smooth 60 fps. See Video.

## Do custom models and Workshop ragdolls work?

Yes, since the first alpha. Find them in the Model Browser or type the path into *Spawn by path*. Models made as SFM “ragdolls” but ported as props can still be posed bone by bone with the gizmo.

## Where are my scenes and renders?

In `garrysmod/data/speditor/`. ▸ **Help → Where are my files...** shows every folder and copies its path. See Where your files are.

## Can I save the animation as a video?

Yes: Render window → *The whole animation* → *AVI video*. Remove `.dat` from the file name. For MP4 use the ffmpeg command printed in the console. See Video.

## Can I cut a sound?

Since a0.4: drag the edges of a sound bar or use *Cut in two at the playhead*. See Sounds.

## Can skins and bodygroups change during an animation?

Since a0.4 they are part of every key and switch exactly at the key.

## Does it work with soft lamps?

SPeditor has its own soft lights since a0.2: *source size* gives a real penumbra in the render, plus edge softness and 19 ready light kinds. See Lights.

## My shadows have stripes or float away from objects

Stripes: raise *shadowDepthBias* or *shadowSlopeBias* of the light. Floating shadow: lower them. Only spot lights cast shadows.

## Workshop shaders do not appear in the render

Tick *Engine frame* in the Render window and turn the effect on in ▸ **Windows → Addon shaders**.

## Is it compatible with Stop Motion Helper, Portal Gun, StormFox?

Stop Motion Helper: since a0.5 its saves can be taken into SAeditor and written back (Stop Motion Helper). Portal Gun and StormFox are not tested yet — please report what you find in the Workshop discussions.

# What's new

## a0.7.1 NEW

A small patch made from players' comments.

- **Your view as a camera:** Record gameplay turns what you saw — look, turns, zoom — into a scene camera, and the video can be shot through your eyes; your own body is not drawn in it (more).
- **Doors spawned by tools and addons** are recorded too, not only the doors of the map (more).
- Fixed: viewports looked blurry on big screens (1440p, 4K) or when enlarged, small far objects most of all.

## a0.7

- **Motion Editor:** a time selection with falloff on the timeline; one edit goes over a whole stretch of the animation, as in SFM (more).
- **Motion blur** in the animation render, with a shutter of 90°, 180° or 360° (more).
- **Look-at:** a camera or a light keeps looking at an object while the animation moves it (more).
- **SAeditor is built in:** the mod system is gone; old scenes, layouts and changed keys load as before.
- **Community languages:** language packs in the Workshop, a translation template written by the editor (more).
- **Render window:** Picture and Animation tabs, the shots of the video edited right there, WebM on Windows (more, video).
- **Ambient occlusion (SSAO)** of the editor's own, and the **Render look** of the Primary Viewport (more).
- **A camera records its own moves** (more); **point lights cast shadows**, scene objects keep only the shadows of your lights (more, shadows).
- **PAC3 outfits** and the PAC3 window — a hat or glasses on a bone with sliders (more); **Wear** on any skeleton, TF2 paint and crit glow (more).
- **Stretching bones** ( T ), IK on from the start (more).
- **Record gameplay:** physics of ragdolls and props, doors and lifts of the map (more).
- **Your own sounds** for the interface, notifications and timeline clips (more, clips); your dupes and saves taken into the scene (more); new How to use guides with a video player.
- Fixed: ragdoll limbs that could not be moved when another addon unfreezes ragdolls; characters with wrong physics bones (Enhanced TF2 MvM bots); the interface without the mouse after Alt+Tab; the game HUD that stayed hidden; PAC3 wiping poses; the Render window keeping the previous scene.

## a0.6

Made from players' feedback in the Workshop comments and discussions.

- **Record gameplay (machinima):** play and it becomes animation — you, the NPCs around, their weapons, sounds, tracers, explosions and projectiles (more).
- **Bone keys and ghosts:** every bone moves between its own keys, as in SFM; the pose at the neighbouring keys is drawn see-through (more).
- **Continuous keys:** a new key is a spline the motion flows through; new kinds of motion — with a swing, springy, bouncing, gentle, sharp (more).
- **Camera cuts and the player:** the Cameras row of the timeline, a player in the style of Windows Media Player with a render look, walking around the scene; the video follows the cuts (more).
- **Lip sync** by the phonemes of a WAV or the loudness of any sound (more).
- **Help → How to use:** illustrated guides with live demonstrations inside the game (more).
- **Selection and keys:** Shift + click multi-select, Ctrl + A / C / V in the object list, one-step duplicates, every hotkey can be changed (more, keys).
- **Interface scale** 50–200 %, windows keep their places when it changes (more).
- **Posing:** the rotate gizmo has a trackball and a view ring (more). Several keys selected at once on the timeline; camera depth of field and exposure in keys; Auto key records a camera you fly through.
- **Render:** Steam screenshot format (more); the animation render writes AVI only.
- **Addons:** legacy addons in the Model Browser, animation packs of any addon, ready motions and camera moves (more).
- Fixed: new objects missing from the object tree; sounds kept playing after *Remove all animation*; windows slid down after the interface scale changed; server: a size limit for tables sent by players.

## a0.5

Made from players' feedback in the Workshop comments.

- **Interface settings:** accent colour, font and size, window corners, tab mark, animations, sounds; two new themes, Vanilla UI and GMod XP (more).
- **Notifications:** eight styles, animations, six places, a style editor, sound sets and sounds from any addon (more).
- **Take objects from the map:** Ragdoll Mover poses, Advanced / Easy Bonemerge, Advanced Light Entities, Soft Lamps, stock lamps (more).
- **Posing:** plane handles on the Move gizmo, IK that keeps the feet planted, bones locked to other bones, a click on the body picks the bone, models without physics can be posed (more).
- **Skeleton view** of the object tree, as in Ragdoll Mover (more).

- **Model Browser:** Addons with sorting and search, Games, spawn-menu icons (more).
- **SAeditor:** animations up to an hour long with a zoomable timeline; face keys; Speed up / Slow down motion; a curve editor; procedural breathing and blinking; Stop Motion Helper import and export; sound fades; adding several sounds at once (more, curves).
- **Video:** its own frame rate up to 120 fps (a 24 fps animation as smooth 60 fps) and rendering only chosen frames (more).
- **Help:** Where are my files, Interface language switched at once, a card when every window is closed (more).
- Fixed: an imported animation stretched the neck and shoulders; ragdolls taken over by other addons; windows shrinking by a pixel at every layout save.

## a0.4

- **Wear:** clothing, armour and hair made for the same skeleton follow every bone of the character (more).
- **Scale any bone:** hands, head, arms — not only fingers (more).
- **Fingers in imported animations:** fists clench, hands grip (more).
- **Skins and bodygroups in animation keys** (more).
- **Cutting and trimming sounds** on the timeline, in playback and in the video (more).
- Fixed: the layout broke on screens narrower than ~1430 px; a saved layout that does not fit is replaced by the default.
- Fixed: keys in the Edit menu were drawn with a square box.

## a0.3

- Effects: particle effects of the game and of Workshop addons, frozen at any moment; Emitter and engine effects.
- Effects in SAeditor: the moment is keyed; “Play the effect from this frame”.
- Sounds and voices on the timeline; video with sound; faster animation render; five video sizes.
- Animations from another model with the same skeleton; several characters at once.
- Camera motion: shake and smoothness. Camera depth of field and exposure as in SFM.
- Example scene `shootout_construct`.

## a0.2

- Soft light: source size (penumbra), edge softness, glow, visible beam, 19 light kinds, light masks. The mod system and SAeditor.

# Credits

SPeditor is made for Garry's Mod (Facepunch Studios). Source Filmmaker is a product of Valve; SPeditor is not affiliated with Valve. The GMod theme is a copy of the standard Derma skin by Garry Newman. The bundled interface skins belong to their authors:

| SKIN | AUTHOR |
| --- | --- |
| EcoUI | ELuminumG24 |
| Dark UI Spawnmenu | AquaRex |
| Steam 2004 Derma Skin | Maks1mDiamond |
| GMod98 UI Skin | AwesomeGraczGie21 |
| OneShot - UI Skin | Flarky |
| Lambda UI | □□□□□ |
| Vanilla UI+ Derma Skin (Vanilla UI) | Qude |
| Windows XP Classic UI & HUD (GMod XP) | w4da |

Notification sound sets:

| SOUNDS | AUTHOR |
| --- | --- |
| Half Life UI Sounds | BPATAPb_DINAMO |
| Windows XPerience UI Sounds | Jade |

*Take objects from the map* and the Stop Motion Helper import were built against the sources of Ragdoll Mover, Advanced Bonemerge, Easy Bonemerge, Advanced Light Entities, Soft Lamps and Stop Motion Helper — thanks to their authors.

Workshop shader support (Engine frame) was built and tested with GShader Library by Akabenko. Thanks to everyone who reports bugs and ideas in the Workshop comments.